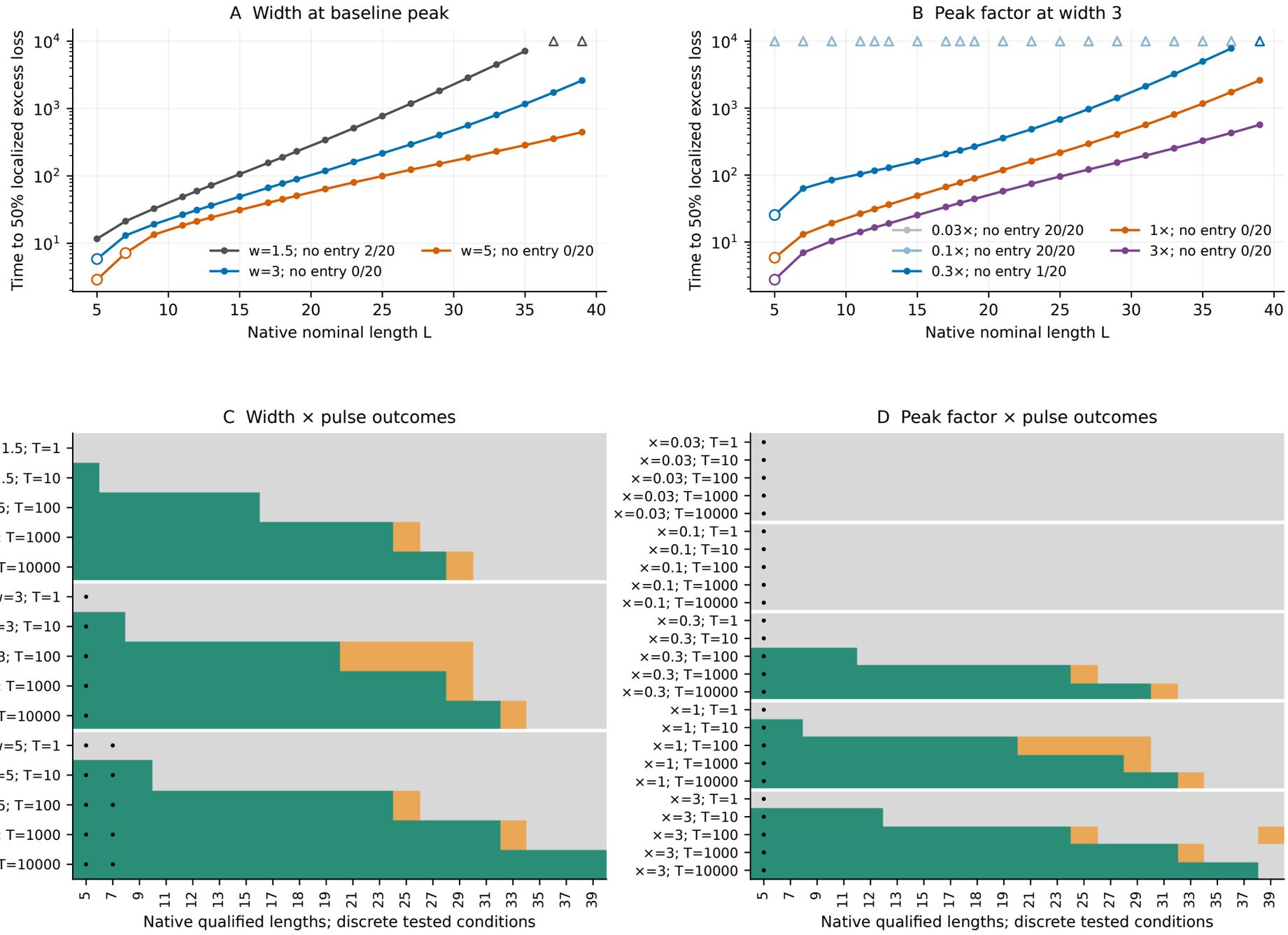
Wave-pinning: complete tested width, strength and pulse series

20 exact qualified basals; 720 protocol rows including 20 valid no-cue controls; 10000 native units after withdrawal.

Observed times are saved-entry bracket midpoints; exact brackets and finite no-entry horizons remain in the compact tables.



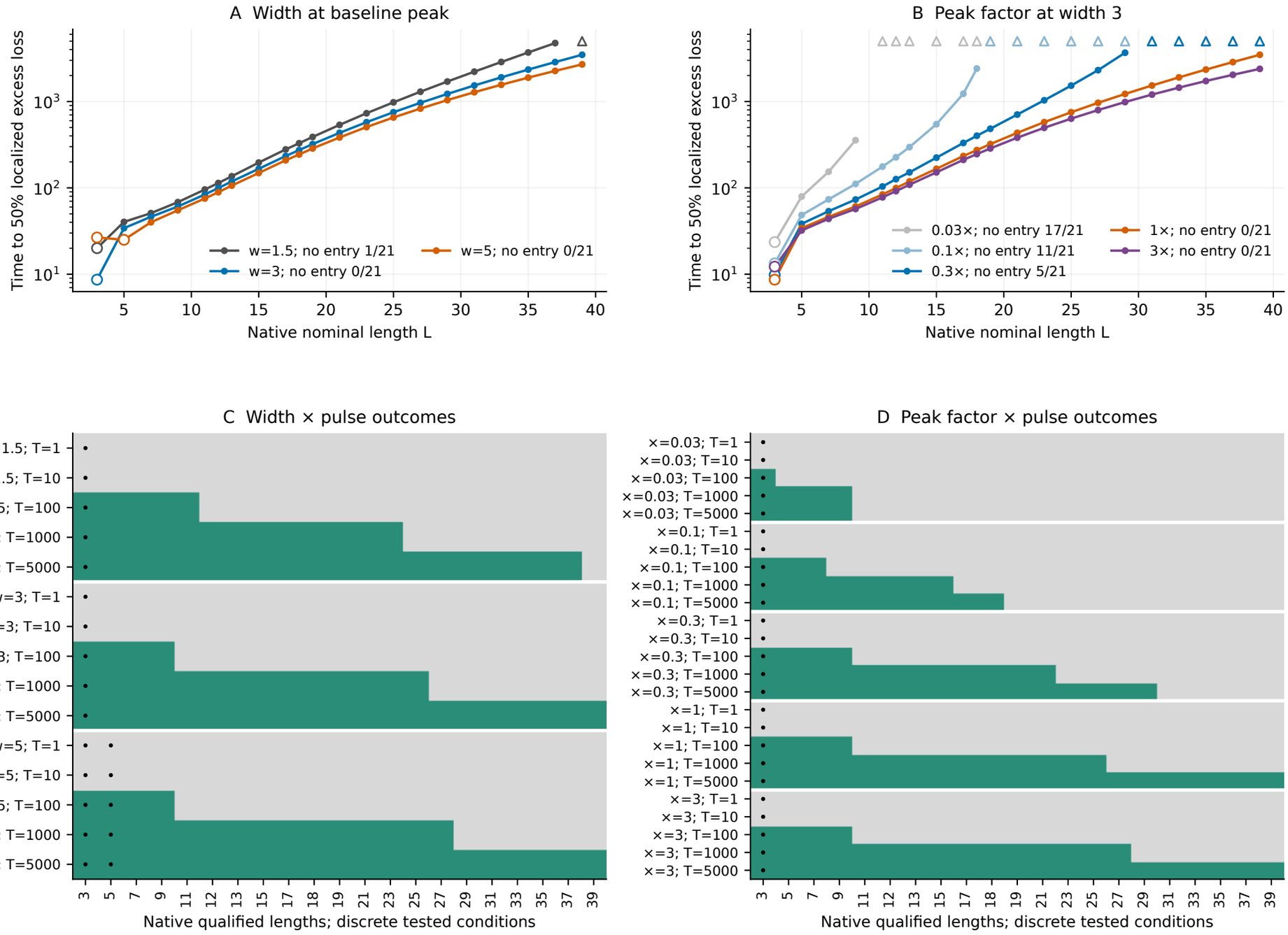
Horizon triangles may coincide: curve legends give no-entry counts. Open circles / map dots mark initial cue/core overlap.

Baseline 1× / middle width is displayed in both contexts, but is one protocol identity. Scalar gates do not alone establish spatial replacement.

Goryachev: complete tested width, strength and pulse series

21 exact qualified basals; 756 protocol rows including 21 valid no-cue controls; 10000 native units after withdrawal.

Observed times are saved-entry bracket midpoints; exact brackets and finite no-entry horizons remain in the compact tables.



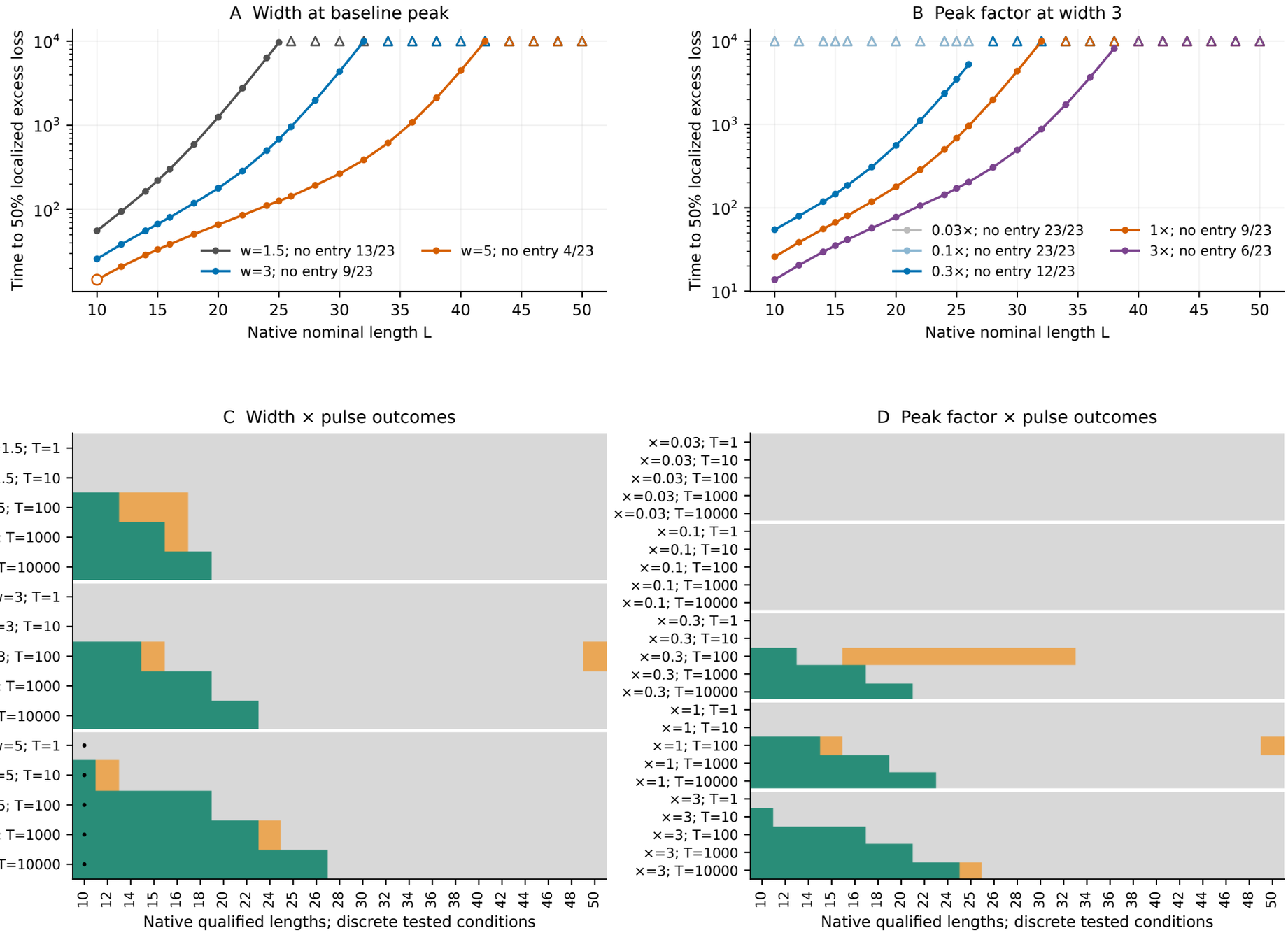
Horizon triangles may coincide: curve legends give no-entry counts. Open circles / map dots mark initial cue/core overlap.

Baseline 1× / middle width is displayed in both contexts, but is one protocol identity. Scalar gates do not alone establish spatial replacement.

Holmes 3: complete tested width, strength and pulse series

23 exact qualified basals; 828 protocol rows including 23 valid no-cue controls; 10000 native units after withdrawal.

Observed times are saved-entry bracket midpoints; exact brackets and finite no-entry horizons remain in the compact tables.



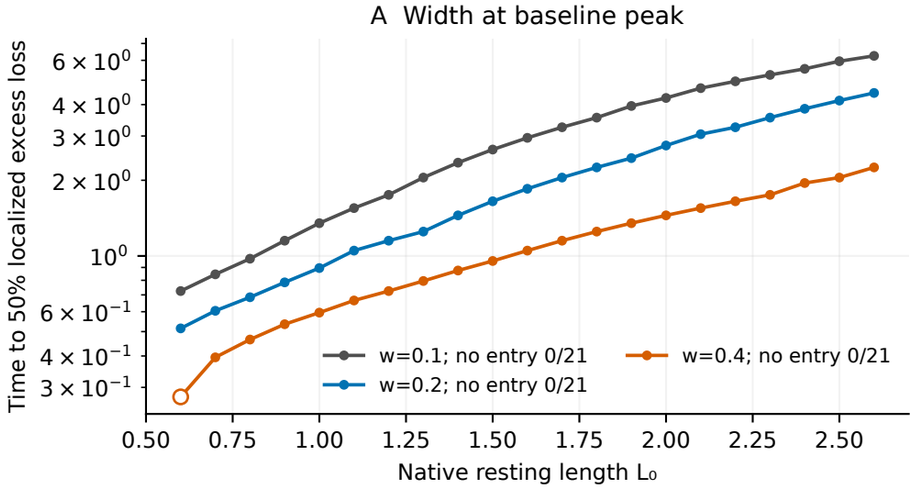
Horizon triangles may coincide: curve legends give no-entry counts. Open circles / map dots mark initial cue/core overlap.

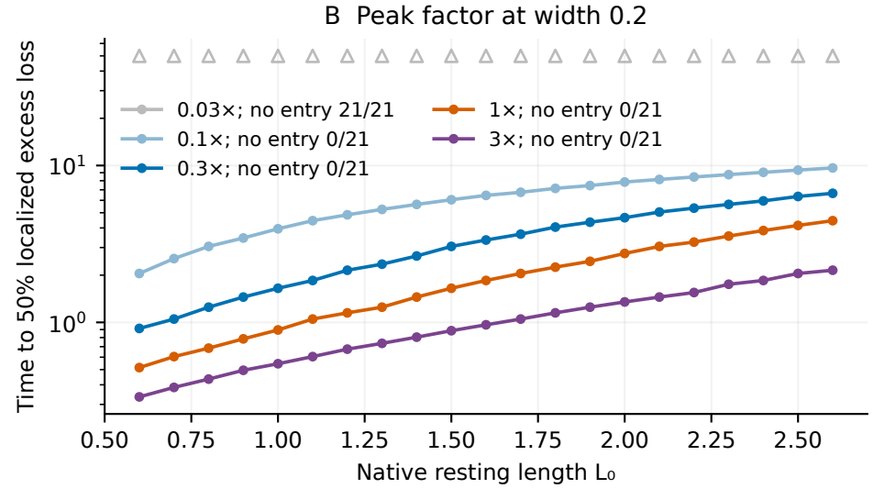
Baseline 1× / middle width is displayed in both contexts, but is one protocol identity. Scalar gates do not alone establish spatial replacement.

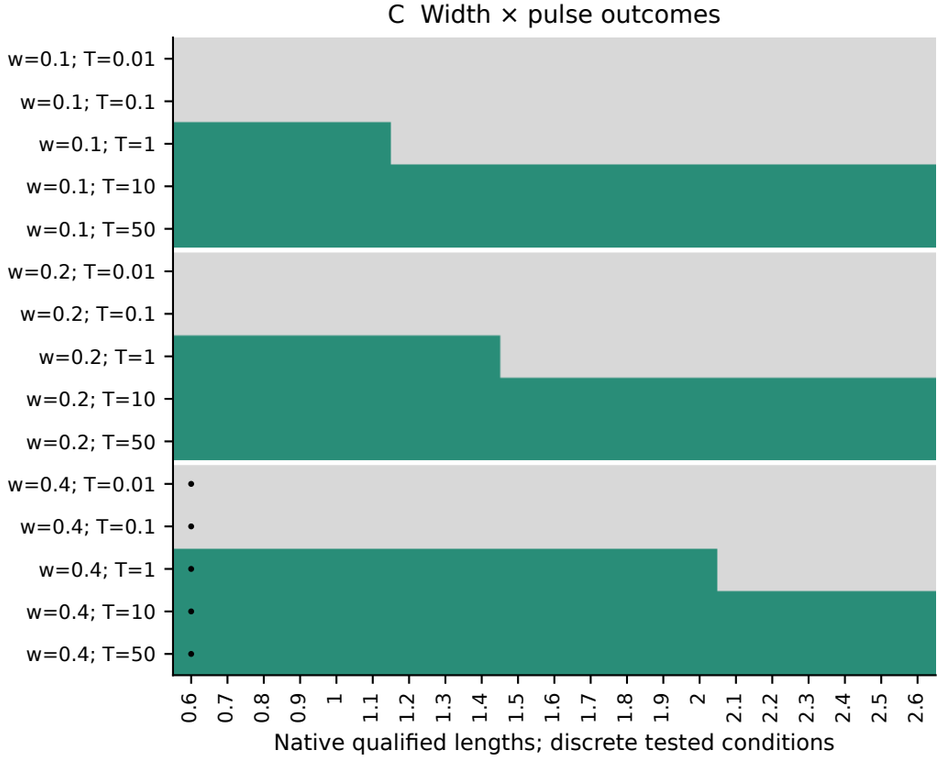
Spring: complete tested width, strength and pulse series

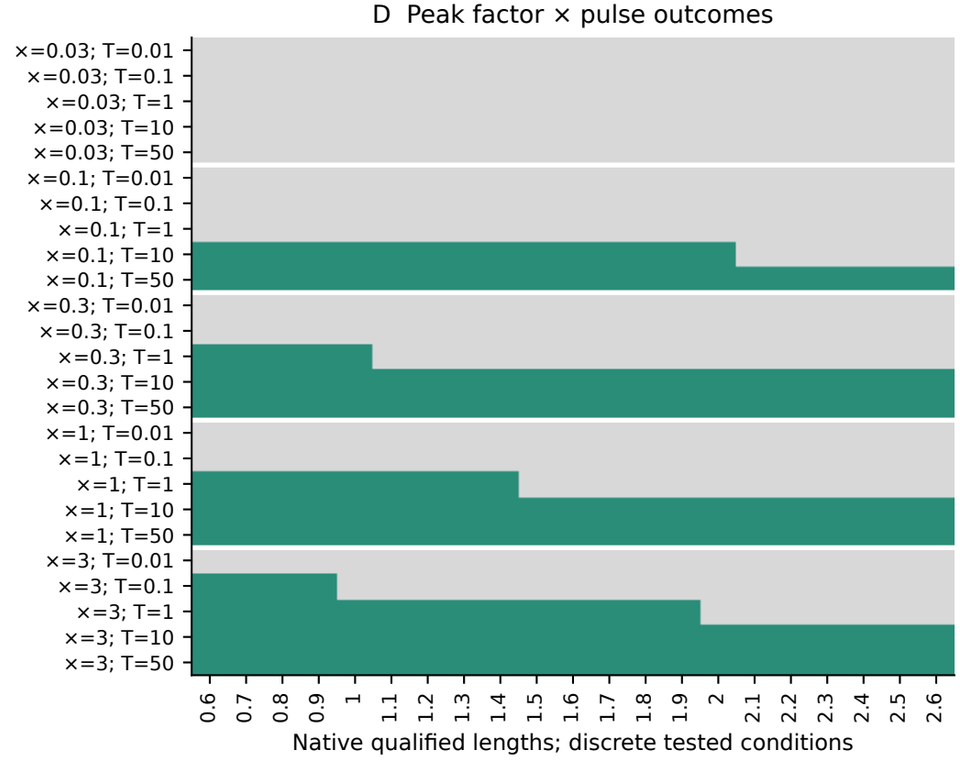
21 exact qualified basals; 756 protocol rows including 21 valid no-cue controls; 10000 native units after withdrawal.

Observed times are saved-entry bracket midpoints; exact brackets and finite no-entry horizons remain in the compact tables.









Scalar full strength: fail; field tail: pass
Scalar full strength + field tail: pass
Field tail: fail (scalar flag separately retained)

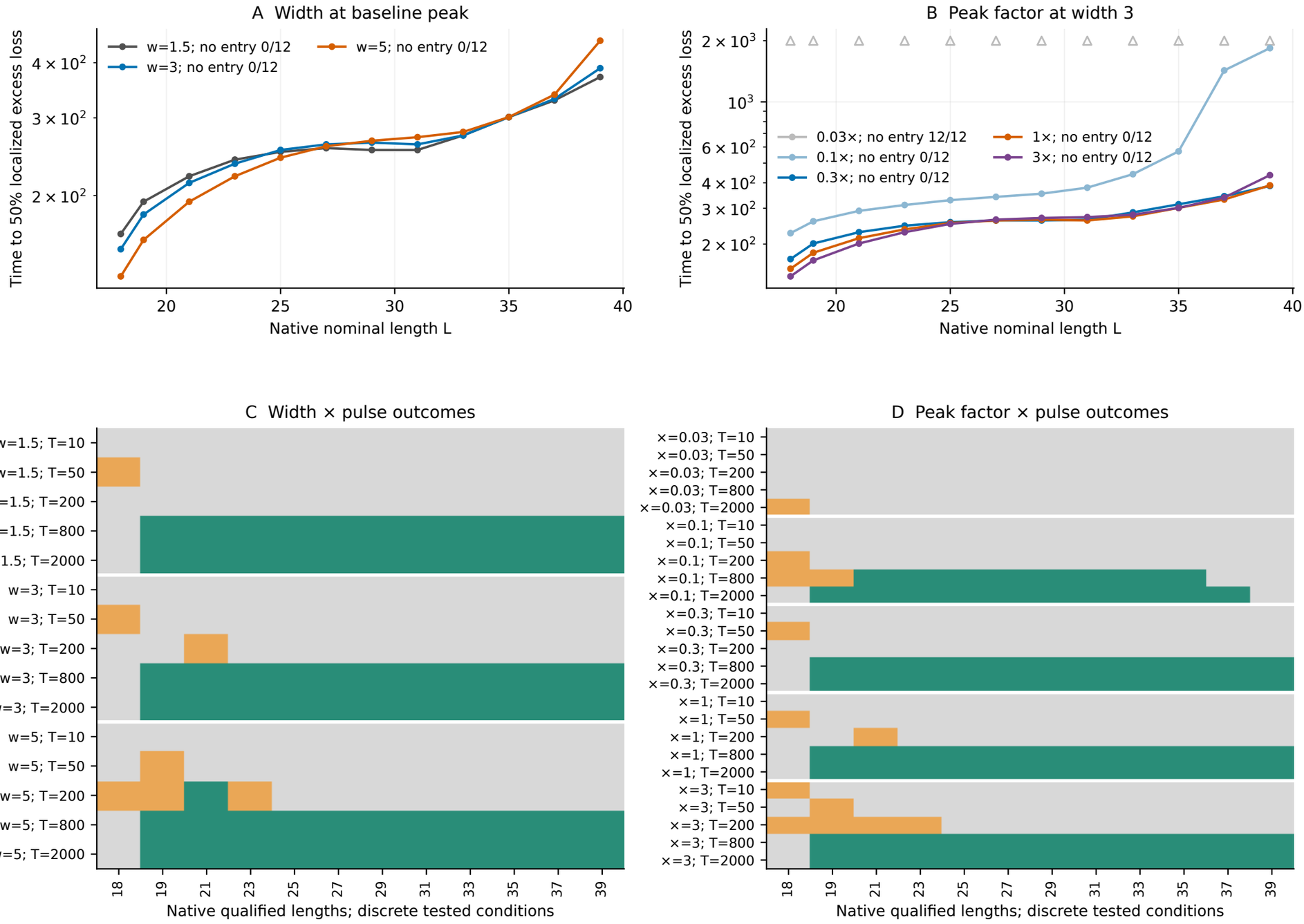
Horizon triangles may coincide: curve legends give no-entry counts. Open circles / map dots mark initial cue/core overlap.

Baseline 1× / middle width is displayed in both contexts, but is one protocol identity. Scalar gates do not alone establish spatial replacement.

Corrected linear deBelly: complete tested width, strength and pulse series

12 exact qualified basals; 432 protocol rows including 12 valid no-cue controls; 10000 native units after withdrawal.

Observed times are saved-entry bracket midpoints; exact brackets and finite no-entry horizons remain in the compact tables.

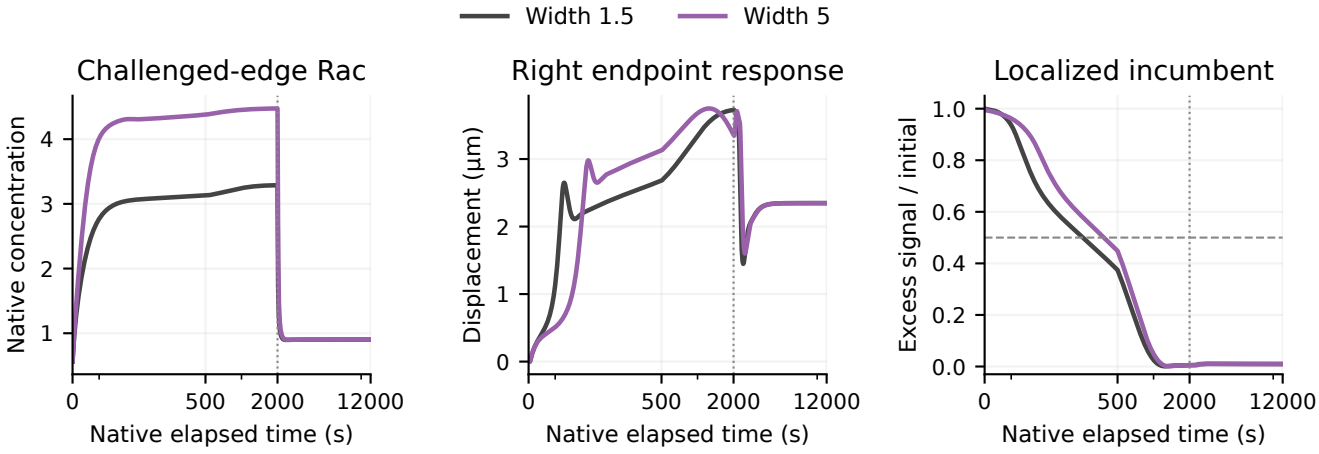


Horizon triangles may coincide: curve legends give no-entry counts. Open circles / map dots mark initial cue/core overlap.

Baseline 1× / middle width is displayed in both contexts, but is one protocol identity. Scalar gates do not alone establish spatial replacement.

More local activation need not mean earlier incumbent loss

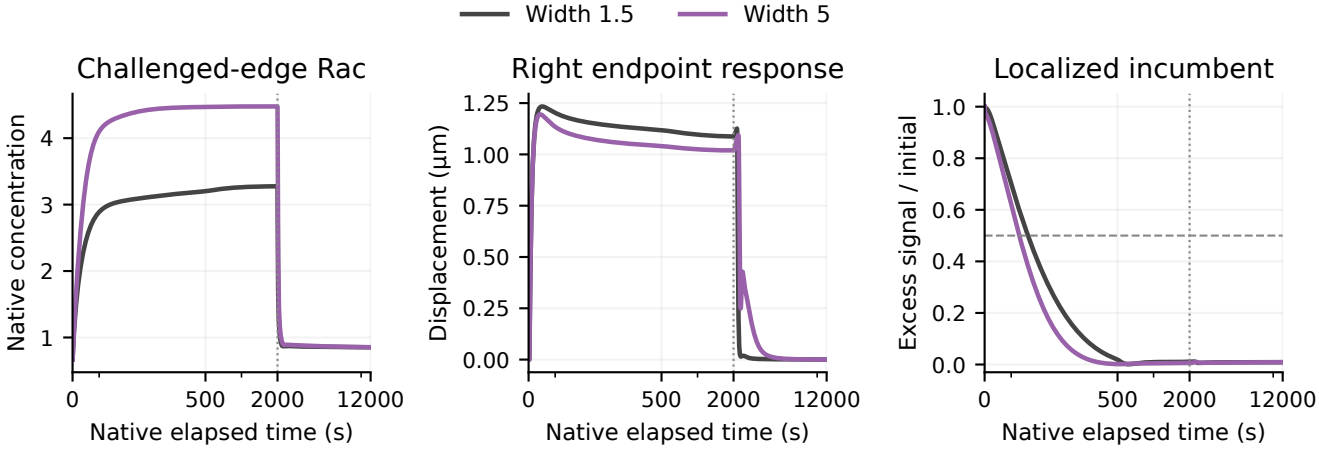
Corrected linear deBelly L39; cue off at 2000 s, followed by 10000 s off



Time axis is linear through 500 s, logarithmic thereafter; dotted line marks withdrawal.

More local activation need not mean earlier incumbent loss

Corrected linear deBelly L18; cue off at 2000 s, followed by 10000 s off

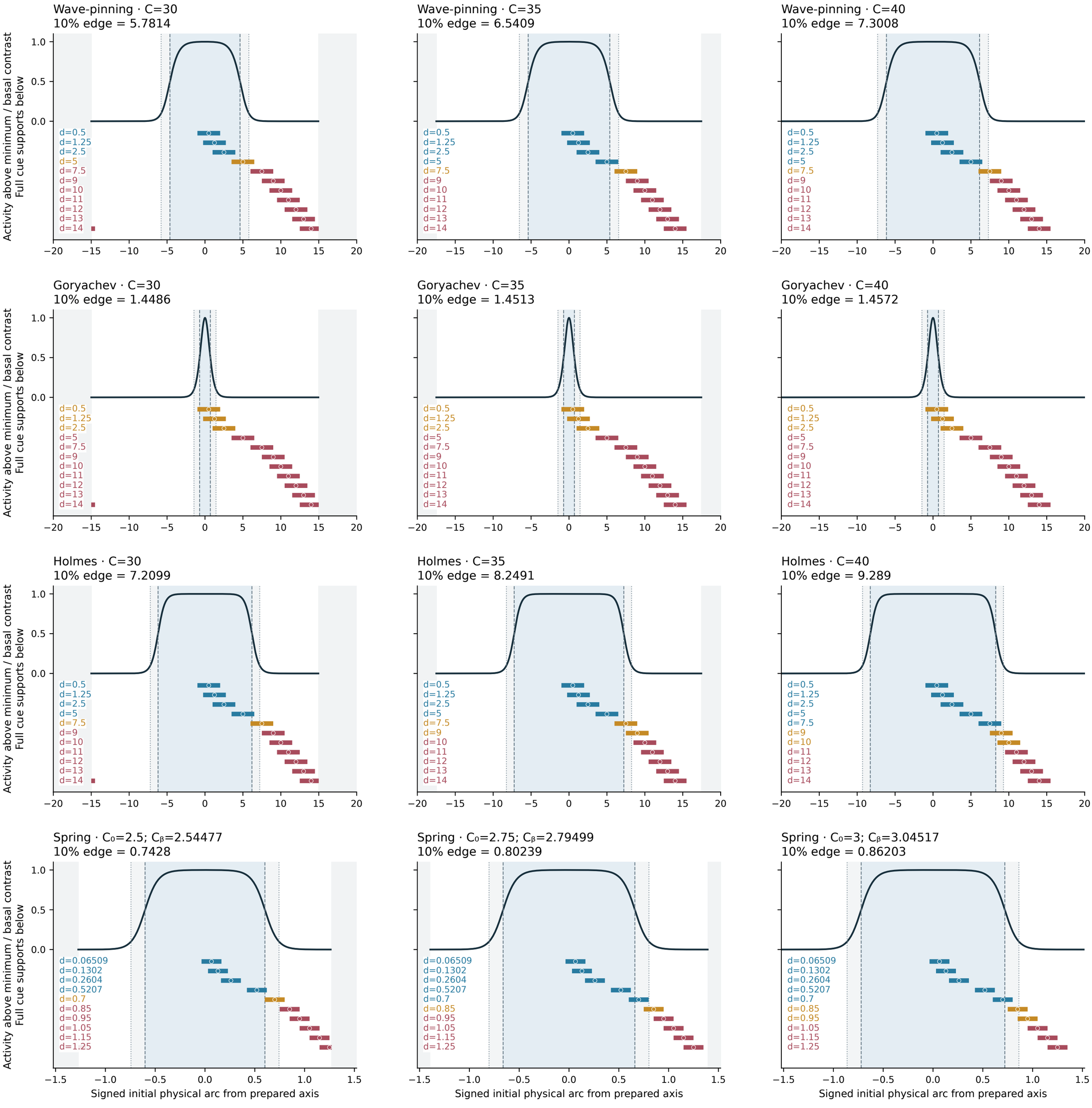


Time axis is linear through 500 s, logarithmic thereafter; dotted line marks withdrawal.

Qualified narrow-range design: internal, edge and outside cues

All 129 frozen placements on the actual 12 basals; 645 cue-duration conditions. No parameters or basals are retuned.

Bars: full physical support. Dots: cue centres. Colours classify the whole footprint against the initial 10%-contrast region.



Inside support **Straddling support** **Outside support**

Black profile: [u − min(u)] / [max(u) − min(u)]. Dashed edges: 50%; dotted edges: 10%. These are operational activity thresholds.

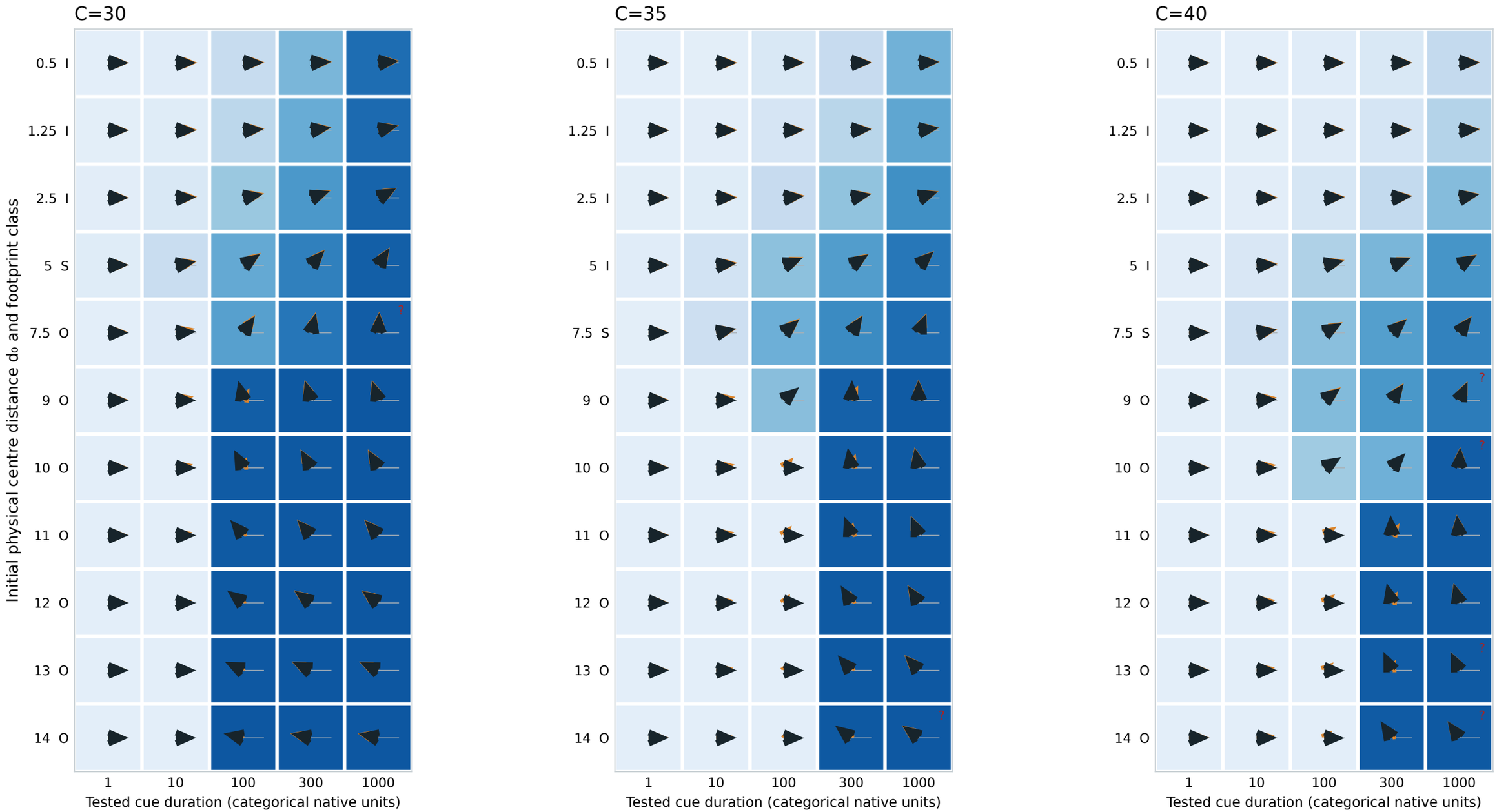
Common physical x scales within model rows; grey outer areas lie beyond smaller rings. Supports wrap periodically near the antipode.

Spring distances are initial physical arcs of stationary material centres; subsequent physical arcs vary with C(t). No site ancestry or cell-motion claim.

Wave-pinning · Every pulse: withdrawal and retained activity vectors

Orange: cue withdrawal. Black: after 13,000 cue-free units. Each arrow is the actual P; one common scale in every cell.

Background: fraction of initial target-angle error removed at the retained endpoint (colour clipped to 0–1 only). Grey: pending / weak final R.



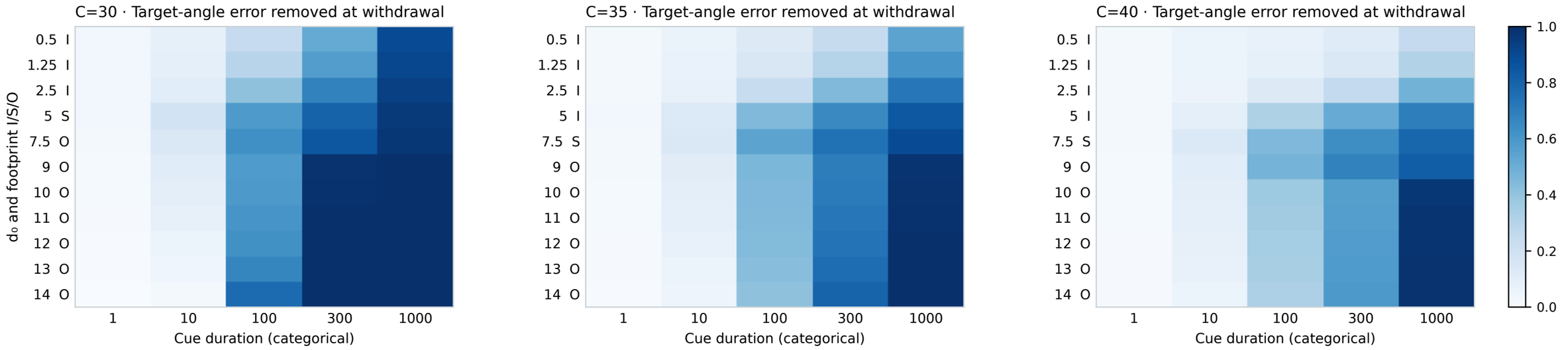
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

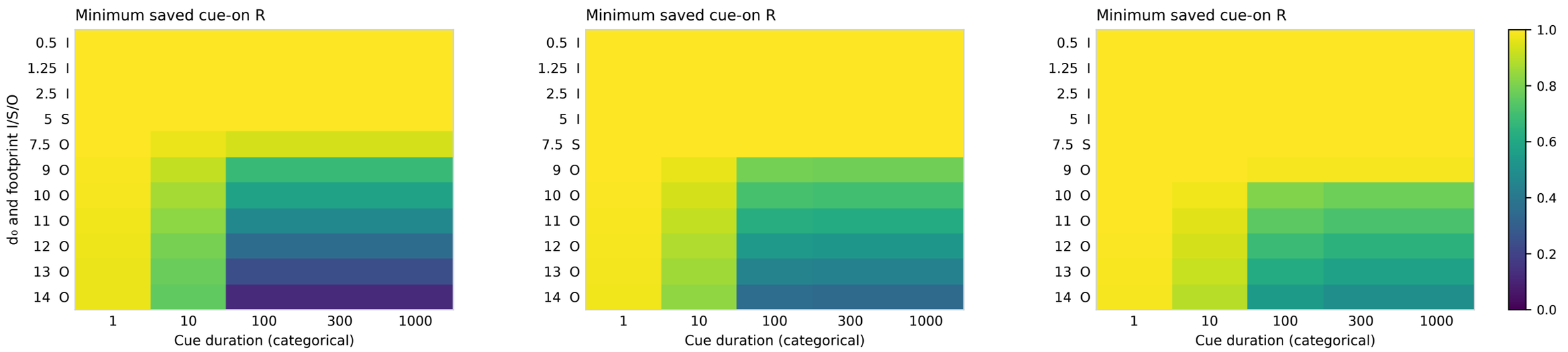
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

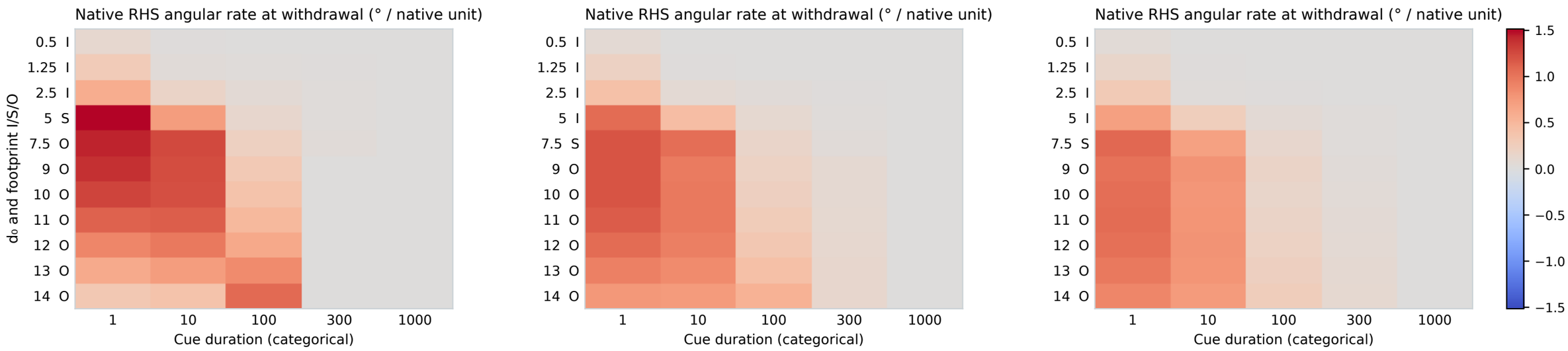
Rightward guide is P(0)=1; circle = measured R<0.1 without a heading. ? full-field tail unavailable; ! failed finite-tail gate. No minimum-pulse inference.

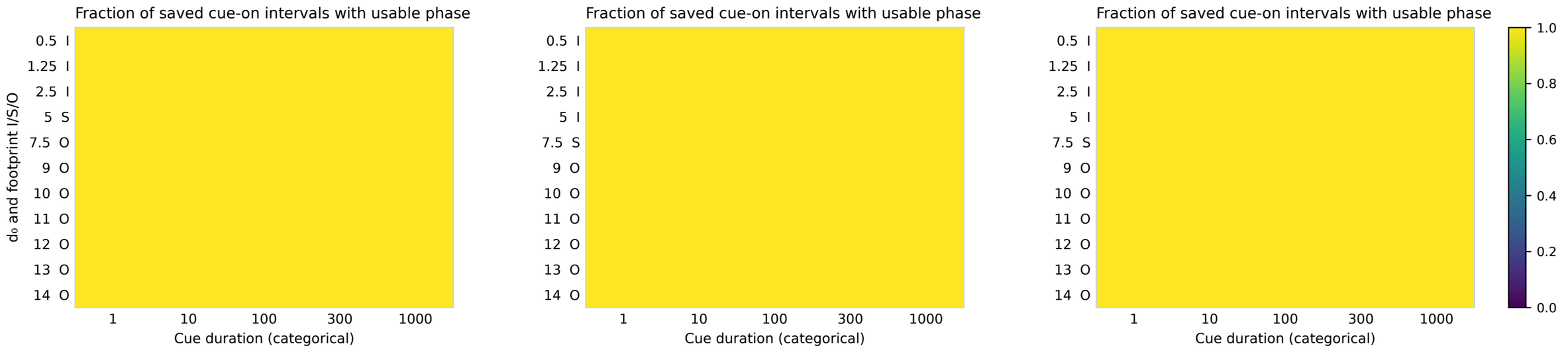
Wave-pinning · Every pulse: cue-on response and sampled coverage

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









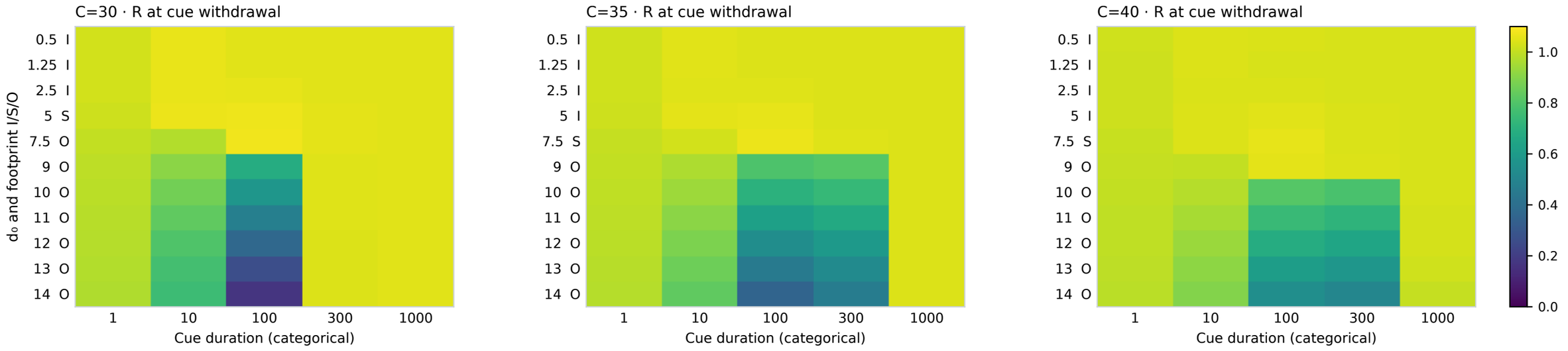
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

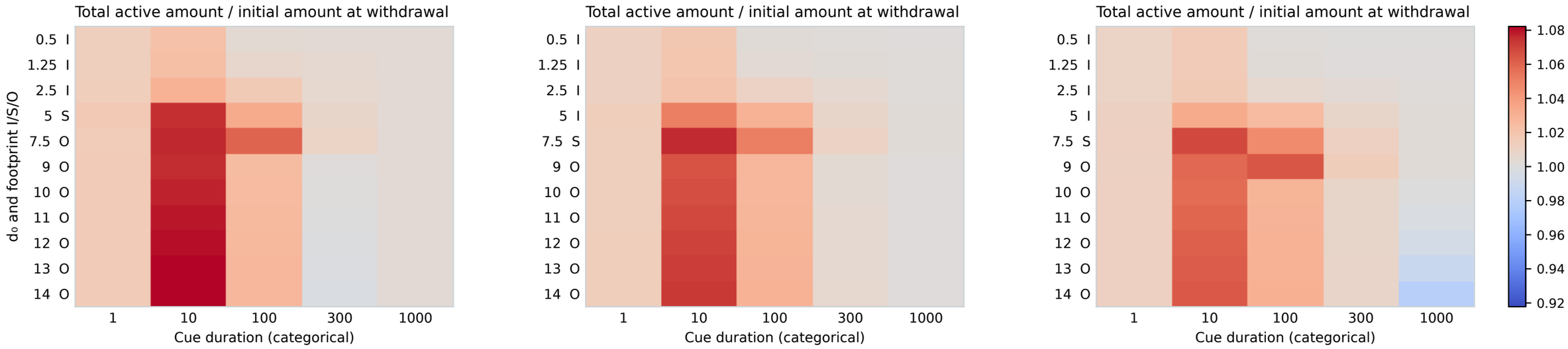
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

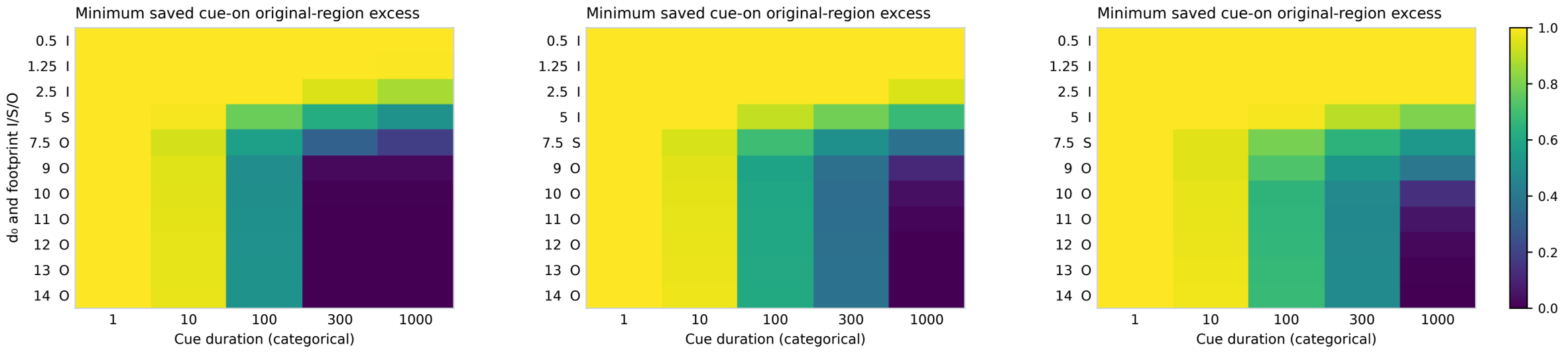
Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

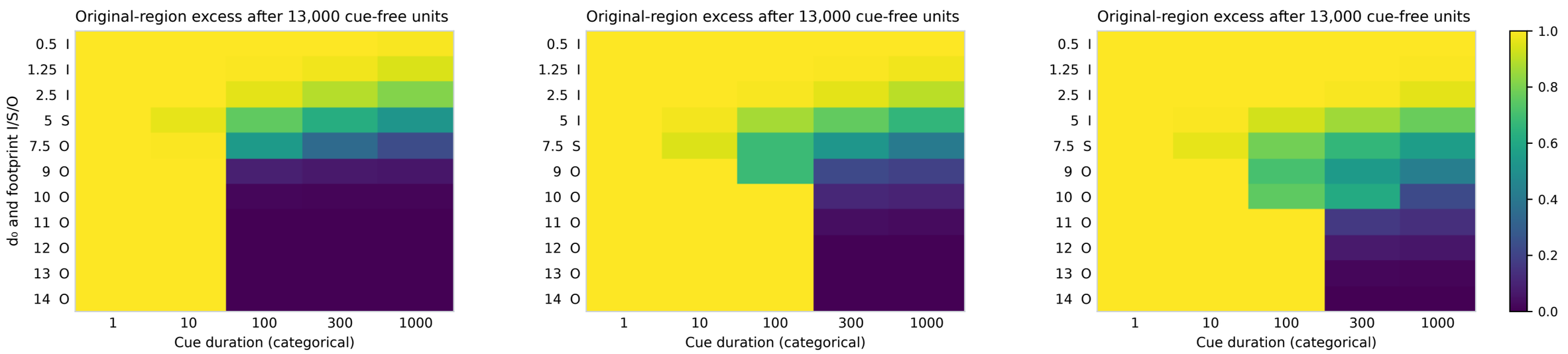
Wave-pinning · Every pulse: vector magnitude, active amount and fixed-region response

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









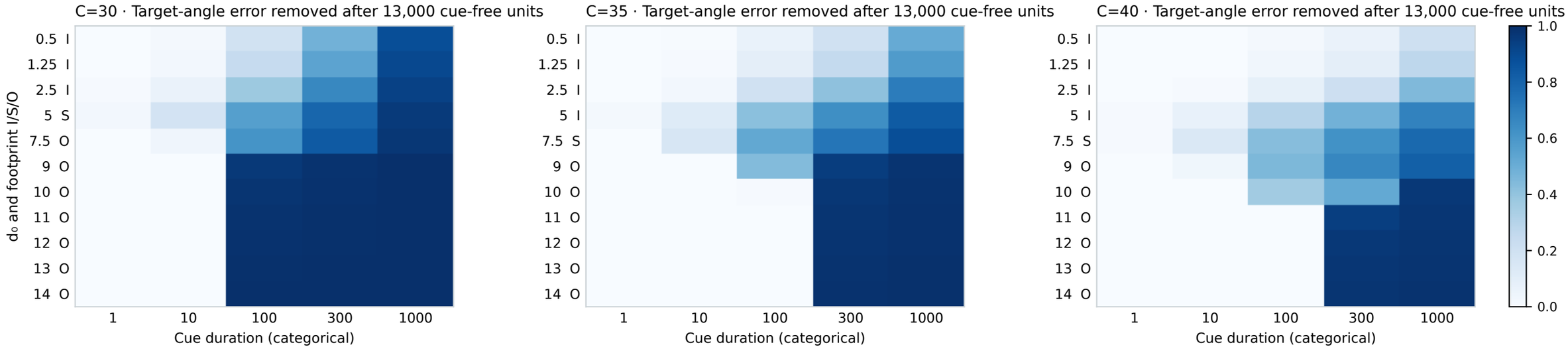
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Wave-pinning · Every pulse: finite cue-free endpoint organization

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.



All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Wave-pinning · Every placement: full longest-cue time courses

d₀=0.5 d₀=1.25 d₀=2.5 d₀=5 d₀=7.5 d₀=9 d₀=10 d₀=11 d₀=12 d₀=13 d₀=14

C=30 · 11/11 histories

C=35 · 11/11 histories

C=40 · 11/11 histories

Heading (degrees)

R = |P|

Total active amount / initial

Fixed original-region excess

Sampled phase secant (degrees / native unit)

Cue-on time (native units)

All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

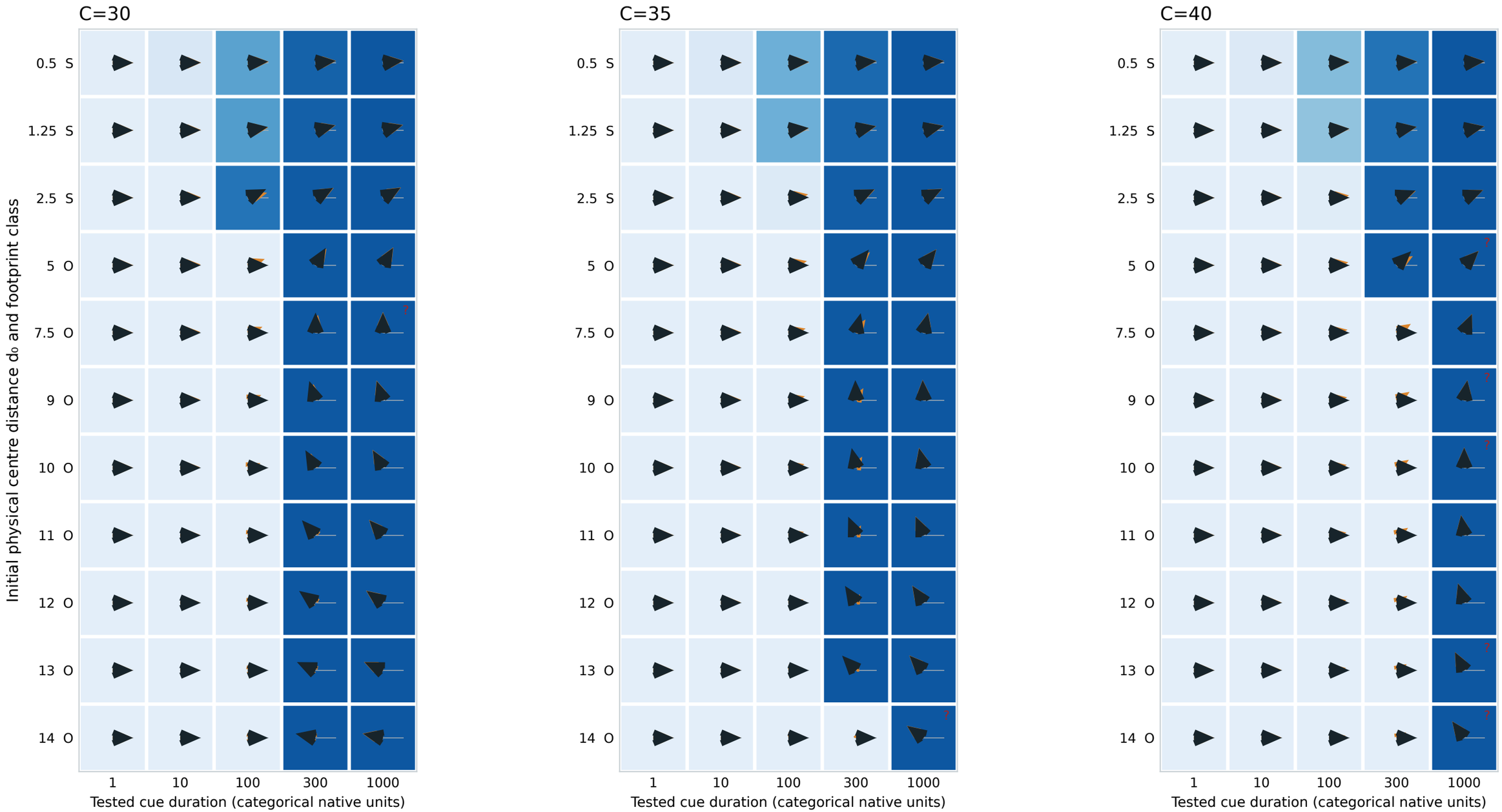
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Solid/dashed/dotted = I/S/O at initial 10% edge. Time linear to 0.1 then log; rates linear near ±0.001 then log. Each rate segment spans its actual saved interval.

Goryachev · Every pulse: withdrawal and retained activity vectors

Orange: cue withdrawal. Black: after 13,000 cue-free units. Each arrow is the actual P; one common scale in every cell.

Background: fraction of initial target-angle error removed at the retained endpoint (colour clipped to 0–1 only). Grey: pending / weak final R.



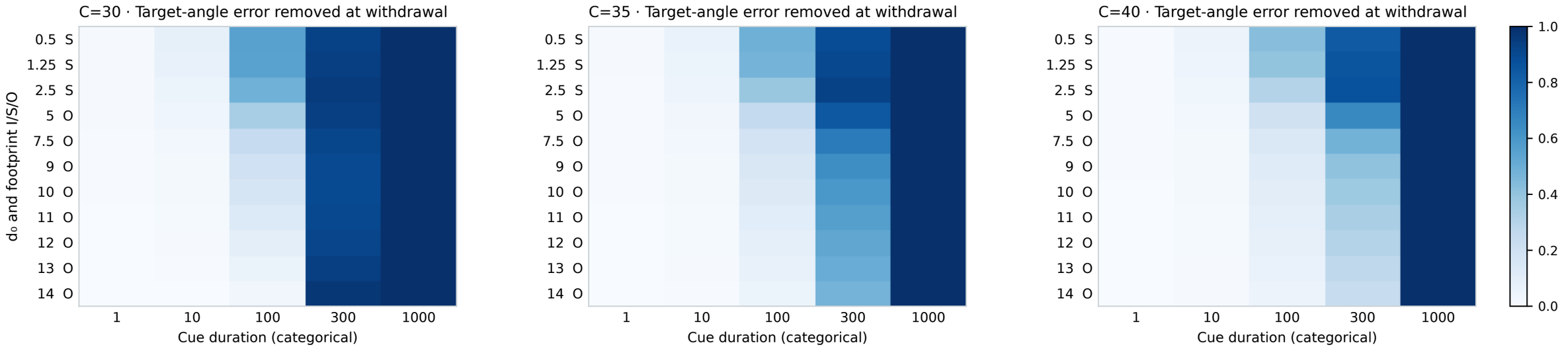
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

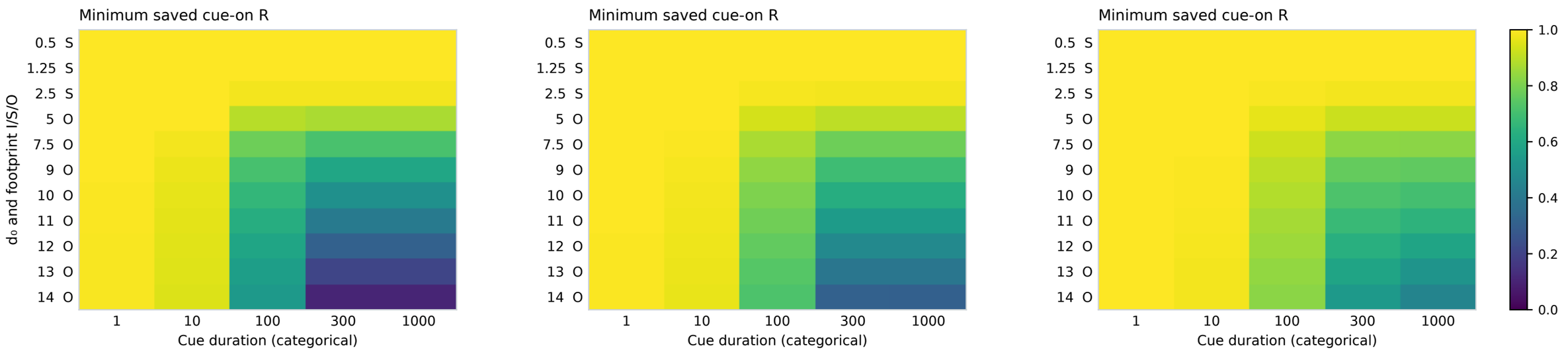
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

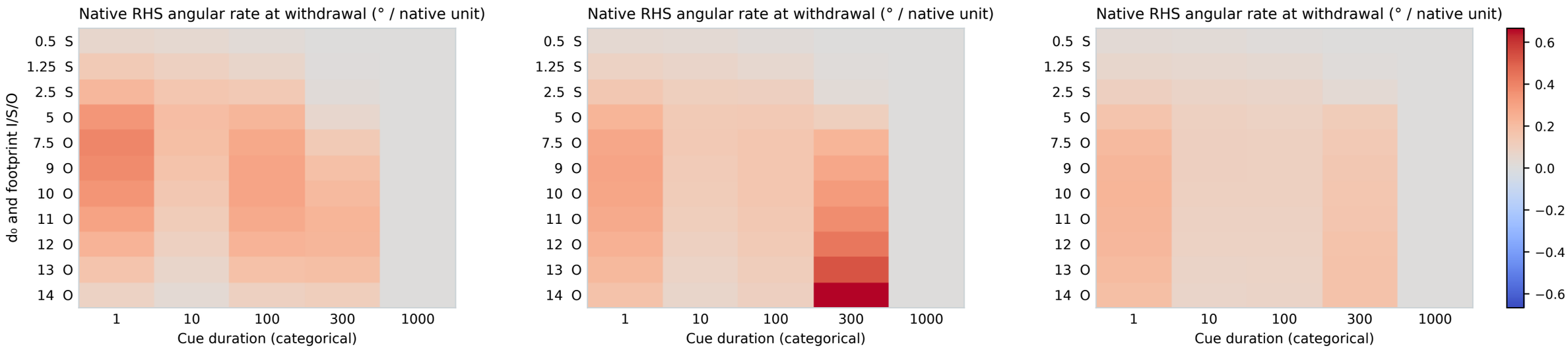
Rightward guide is P(0)=1; circle = measured R<0.1 without a heading. ? full-field tail unavailable; ! failed finite-tail gate. No minimum-pulse inference.

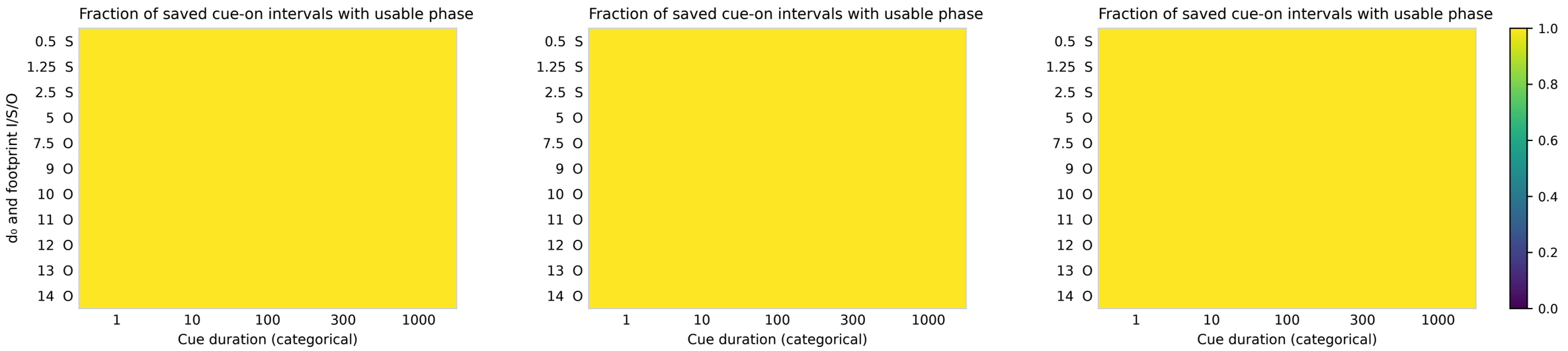
Goryachev · Every pulse: cue-on response and sampled coverage

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









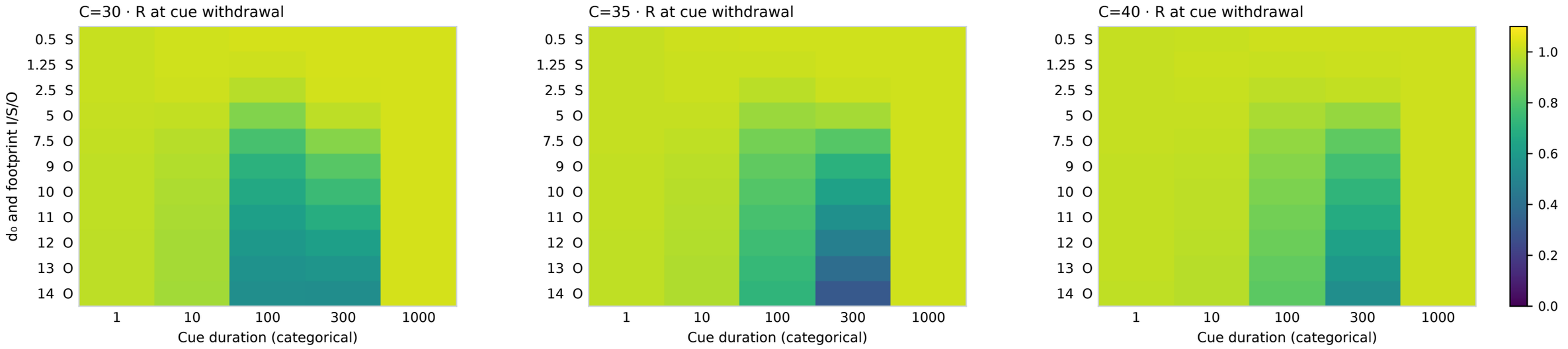
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

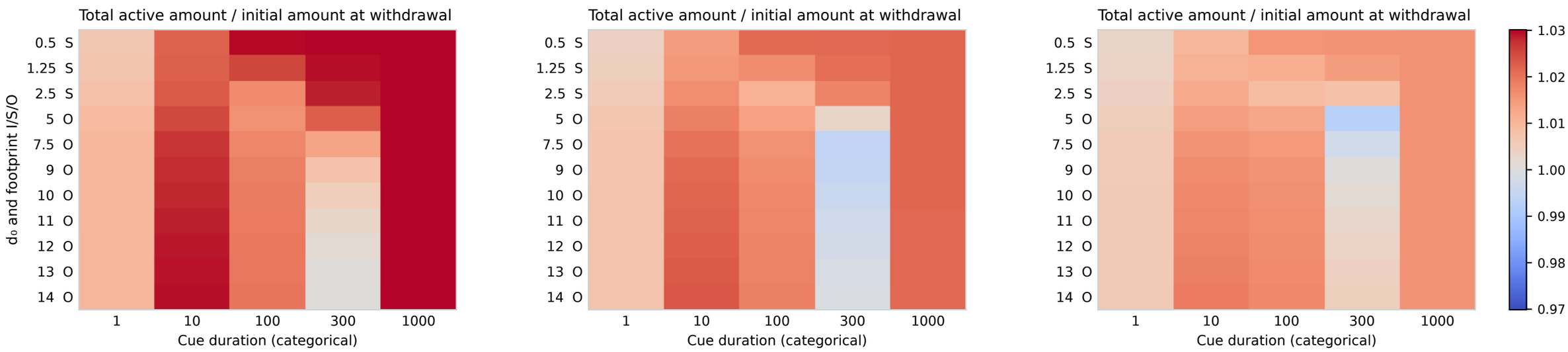
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

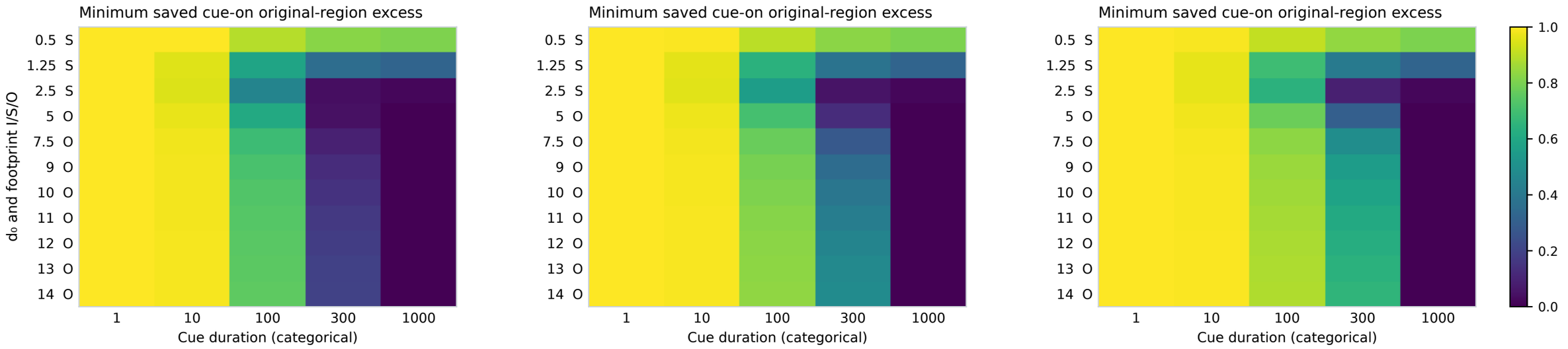
Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

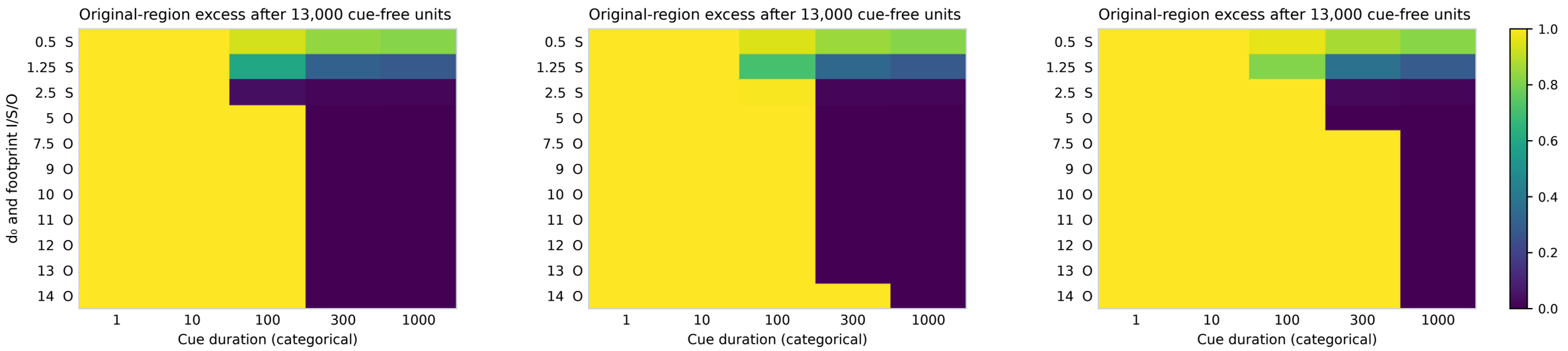
Goryachev · Every pulse: vector magnitude, active amount and fixed-region response

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

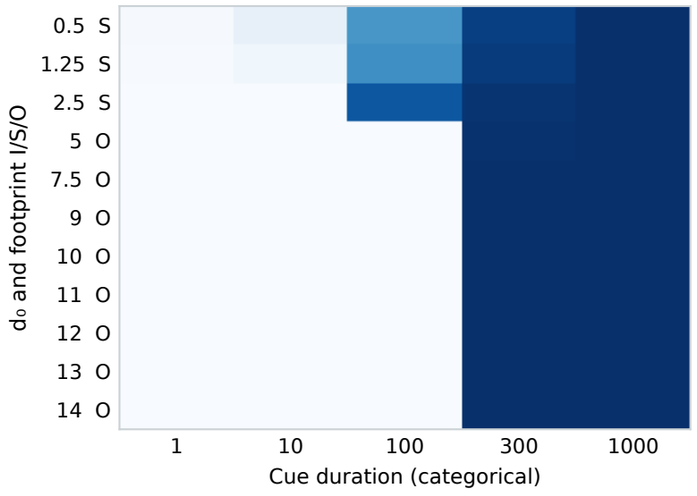
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

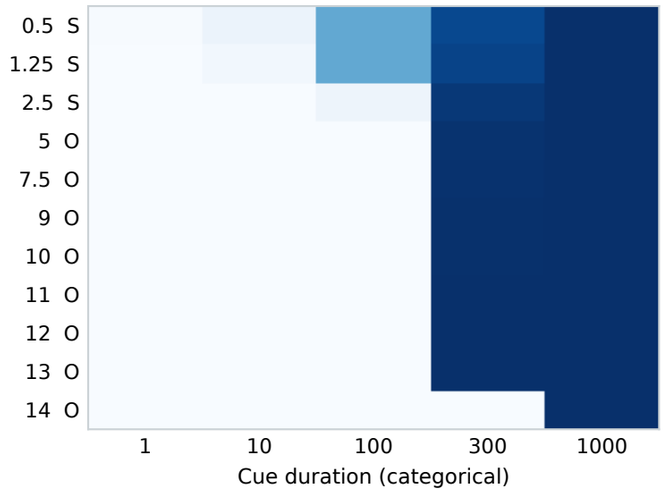
Goryachev · Every pulse: finite cue-free endpoint organization

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.

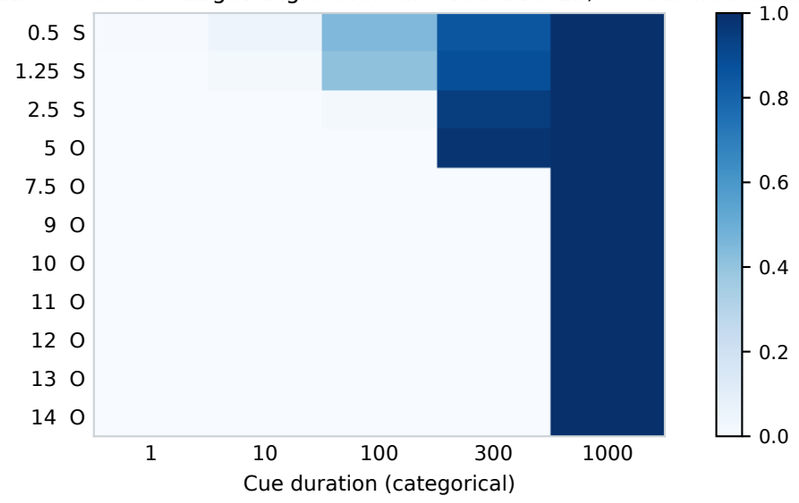
C=30 · Target-angle error removed after 13,000 cue-free units



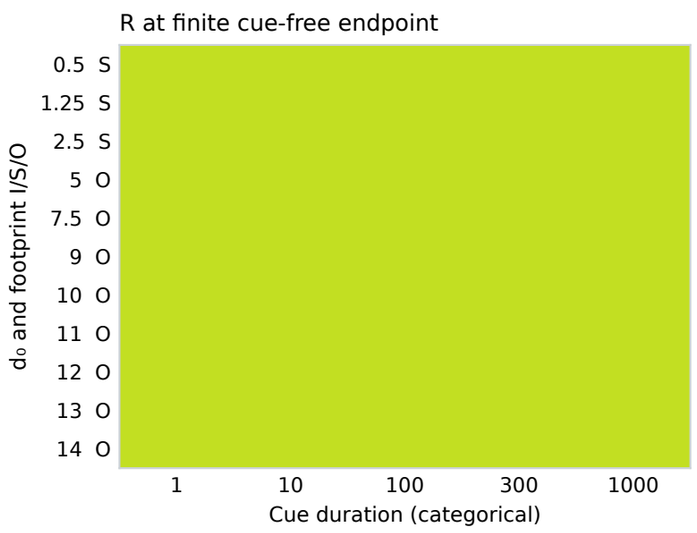
C=35 · Target-angle error removed after 13,000 cue-free units



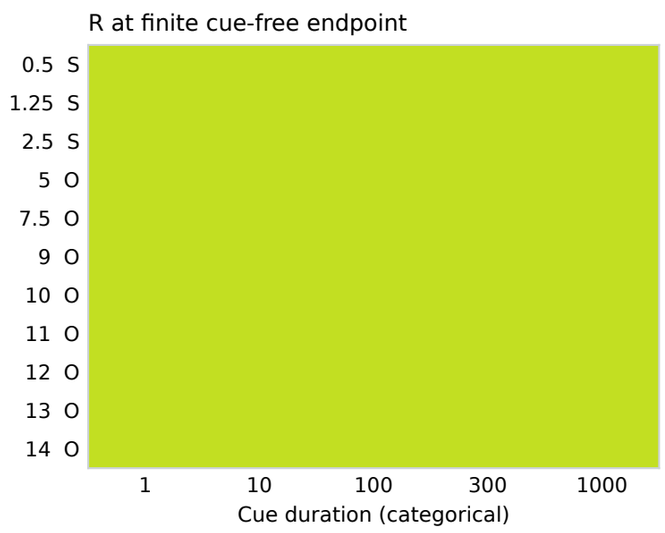
C=40 · Target-angle error removed after 13,000 cue-free units



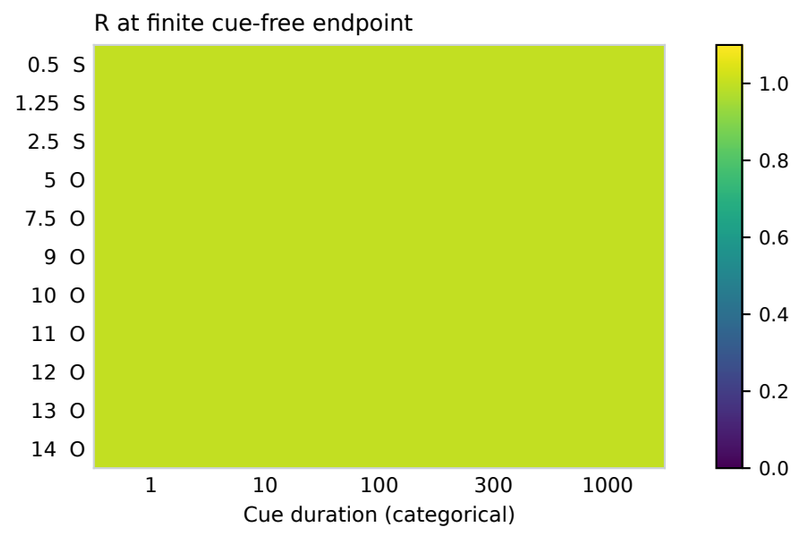
R at finite cue-free endpoint



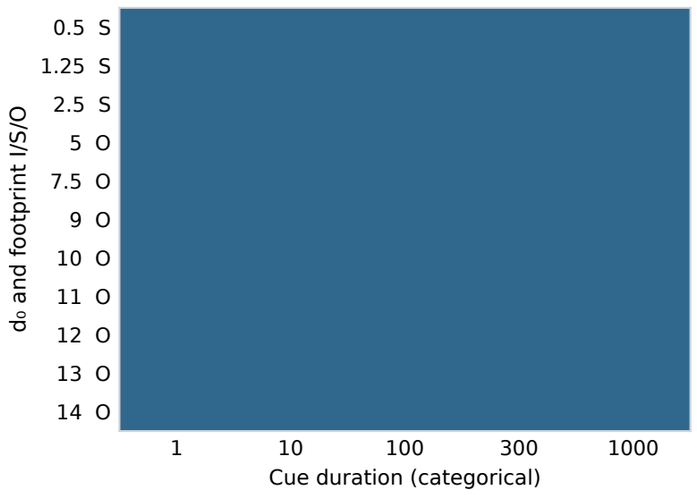
R at finite cue-free endpoint



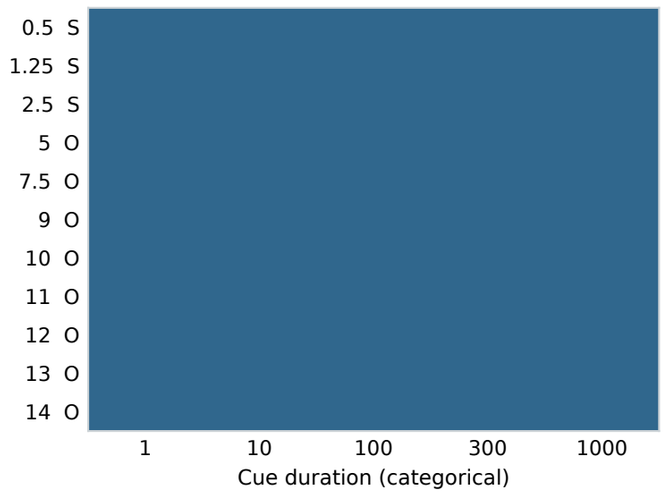
R at finite cue-free endpoint



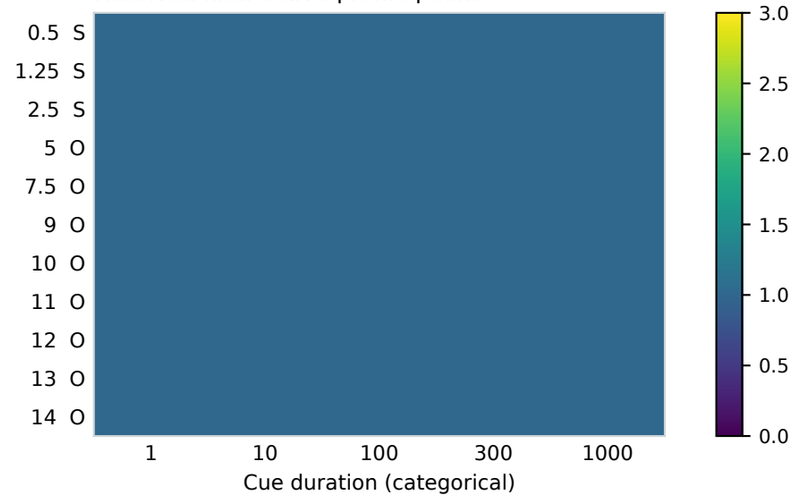
Resolved final active-profile peaks



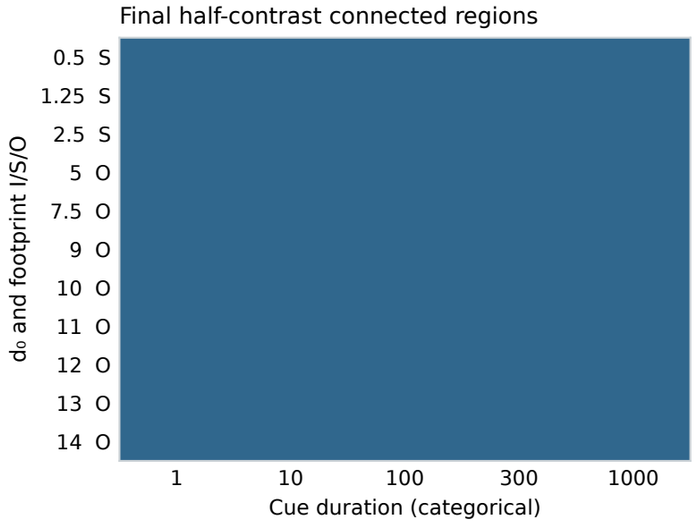
Resolved final active-profile peaks



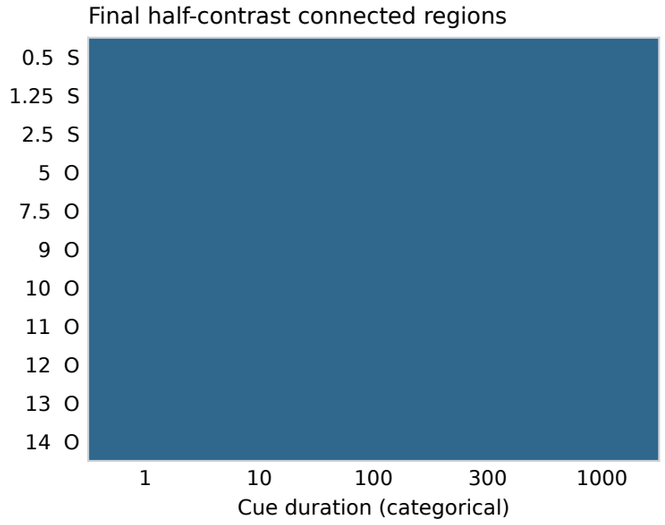
Resolved final active-profile peaks



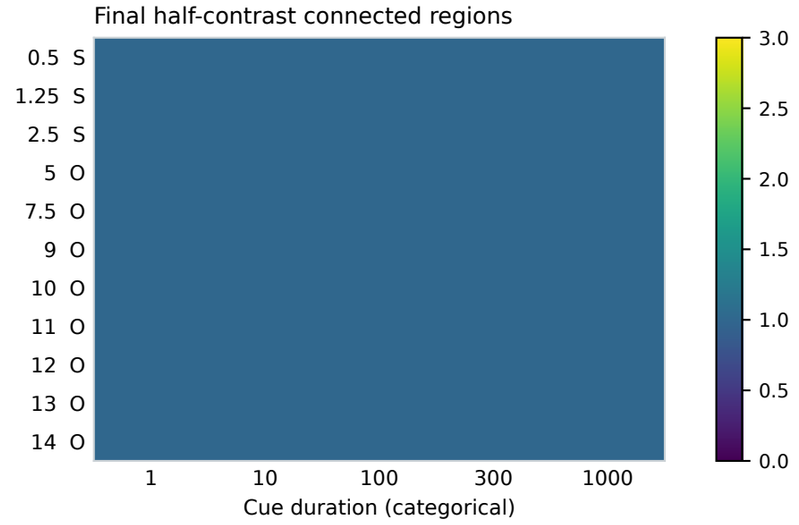
Final half-contrast connected regions



Final half-contrast connected regions



Final half-contrast connected regions



All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Goryachev · Every placement: full longest-cue time courses

d₀=0.5 d₀=1.25 d₀=2.5 d₀=5 d₀=7.5 d₀=9 d₀=10 d₀=11 d₀=12 d₀=13 d₀=14

C=30 · 11/11 histories

C=35 · 11/11 histories

C=40 · 11/11 histories

Heading (degrees)

R = |P|

Total active amount / initial

Fixed original-region excess

Sampled phase secant (degrees / native unit)

Cue-on time (native units)

All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

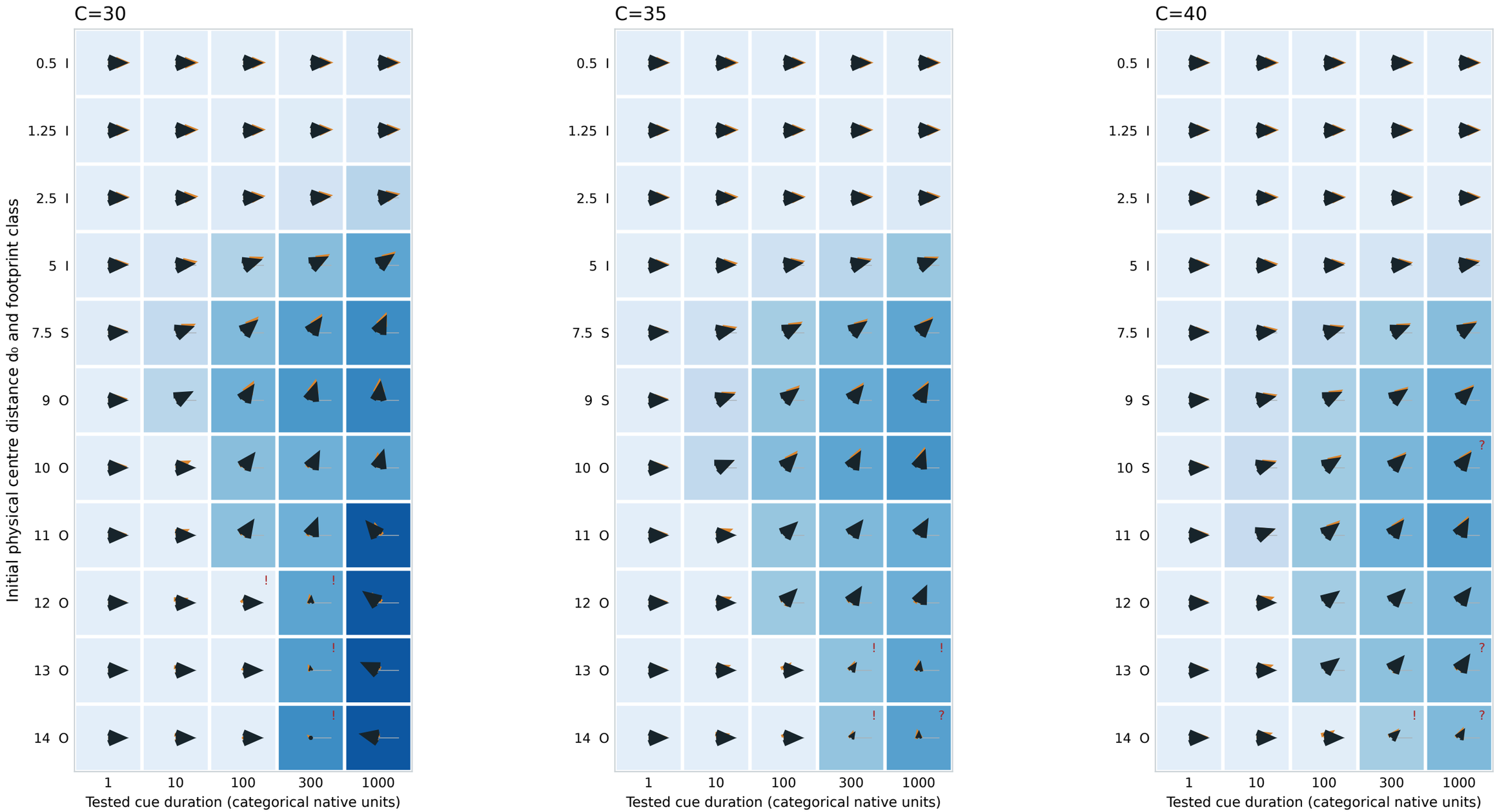
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Solid/dashed/dotted = I/S/O at initial 10% edge. Time linear to 0.1 then log; rates linear near ±0.001 then log. Each rate segment spans its actual saved interval.

Holmes · Every pulse: withdrawal and retained activity vectors

Orange: cue withdrawal. Black: after 13,000 cue-free units. Each arrow is the actual P; one common scale in every cell.

Background: fraction of initial target-angle error removed at the retained endpoint (colour clipped to 0–1 only). Grey: pending / weak final R.



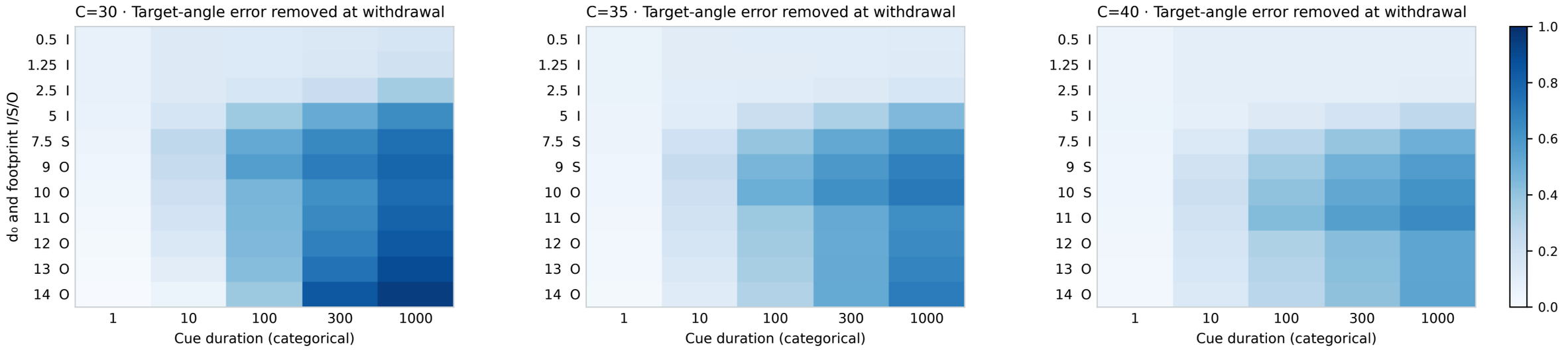
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

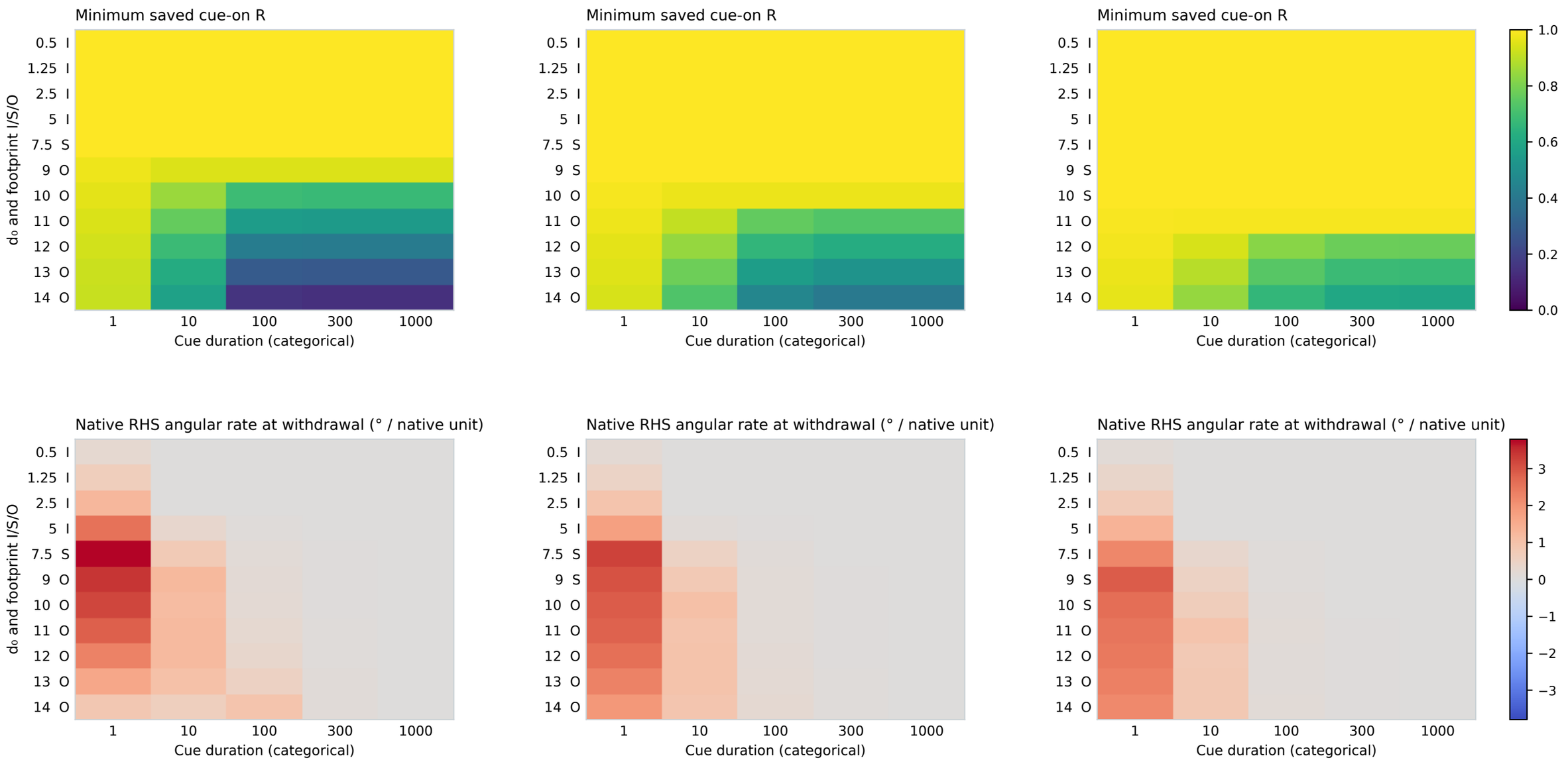
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

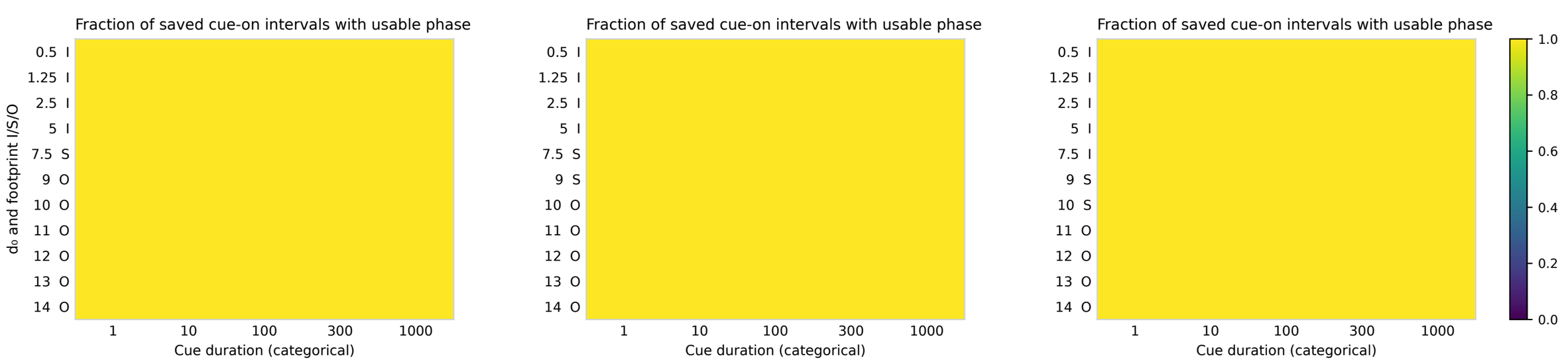
Rightward guide is P(0)=1; circle = measured R<0.1 without a heading. ? full-field tail unavailable; ! failed finite-tail gate. No minimum-pulse inference.

Holmes · Every pulse: cue-on response and sampled coverage

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.







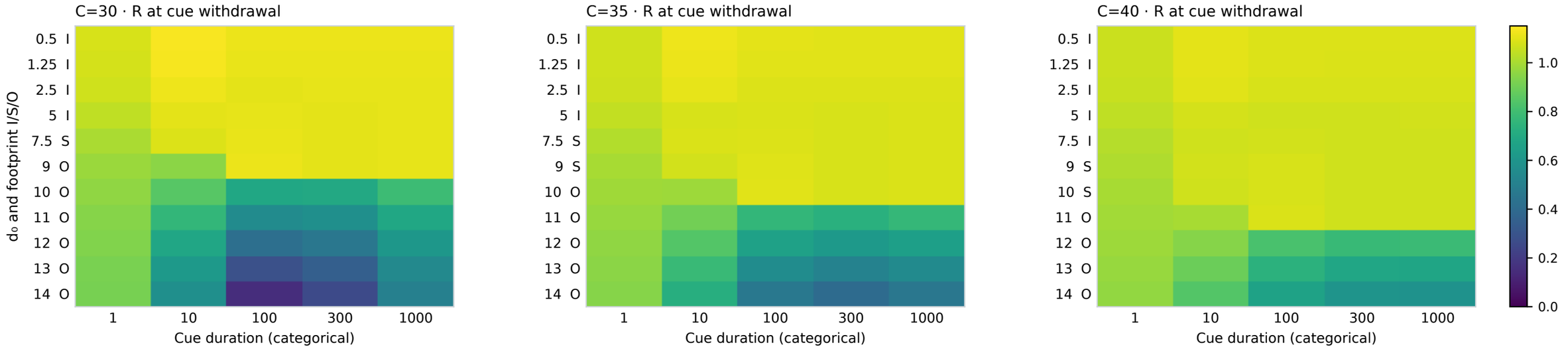
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

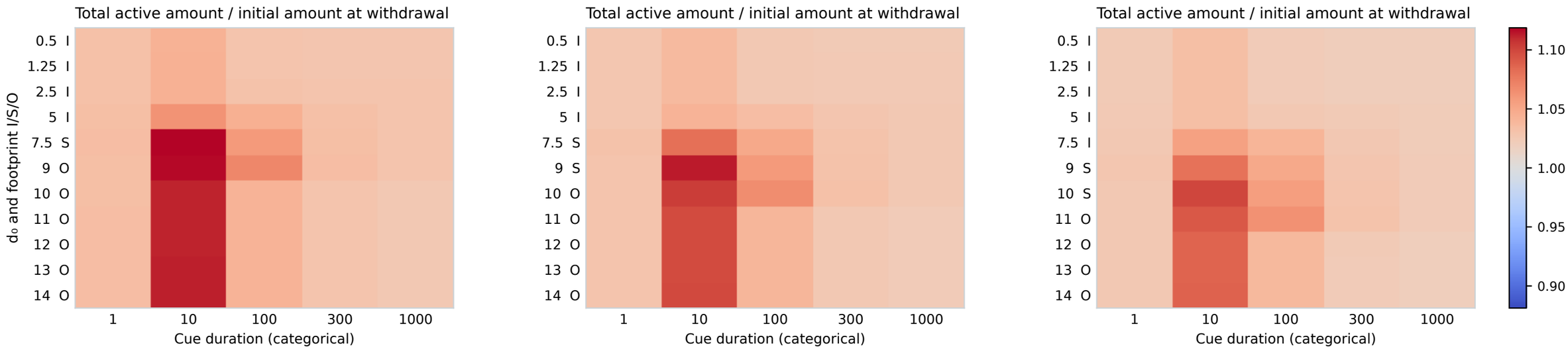
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

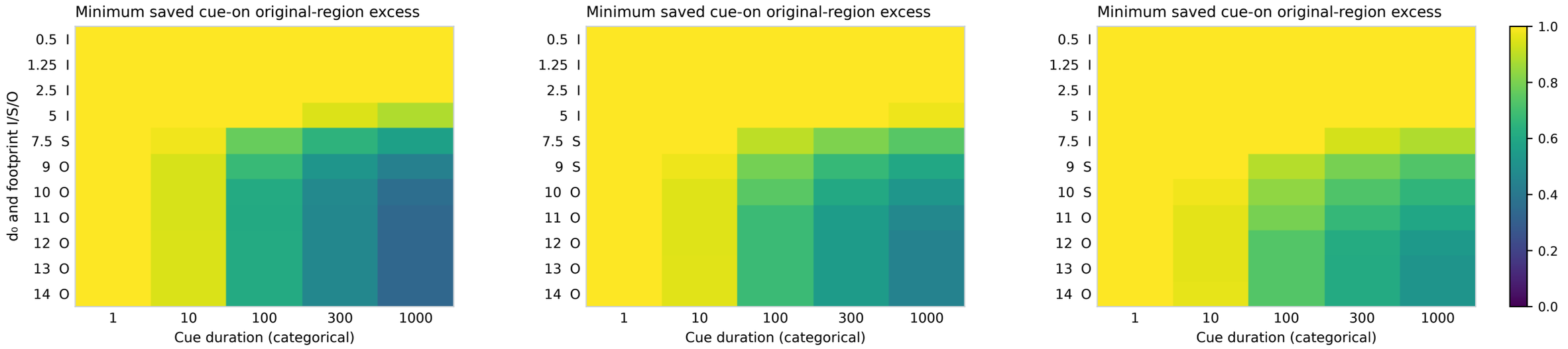
Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

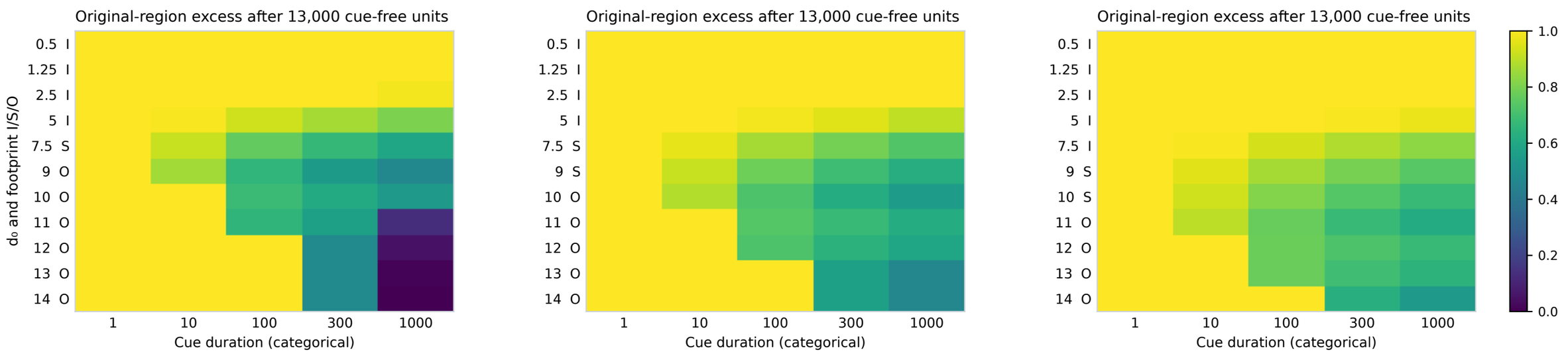
Holmes · Every pulse: vector magnitude, active amount and fixed-region response

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









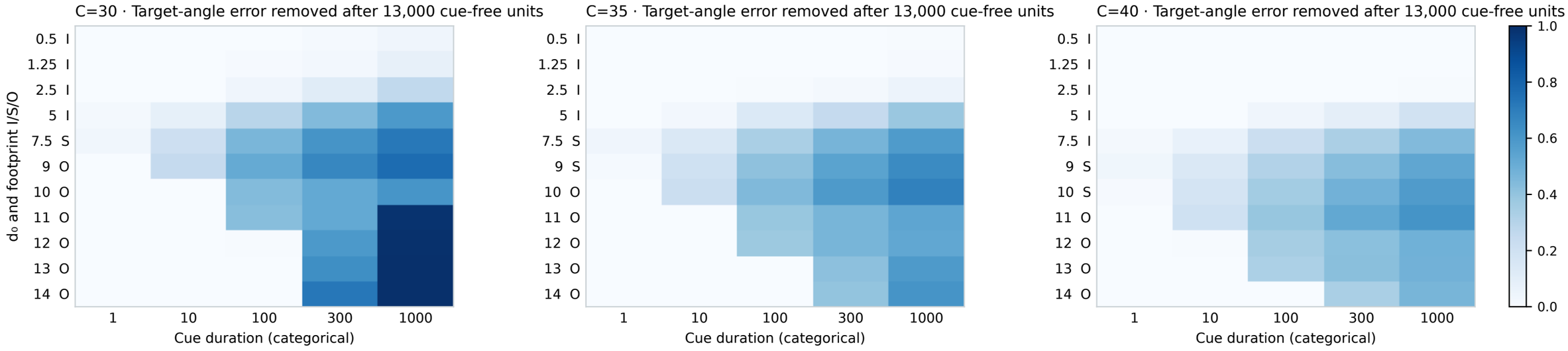
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Holmes · Every pulse: finite cue-free endpoint organization

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.



All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Holmes · Every placement: full longest-cue time courses

d₀=0.5 d₀=1.25 d₀=2.5 d₀=5 d₀=7.5 d₀=9 d₀=10 d₀=11 d₀=12 d₀=13 d₀=14

C=30 · 11/11 histories

C=35 · 11/11 histories

C=40 · 11/11 histories

Heading (degrees)

R = |P|

Total active amount / initial

Fixed original-region excess

Sampled phase secant (degrees / native unit)

Cue-on time (native units)

All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

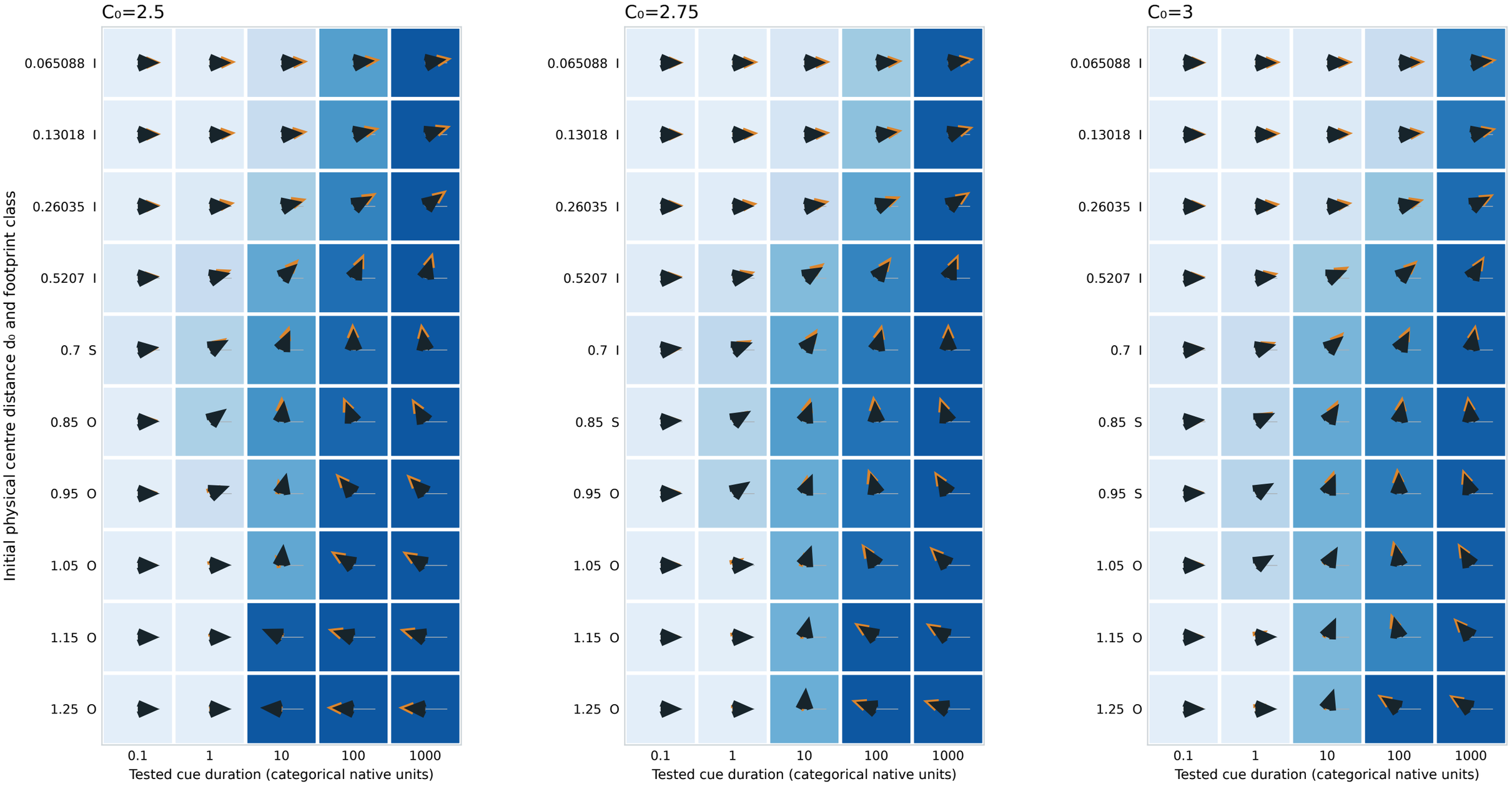
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Solid/dashed/dotted = I/S/O at initial 10% edge. Time linear to 0.1 then log; rates linear near ±0.001 then log. Each rate segment spans its actual saved interval.

Spring · Every pulse: withdrawal and retained activity vectors

Orange: cue withdrawal. Black: after 13,000 cue-free units. Each arrow is the actual P; one common scale in every cell.

Background: fraction of initial target-angle error removed at the retained endpoint (colour clipped to 0–1 only). Grey: pending / weak final R.



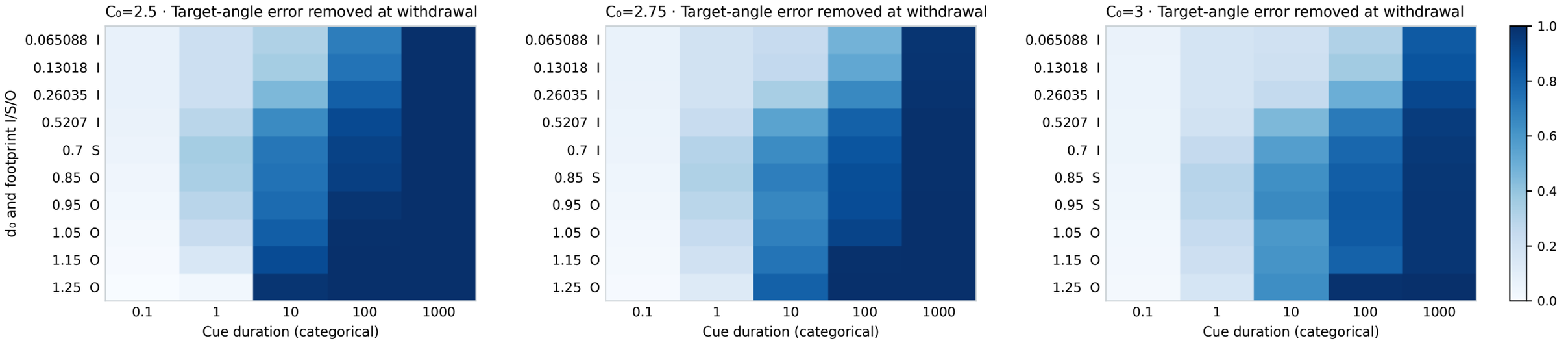
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

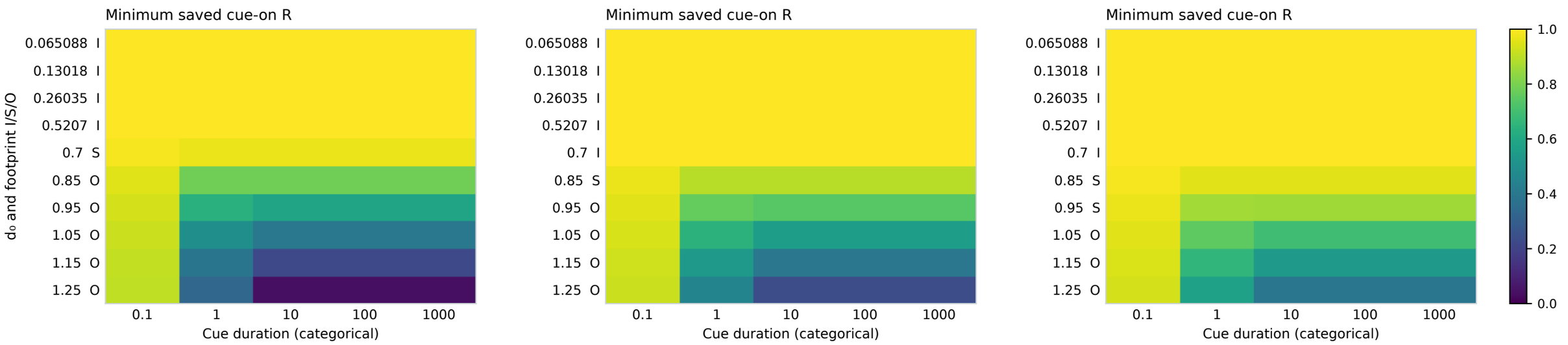
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

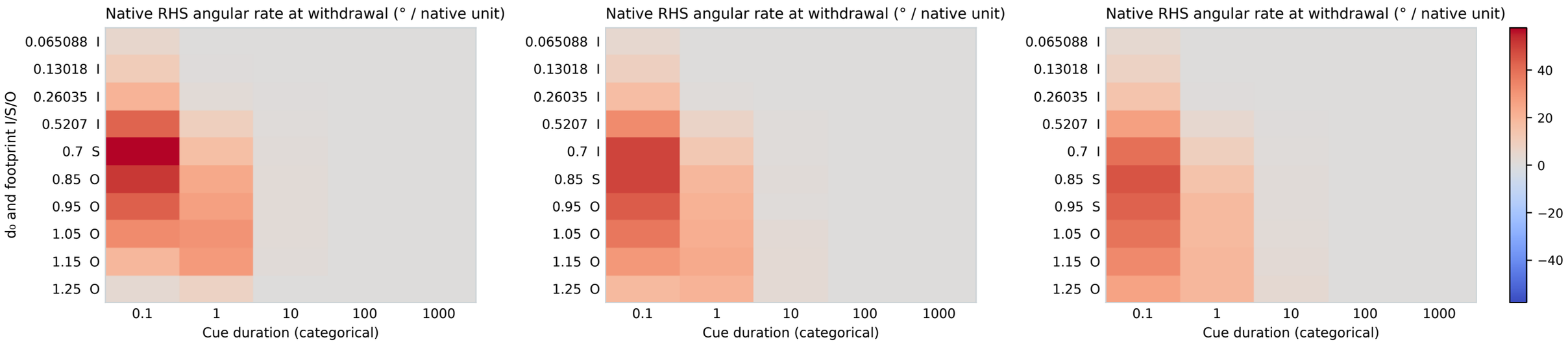
Rightward guide is P(0)=1; circle = measured R<0.1 without a heading. ? full-field tail unavailable; ! failed finite-tail gate. No minimum-pulse inference.

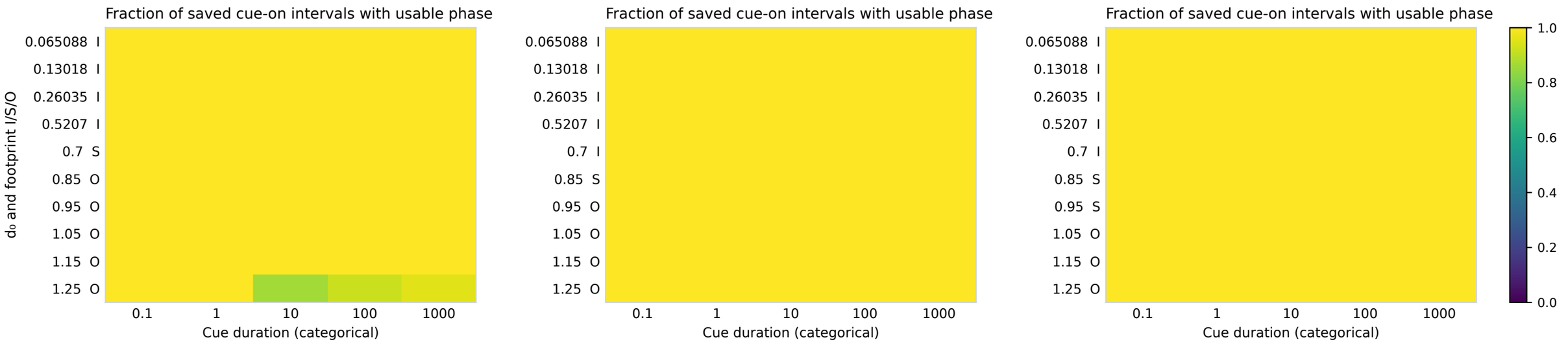
Spring · Every pulse: cue-on response and sampled coverage

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

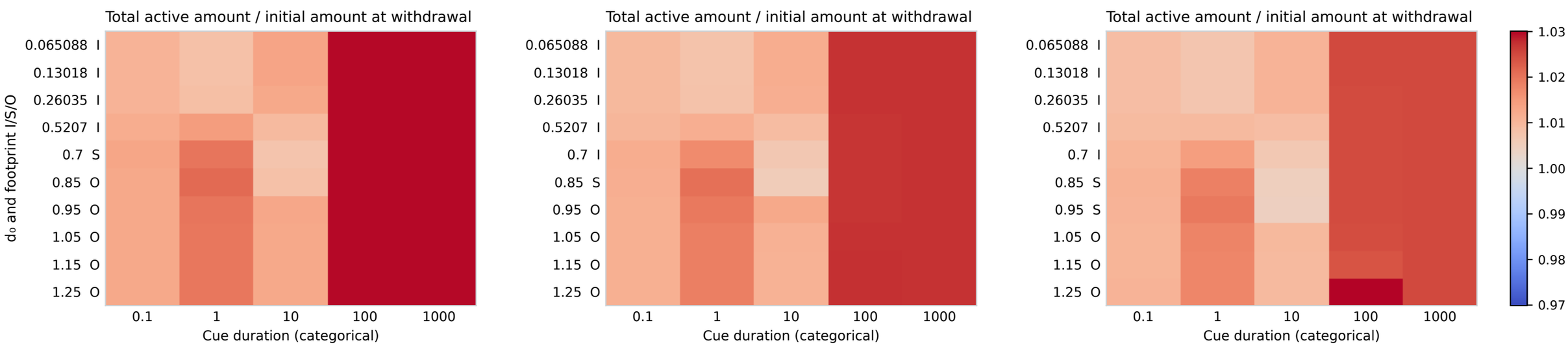
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

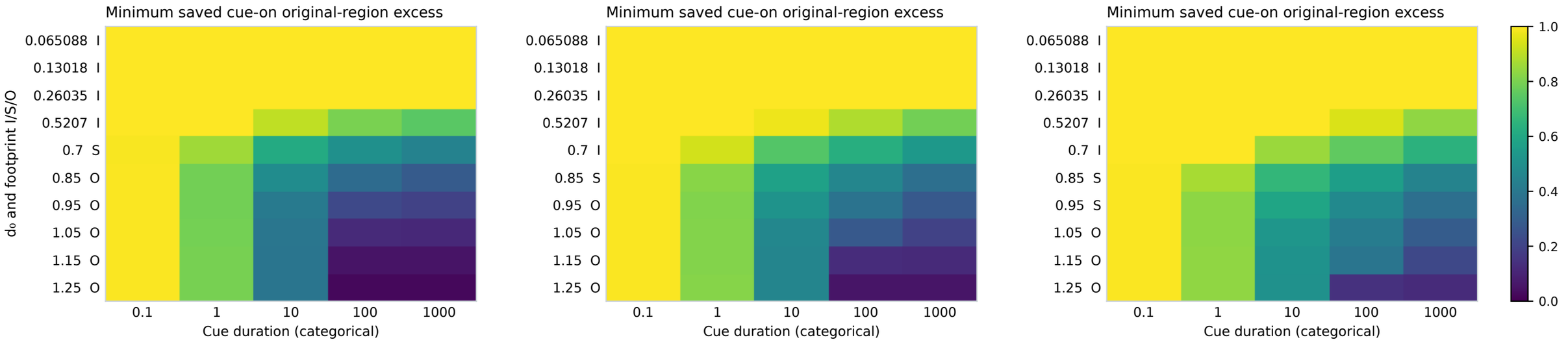
Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Spring · Every pulse: vector magnitude, active amount and fixed-region response

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









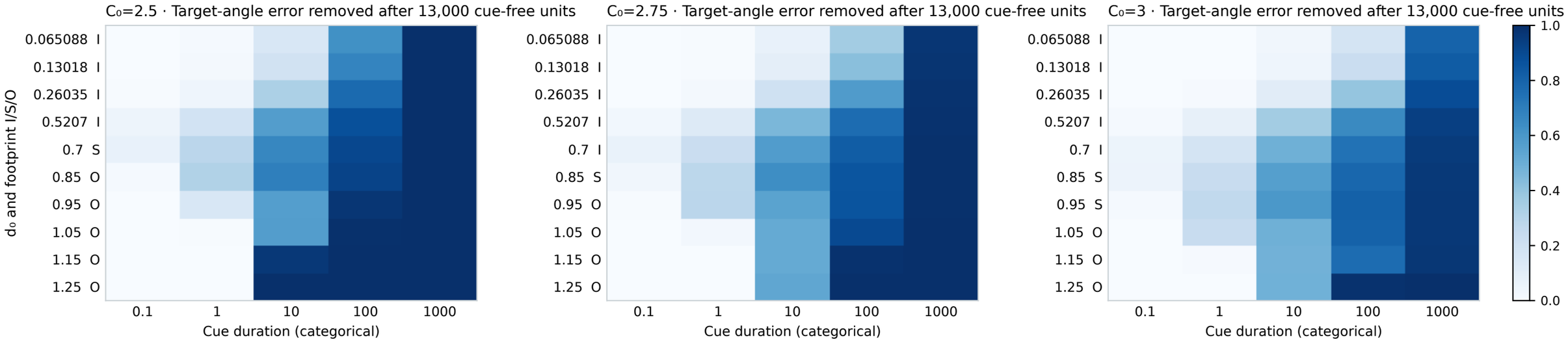
All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

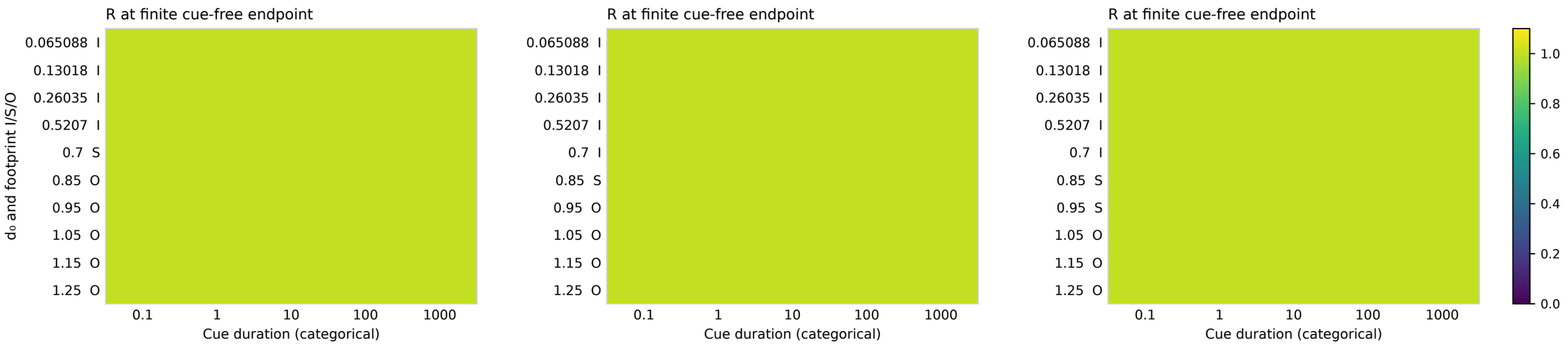
I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

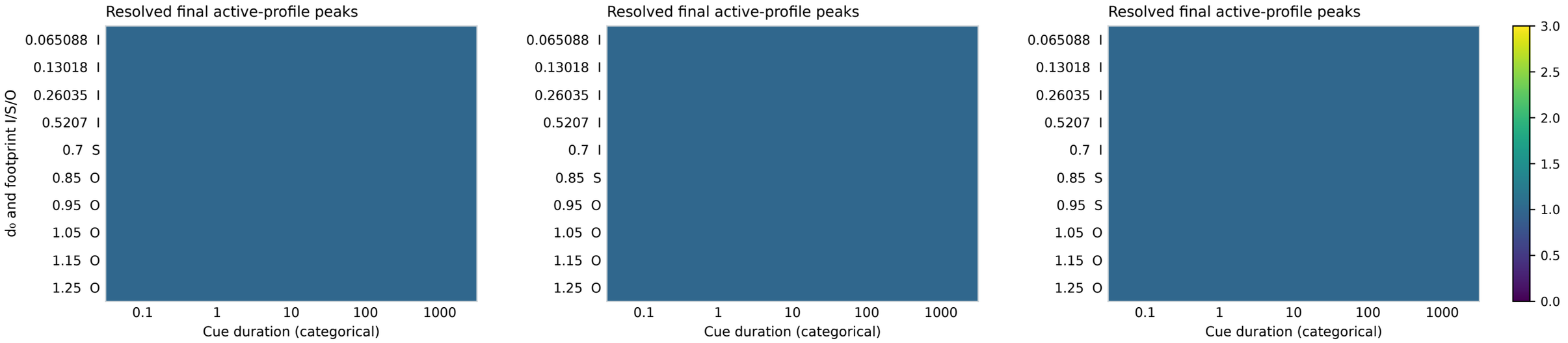
Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

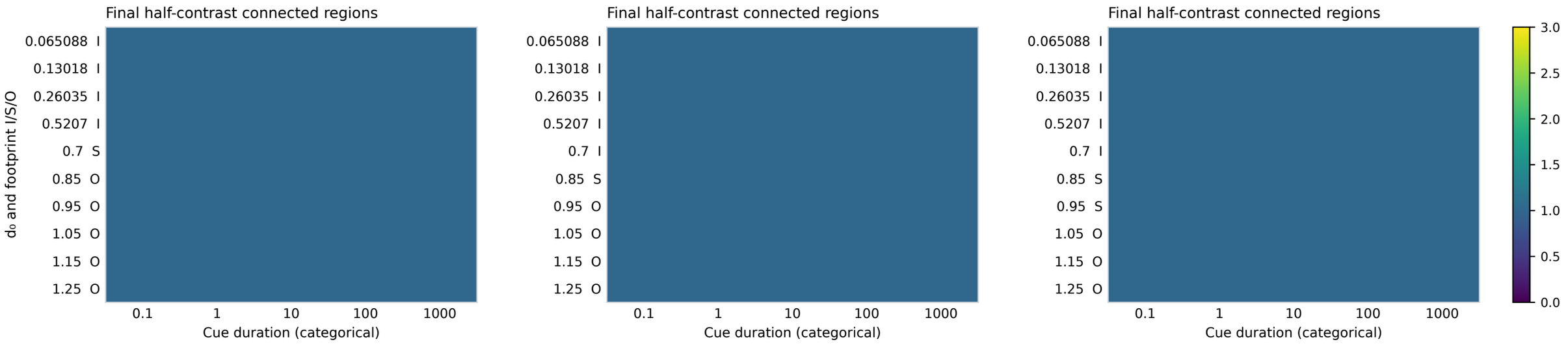
Spring · Every pulse: finite cue-free endpoint organization

Five tested cue durations × every prescribed distance × three exact qualified basals. Grey = pending or the specific quantity is unavailable.









All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Error-closure colours clip to 0–1; exact raw values remain in CSV. Fixed original-region loss is redistribution-compatible, not site ancestry or destruction.

Spring · Every placement: full longest-cue time courses

d₀=0.065088 d₀=0.13018 d₀=0.26035 d₀=0.5207 d₀=0.7 d₀=0.85 d₀=0.95 d₀=1.05 d₀=1.15 d₀=1.25

C₀=2.5 · 10/10 histories

C₀=2.75 · 10/10 histories

C₀=3 · 10/10 histories

Heading (degrees)

R = |P|

Total active amount / initial

Fixed original-region excess

Sampled phase secant (degrees / native unit)

Cue-on time (native units)

All requested placements and durations retained · 645/645 conditions available · data 2026-10-19T20:51:57 UTC

I / S / O = entire initial cue footprint inside / straddling / outside the 10%-contrast region. Centre membership is recorded separately.

Solid/dashed/dotted = I/S/O at initial 10% edge. Time linear to 0.1 then log; rates linear near ±0.001 then log. Each rate segment spans its actual saved interval.

Cue-on reorientation at a common external physical offset

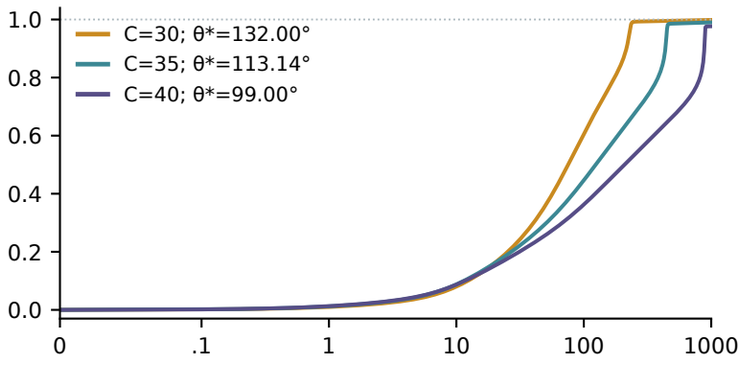
T=1000 · chemical d_0=11; Spring d_0=1.05 · full supports outside every initial 10%-contrast region

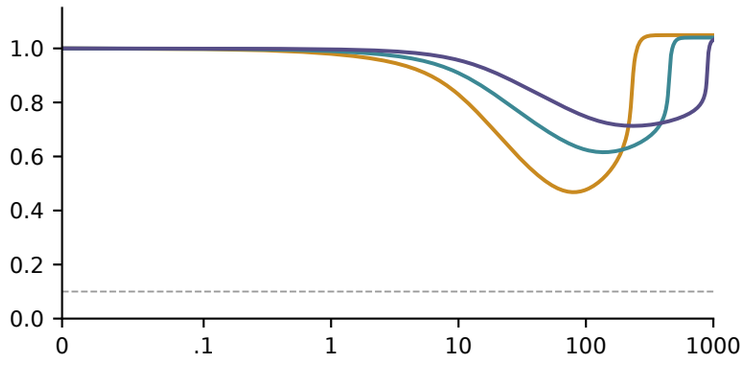
Angular target error removed

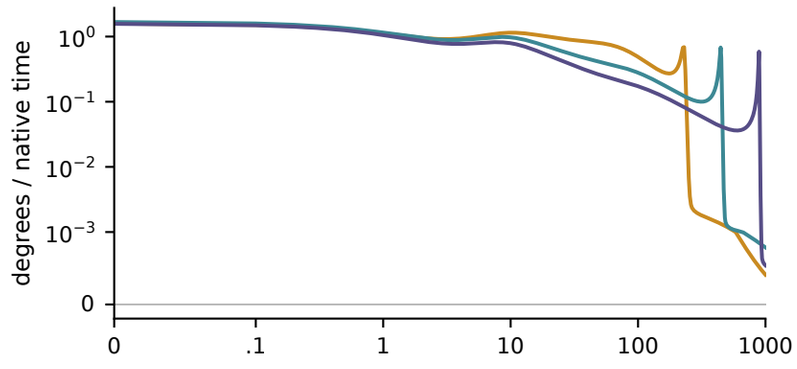
Vector magnitude R

Native instantaneous angular rate

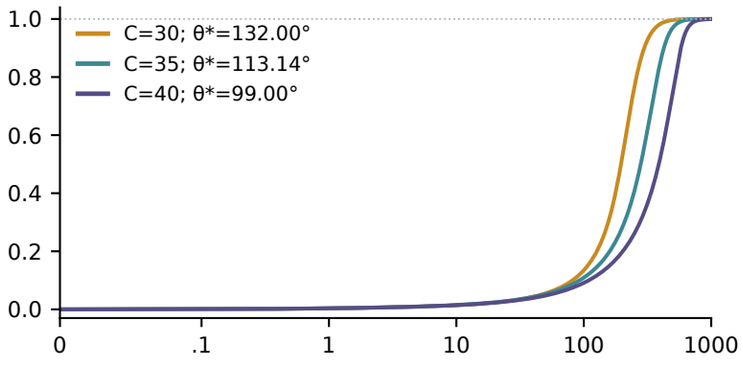
Wave-pinning

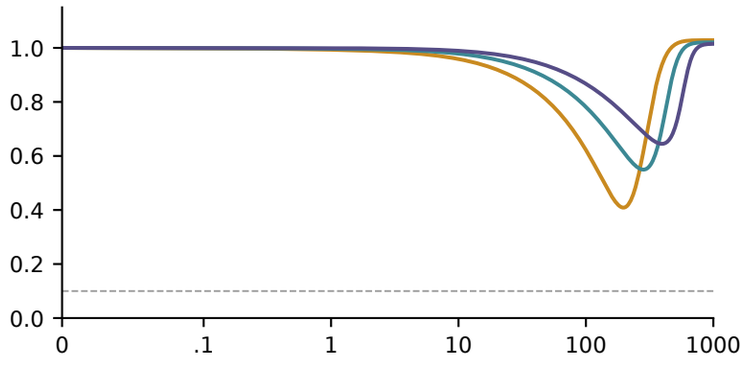


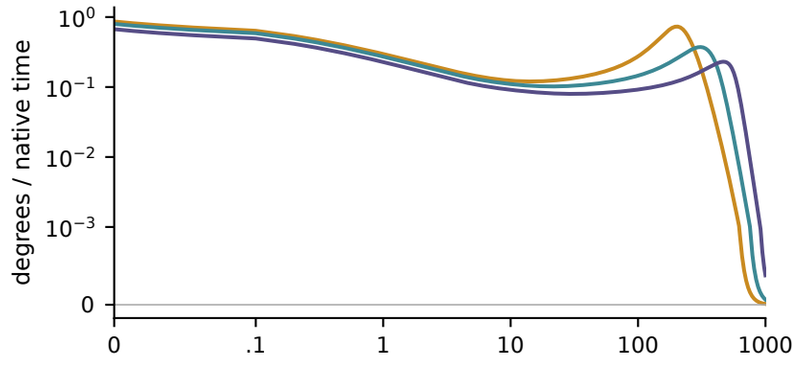




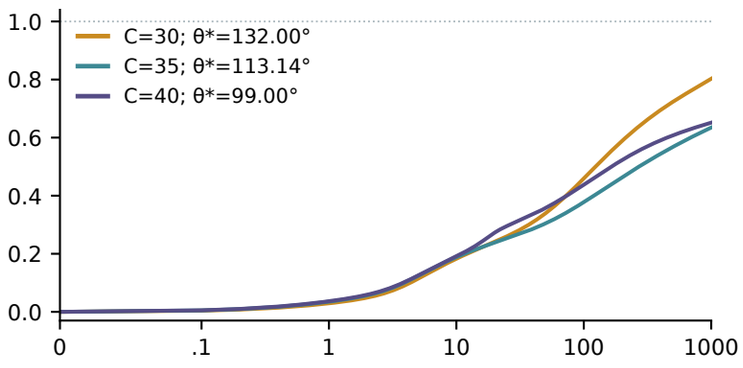
Goryachev

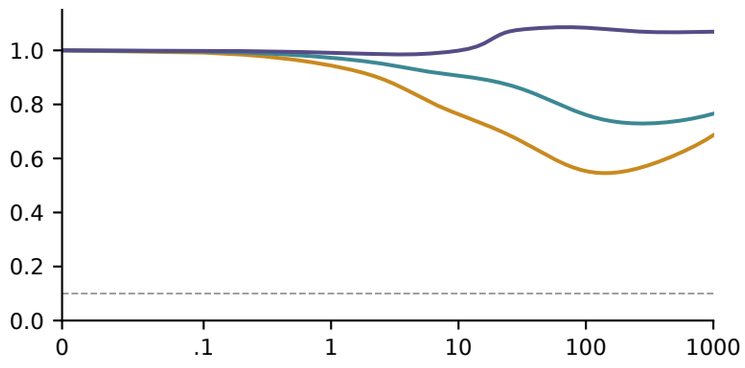


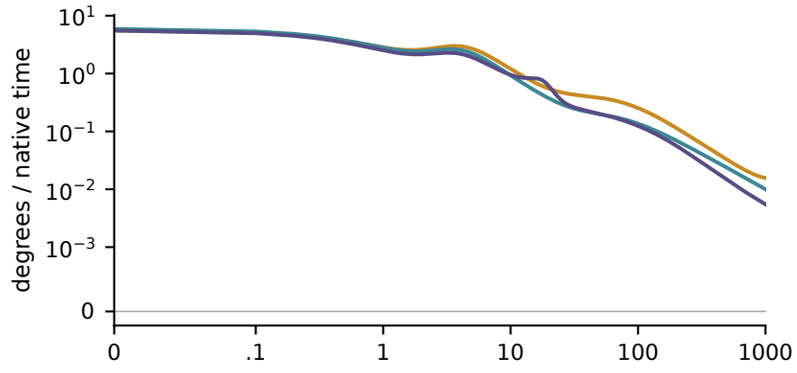




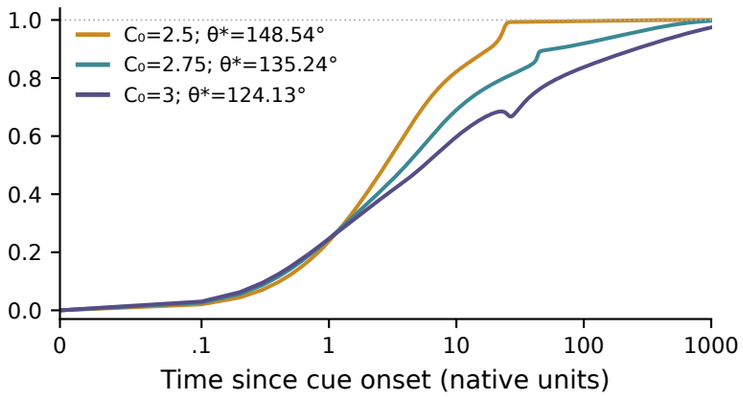
Holmes

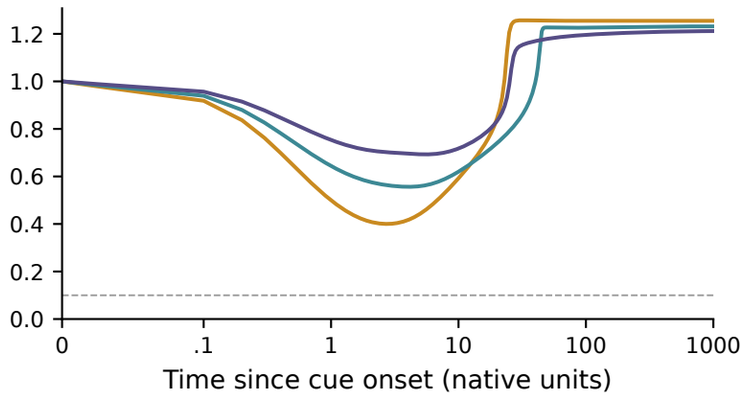


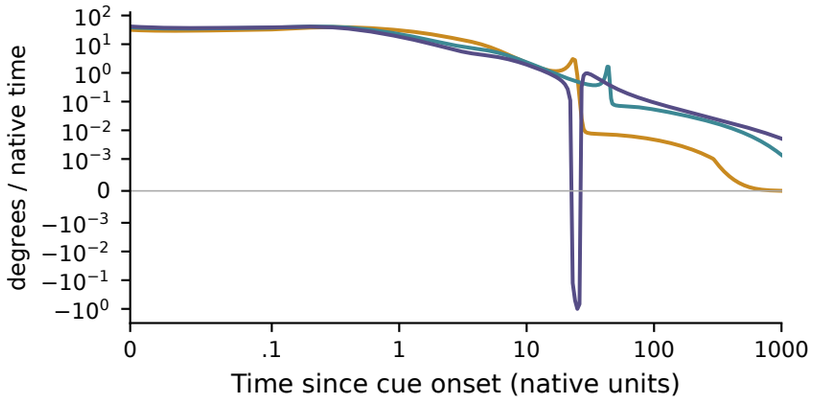




Spring







Twelve actual full-state histories. Error removal uses each case's actual target; raw headings and physical geometry remain in the supporting table.

Time is linear to 0.1 then logarithmic. Rate axes are linear within ±0.001 then logarithmic in magnitude; this is not a calibrated cross-model clock.

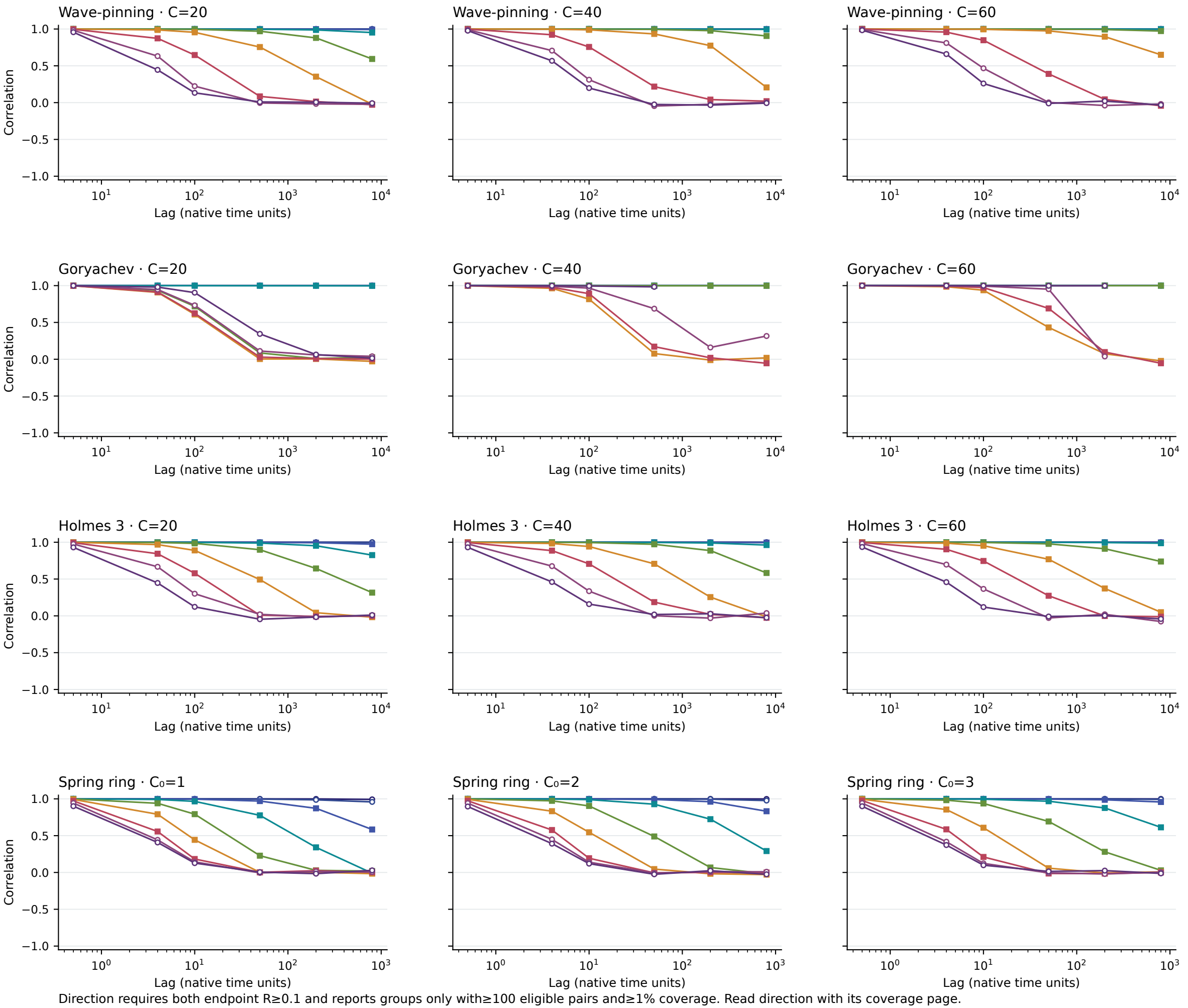
Global angular rate includes finite-width input projection and initial moment magnitude. Native profiles and full outcome/coverage maps are linked in the supplement.

Conditional direction memory: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

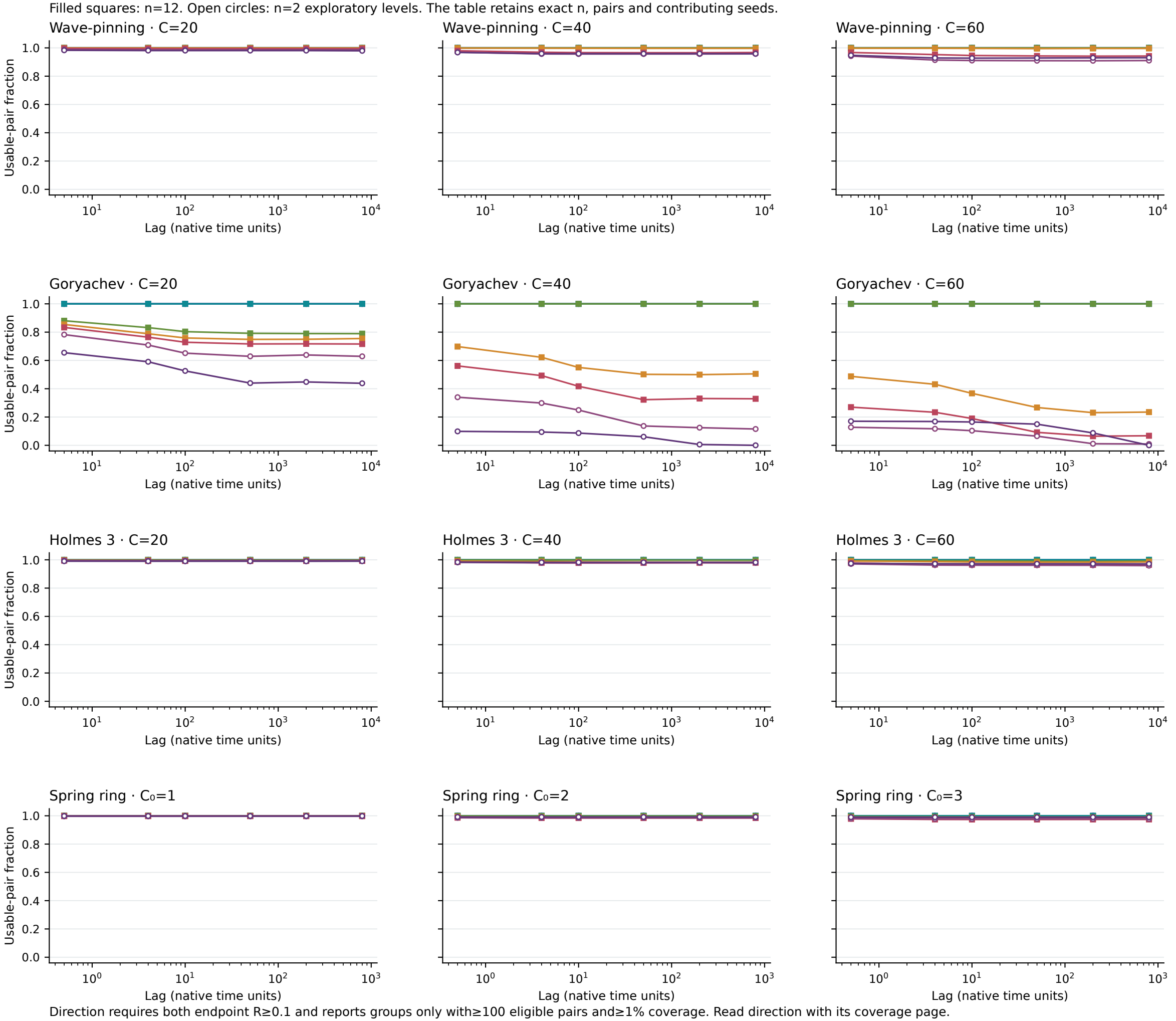
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

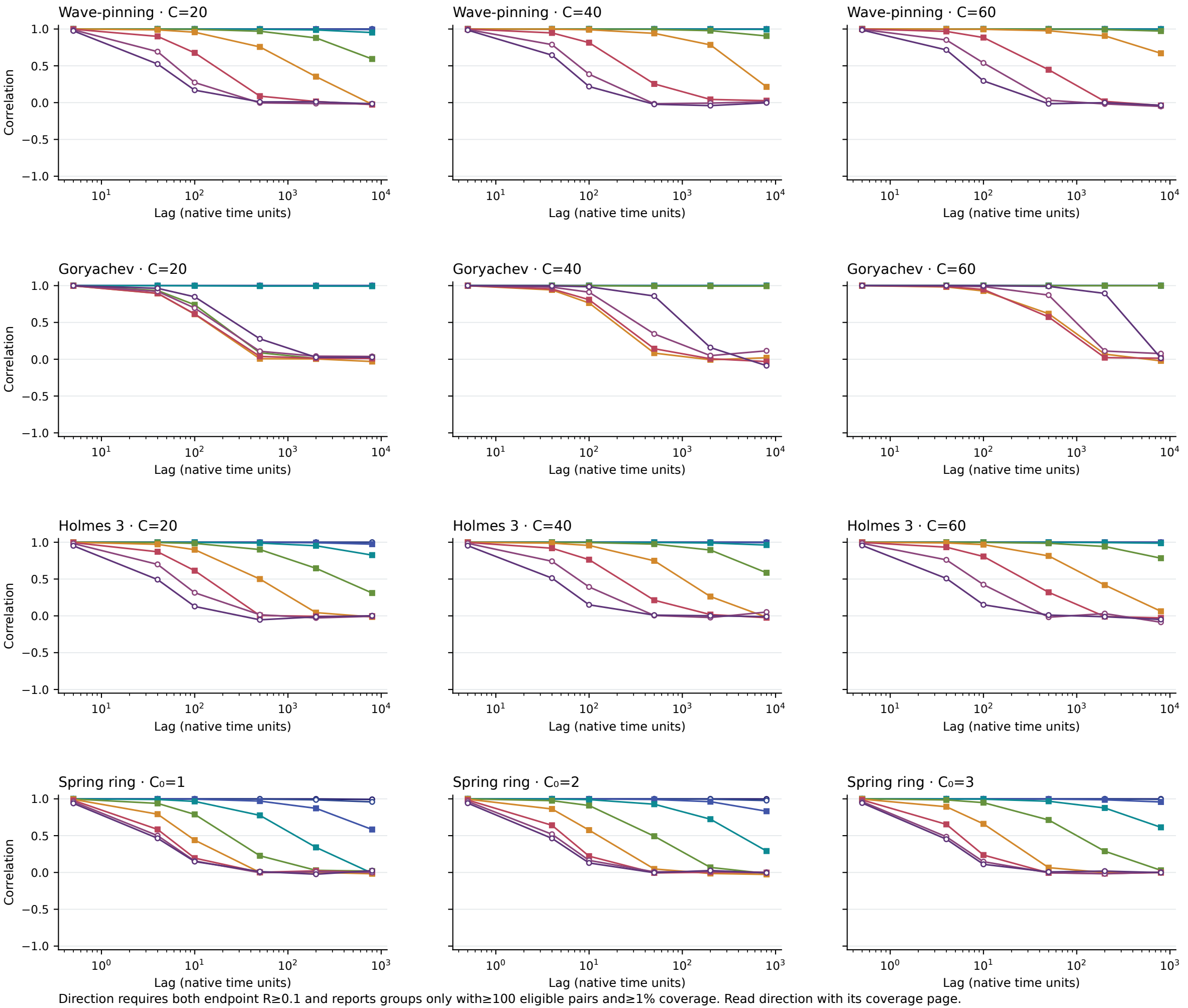
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Uncentered first-moment correlation: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

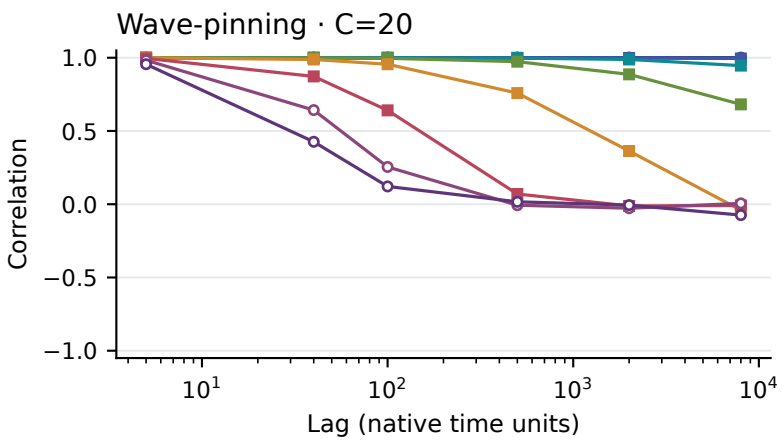
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

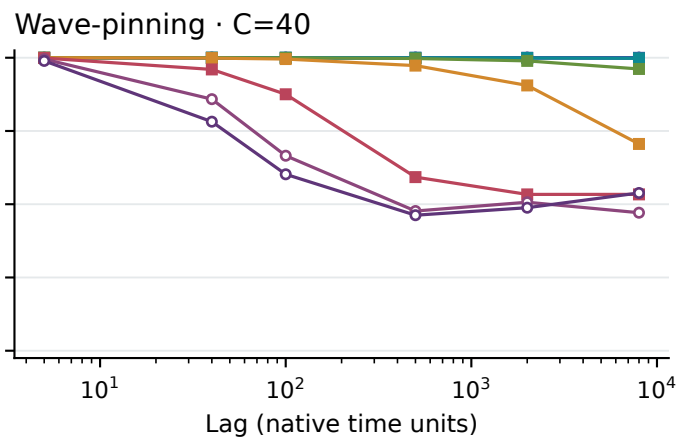
Conditional direction memory: all nine strengths

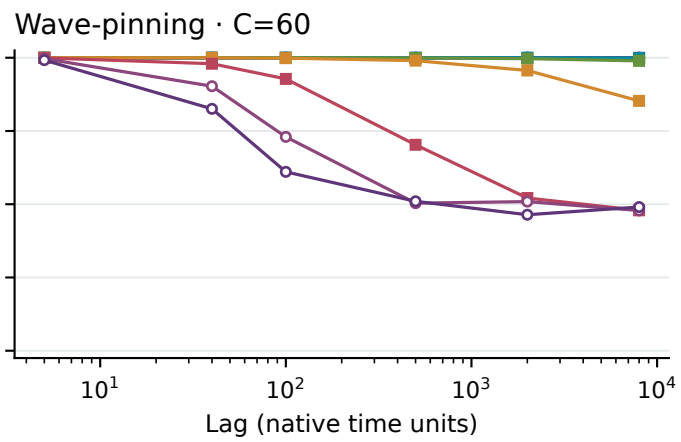
First half · chemical0–16000; Spring0–1600

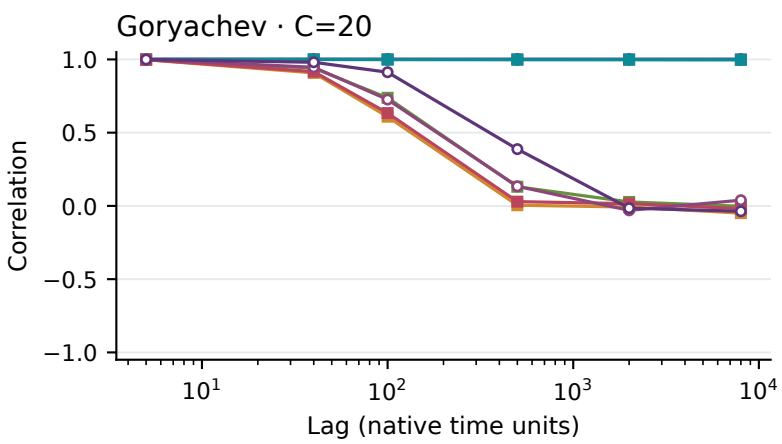
f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

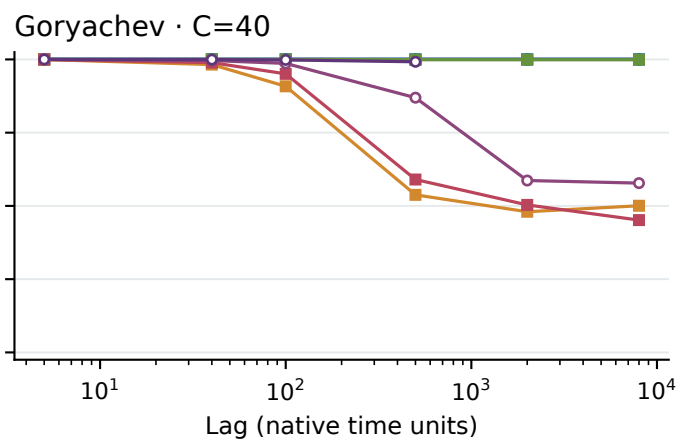
Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.

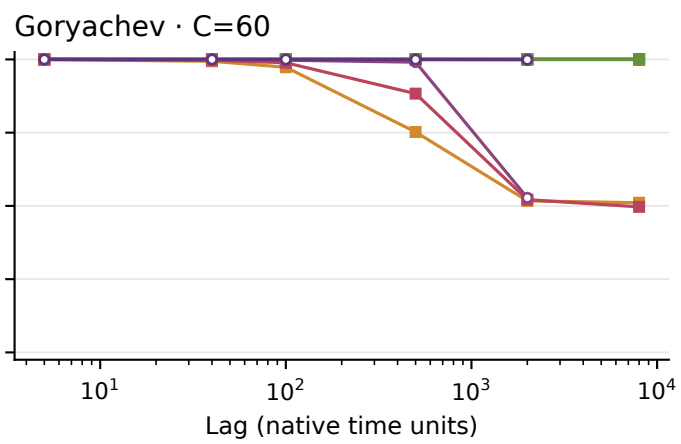


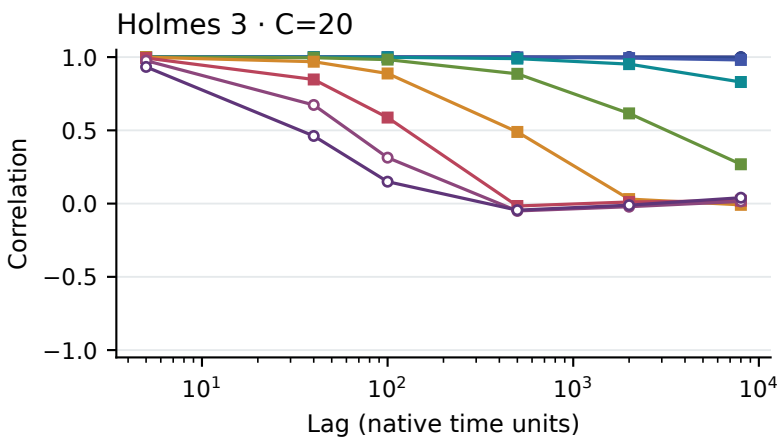


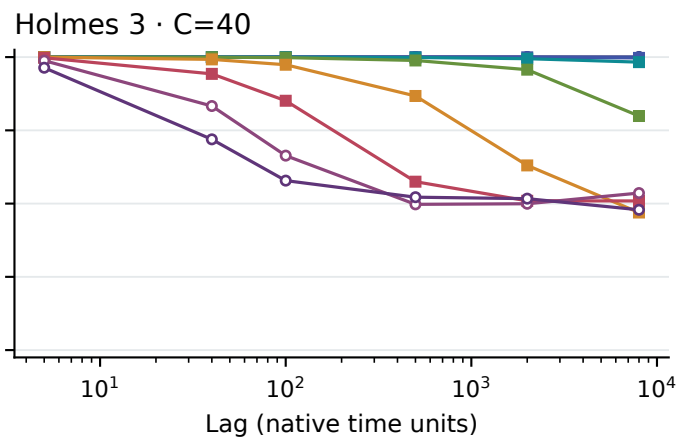


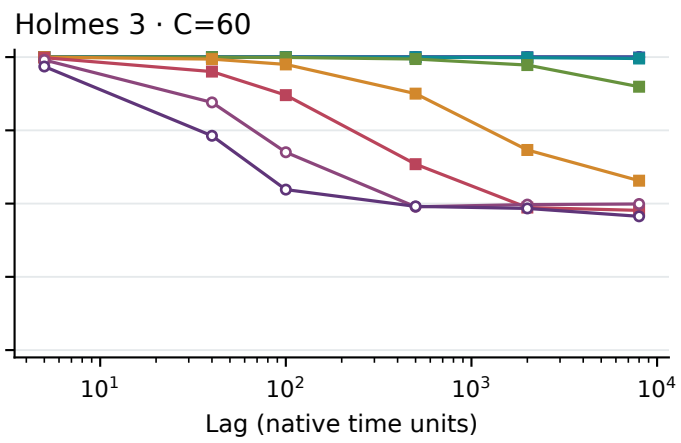


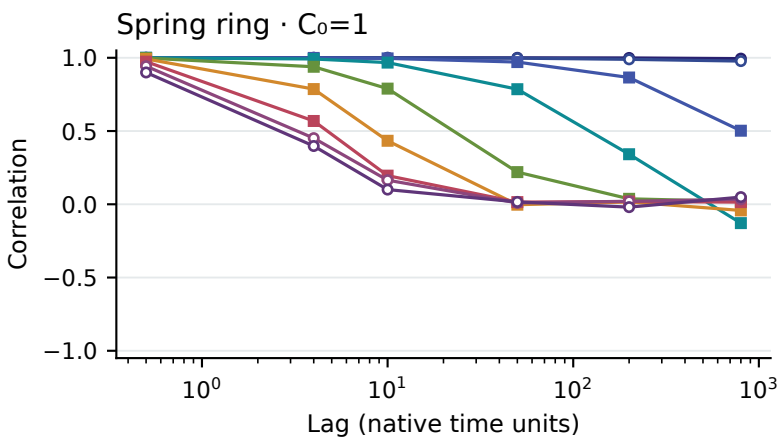


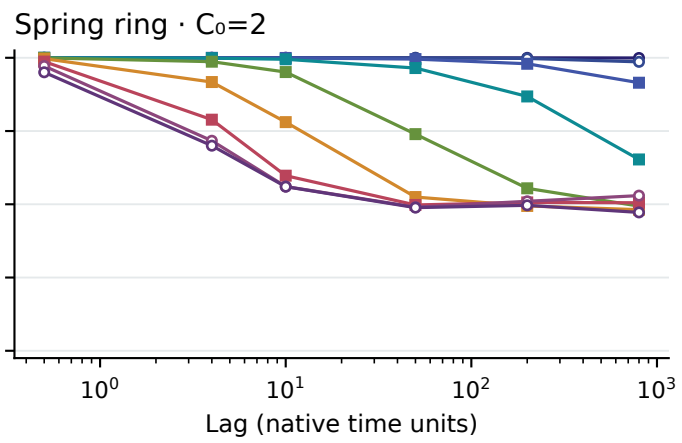


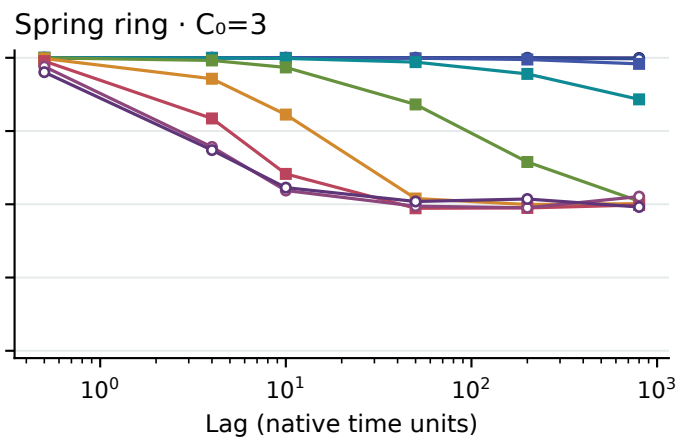












Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

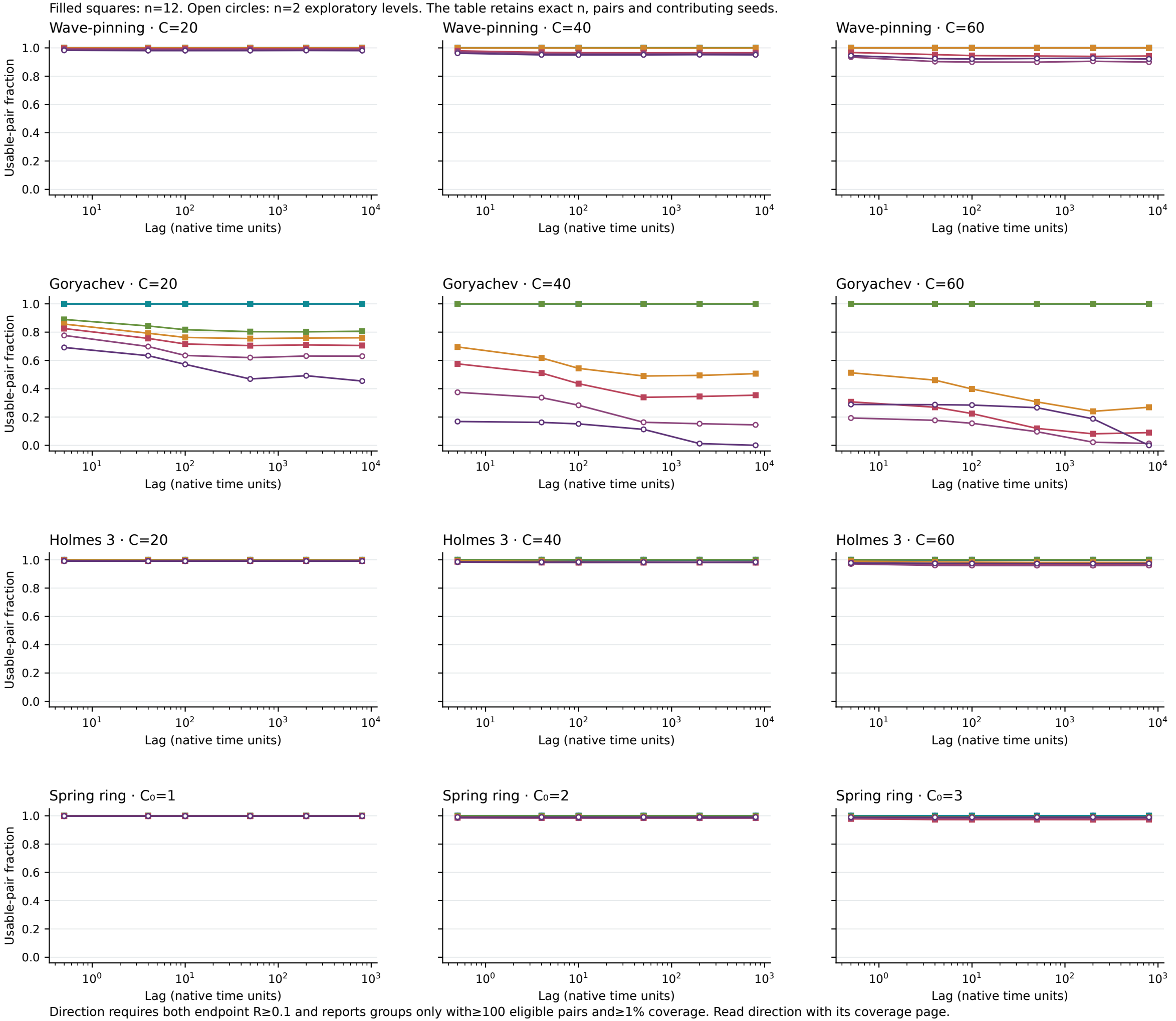
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

First half · chemical0–16000; Spring0–1600

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

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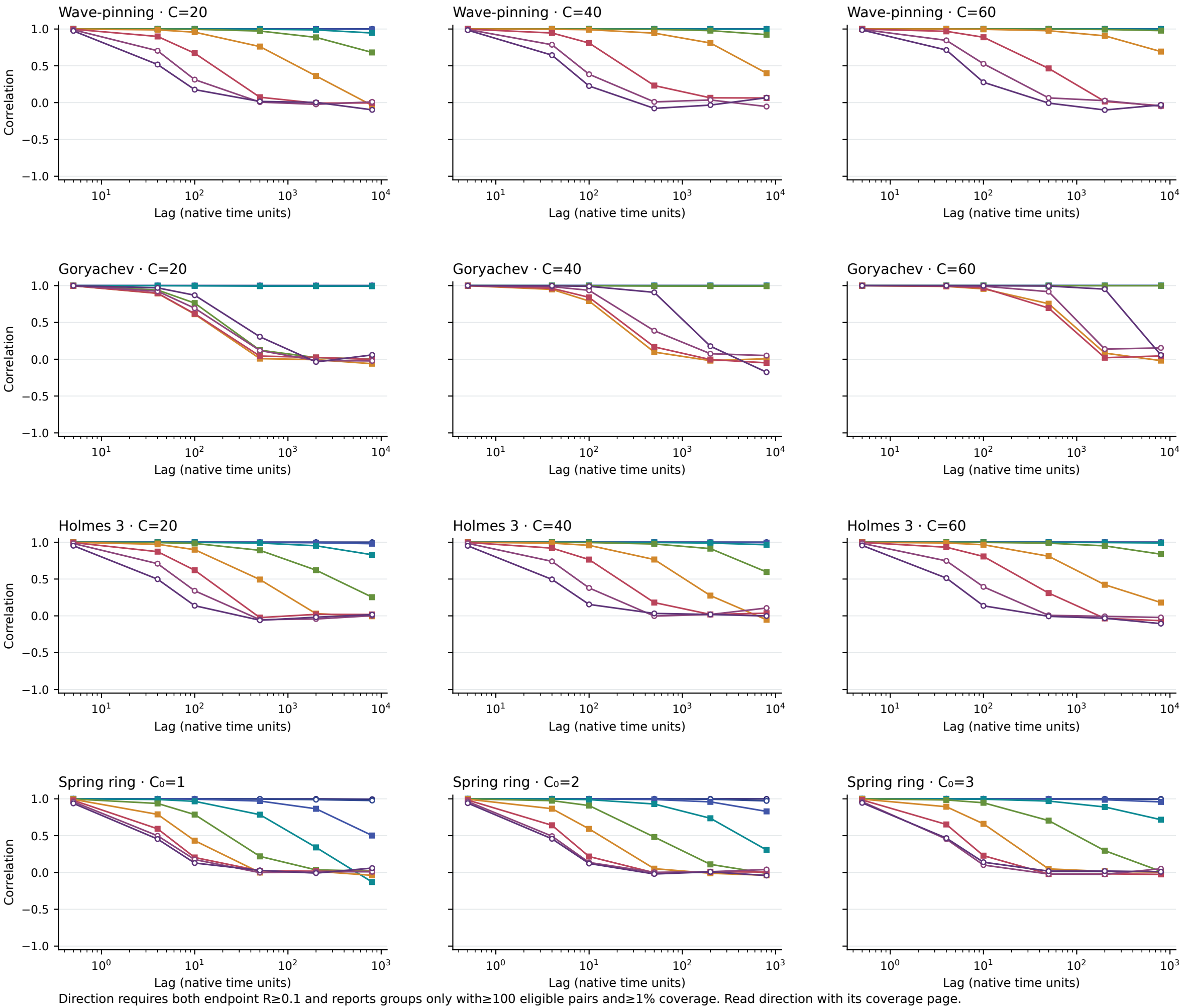
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Uncentered first-moment correlation: all nine strengths

First half · chemical0–16000; Spring0–1600

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



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Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

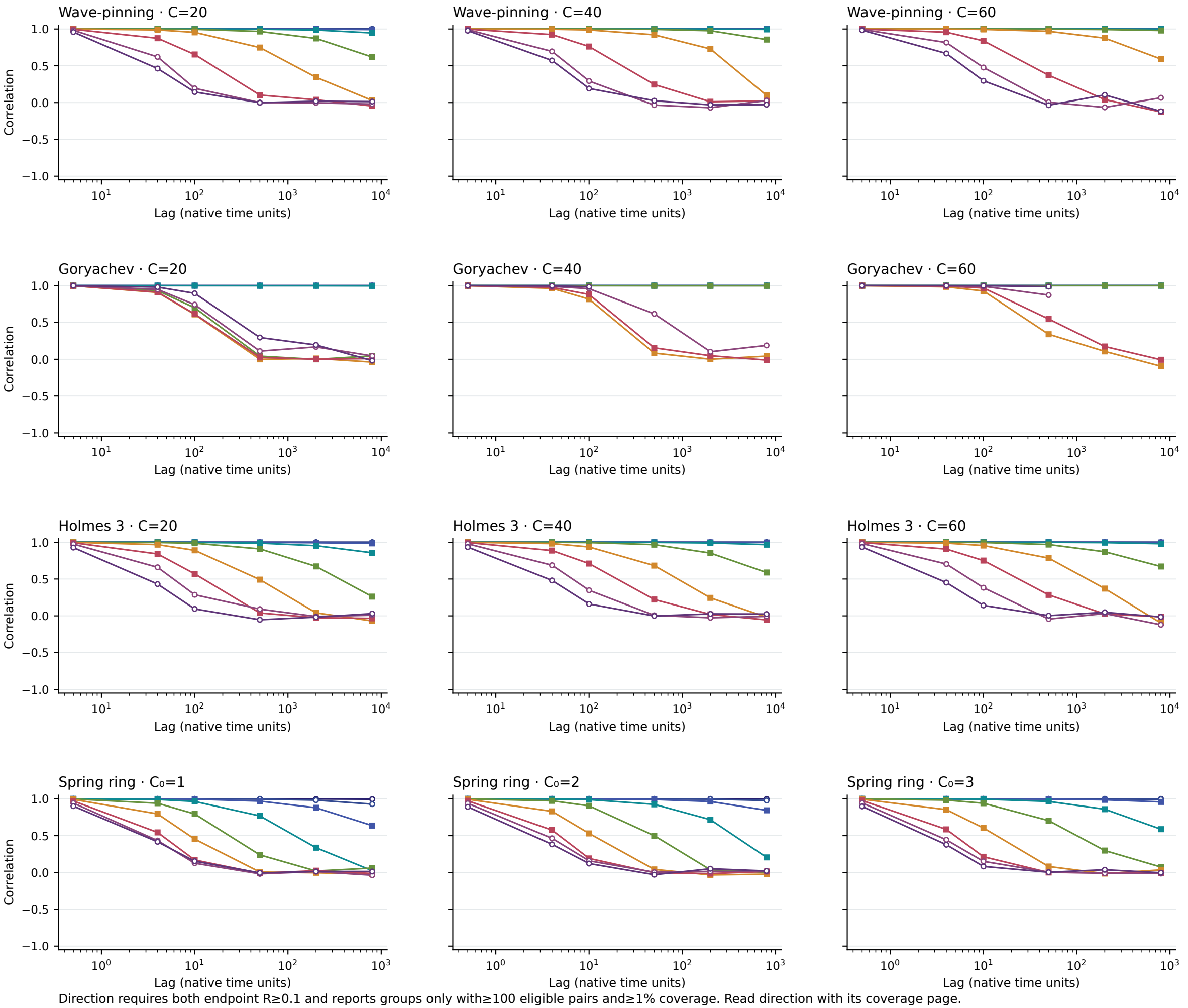
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Conditional direction memory: all nine strengths

Second half after discarding the first half · chemical16000–32000; Spring1600–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

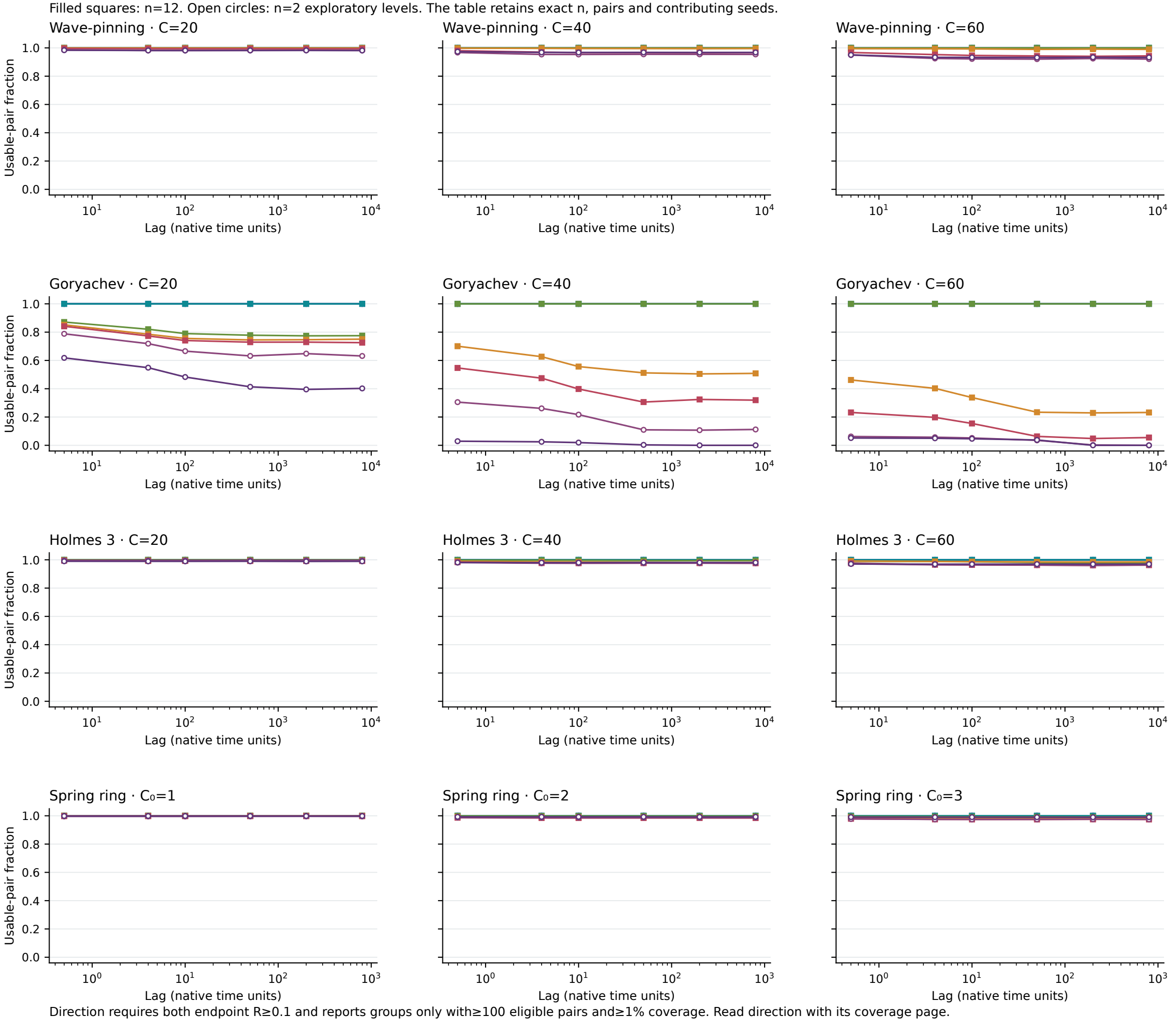
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

Second half after discarding the first half · chemical16000–32000; Spring1600–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



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Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

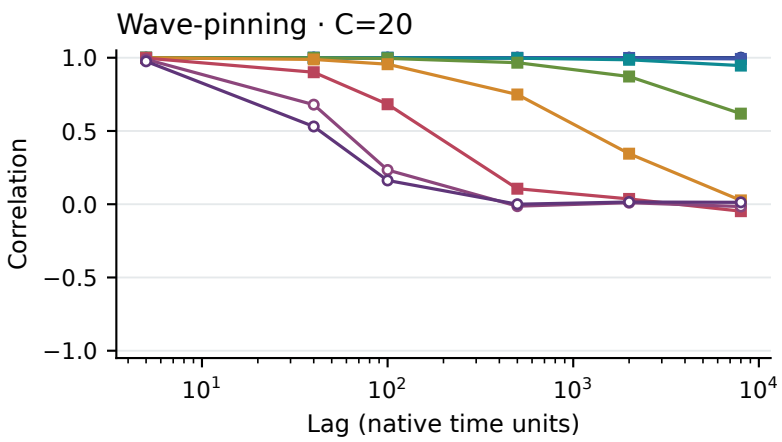
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

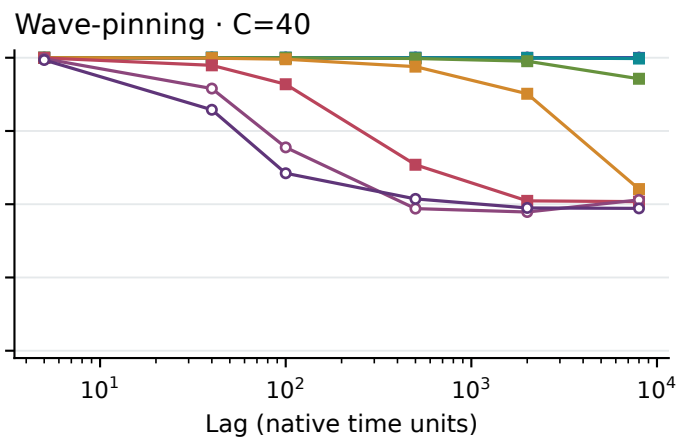
Uncentered first-moment correlation: all nine strengths

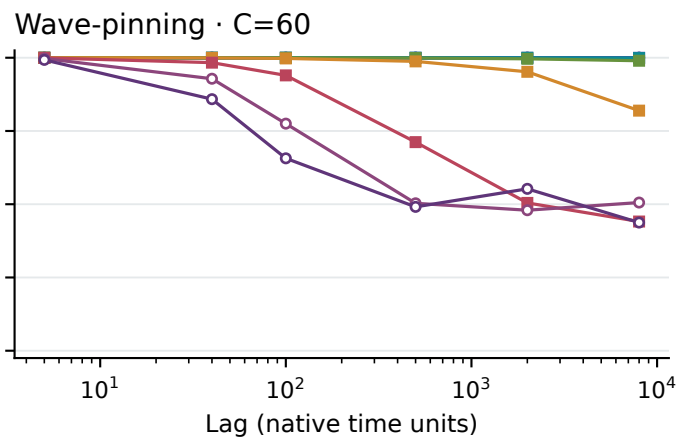
Second half after discarding the first half · chemical16000–32000; Spring1600–3200

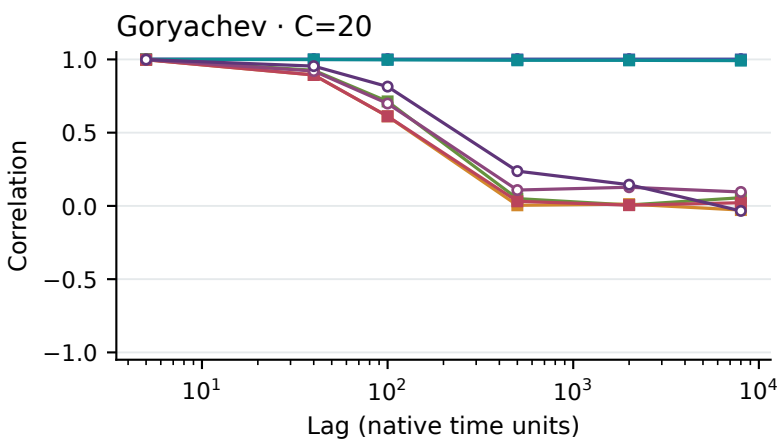
f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

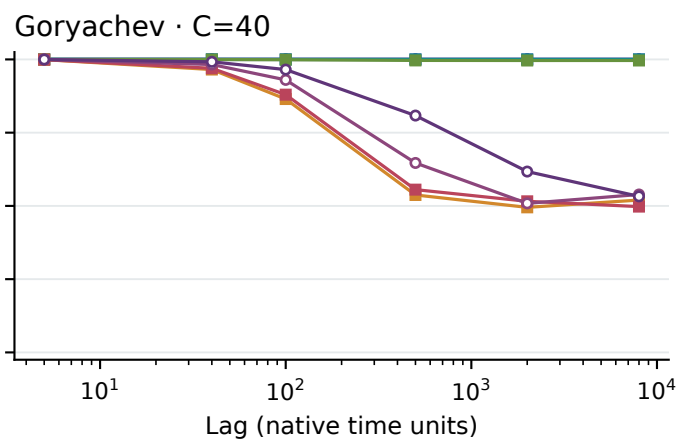
Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.

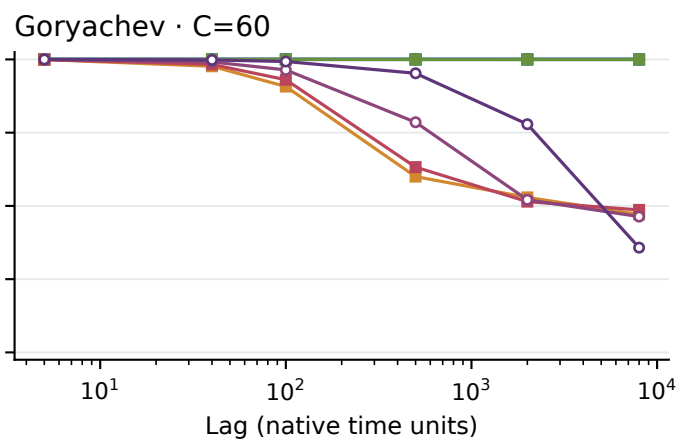


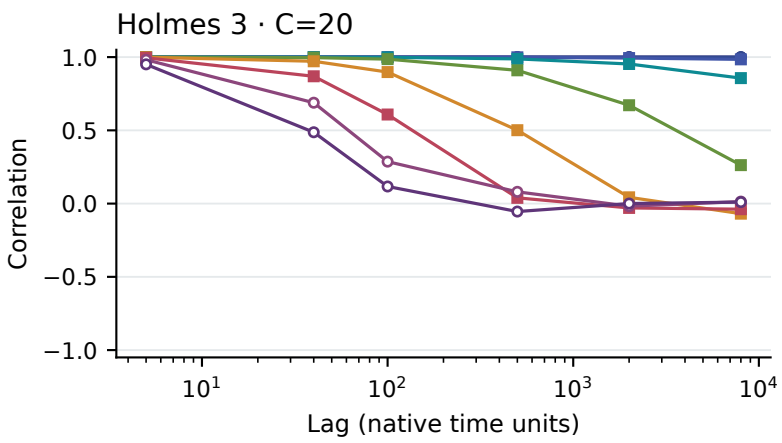


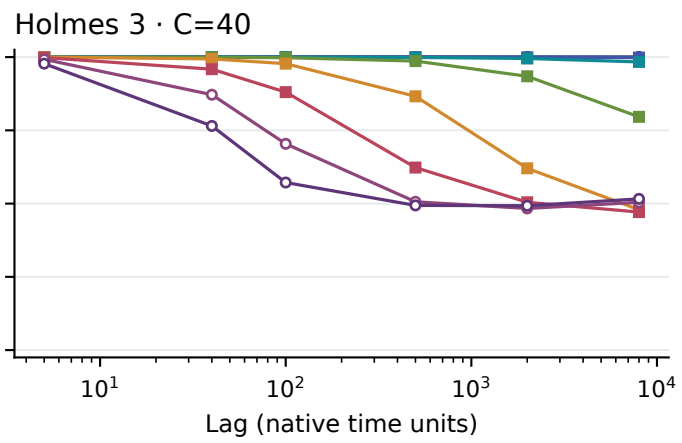


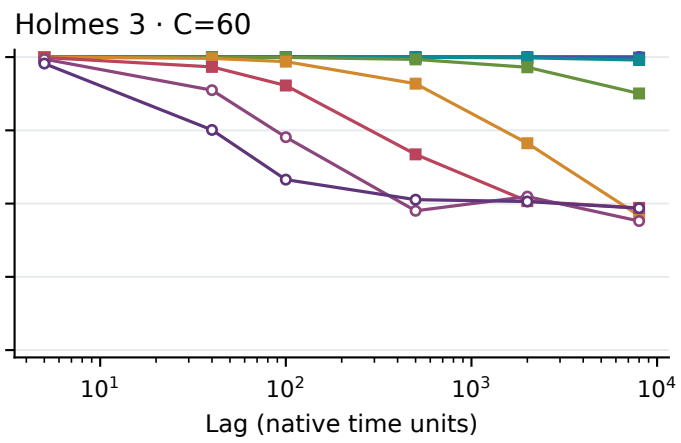


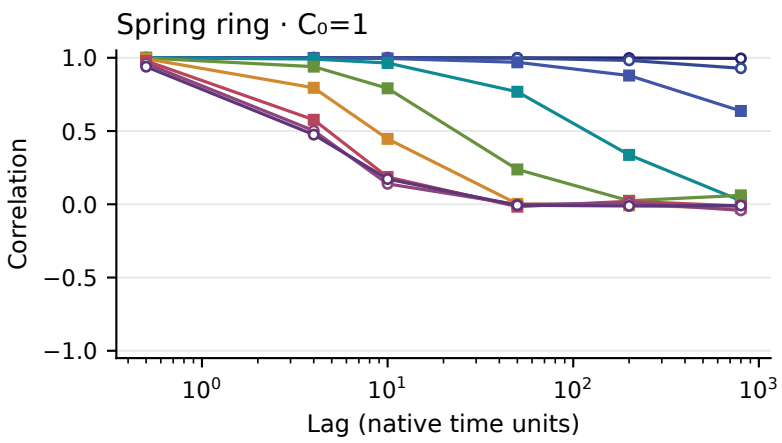


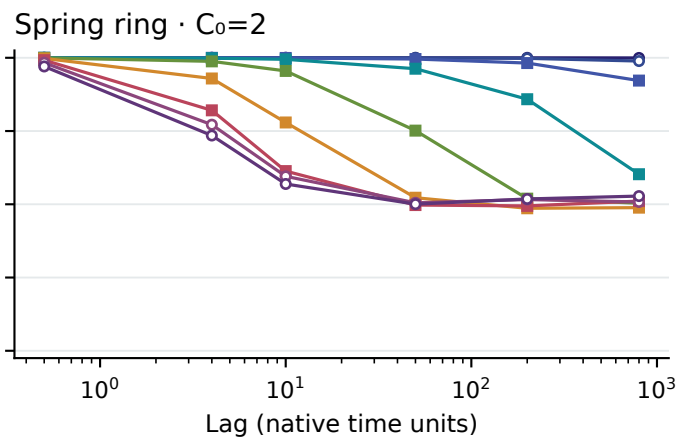


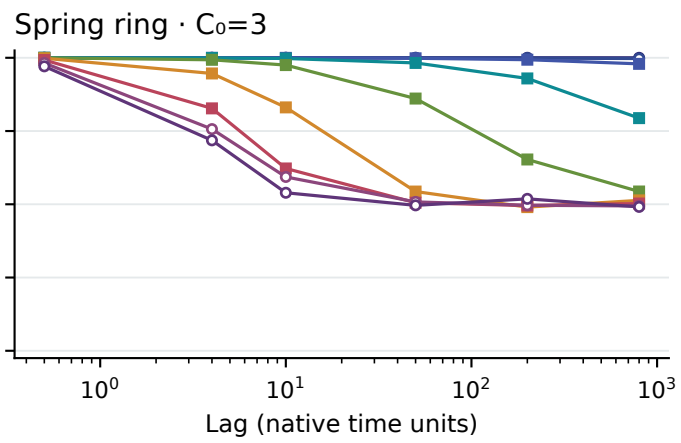












Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

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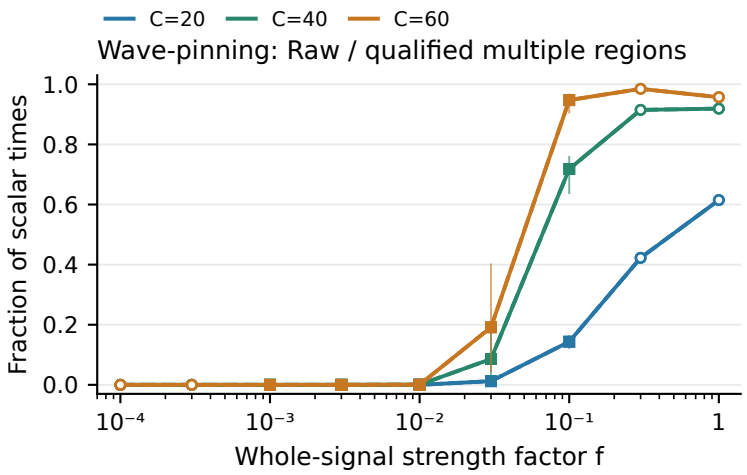
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

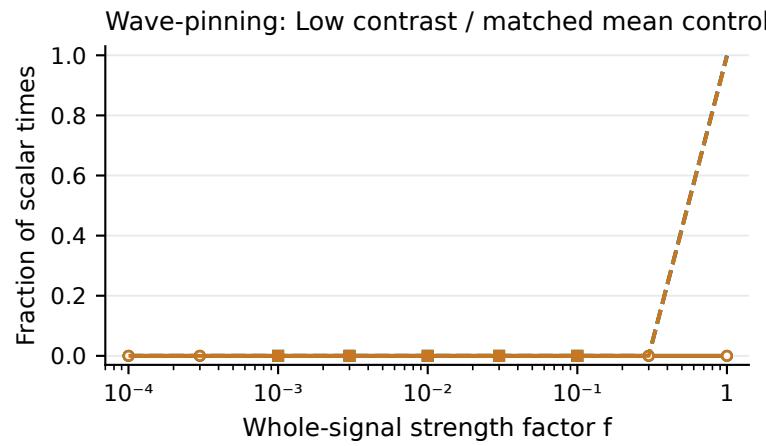
Organization, contrast and global magnitude: all sizes and strengths

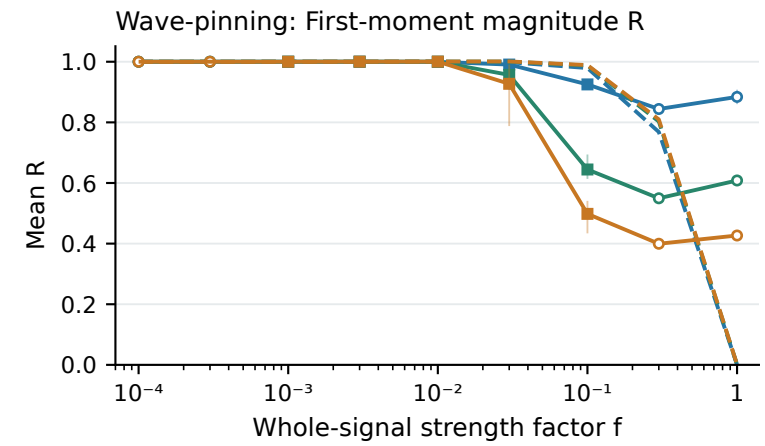
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

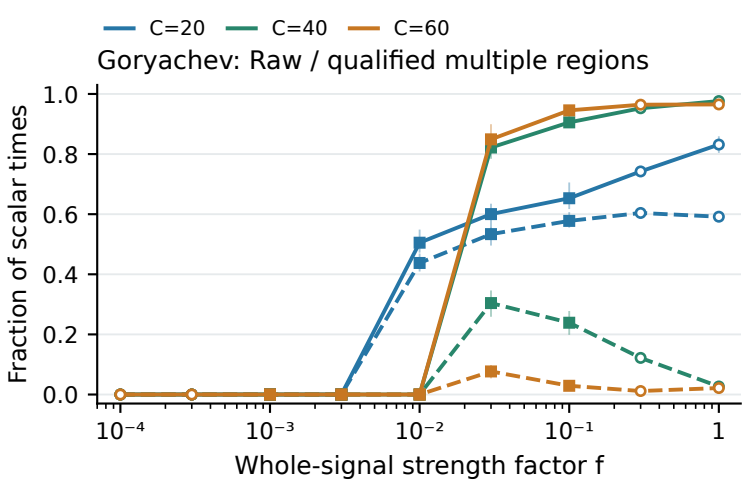
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

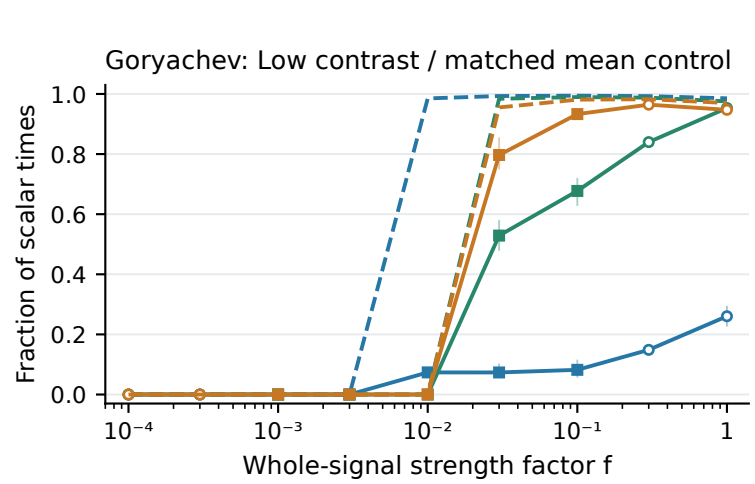
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

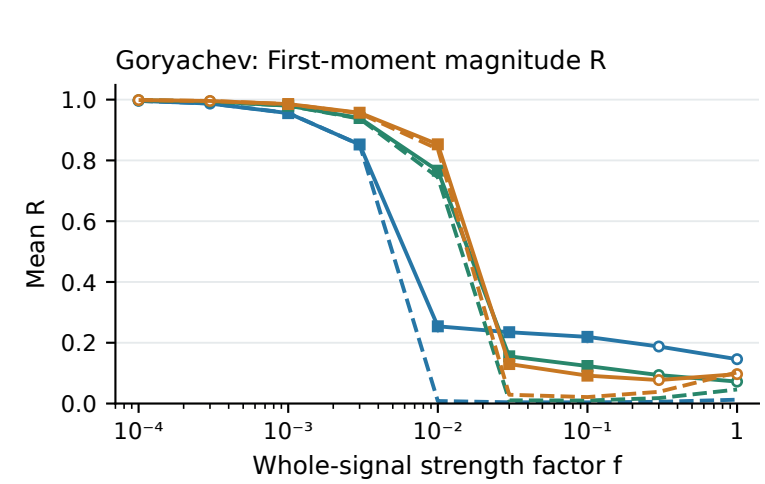


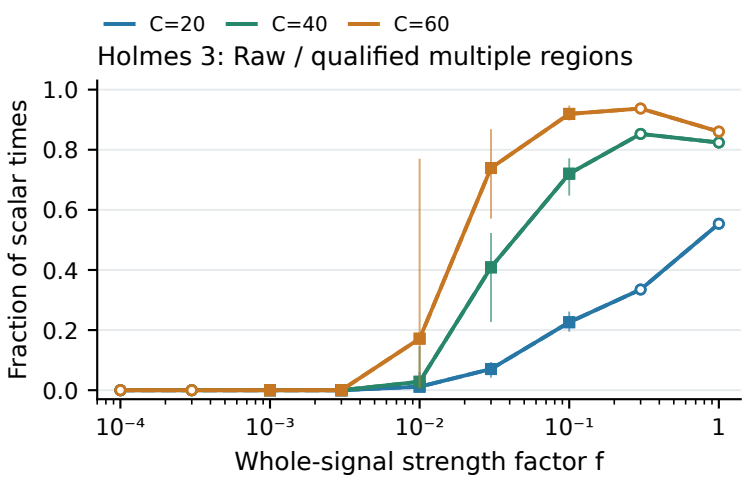


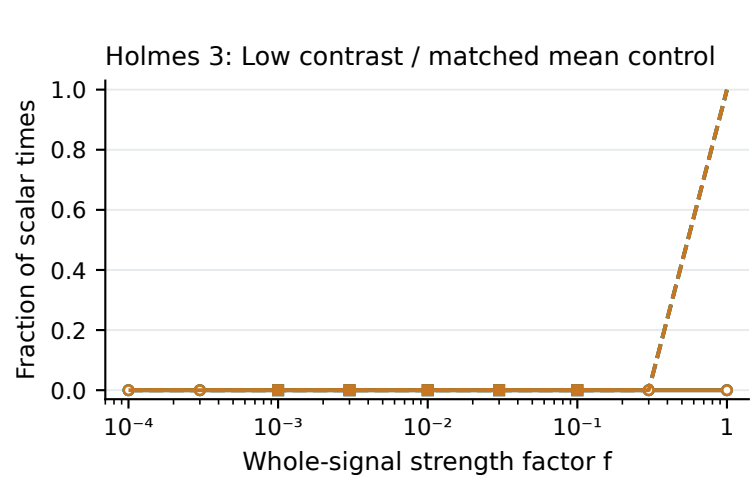


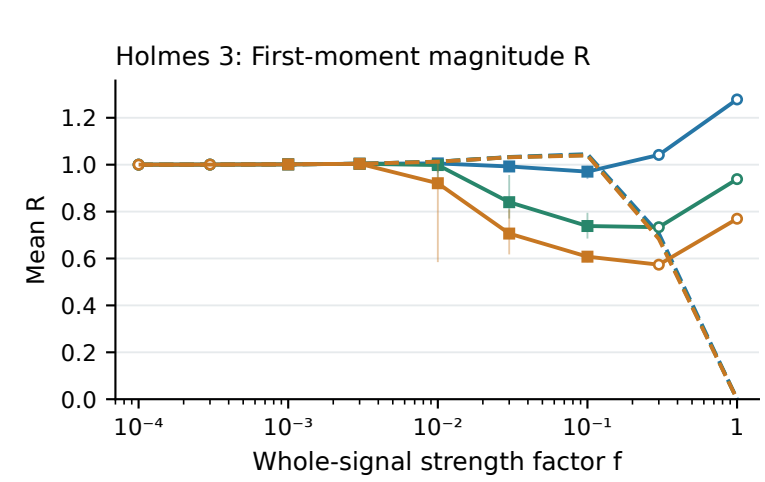


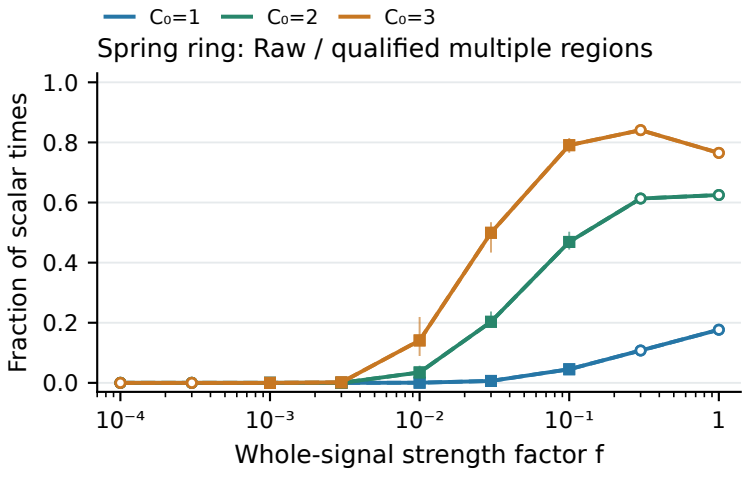


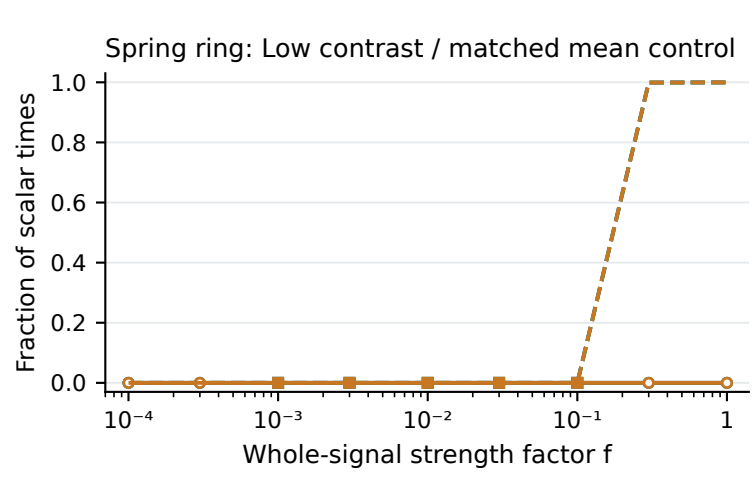


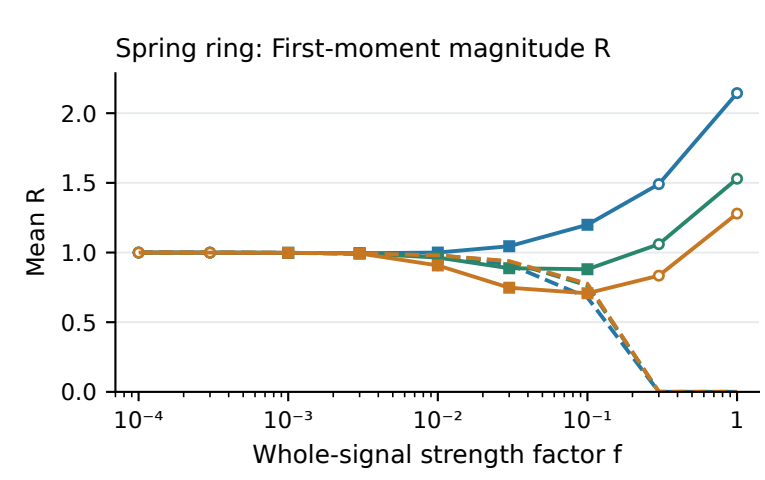












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

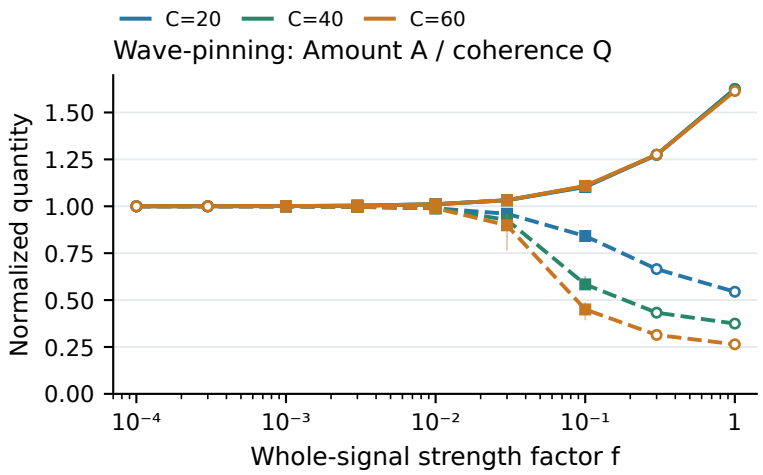
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

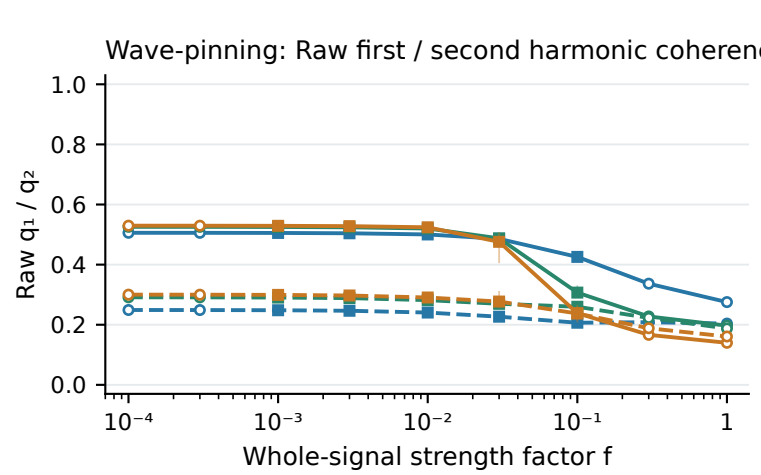
Active amount, coherence and spatial harmonics: all sizes and strengths

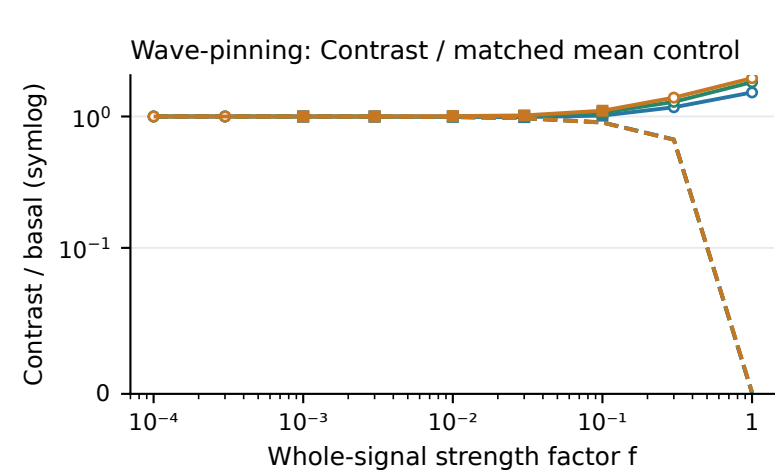
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

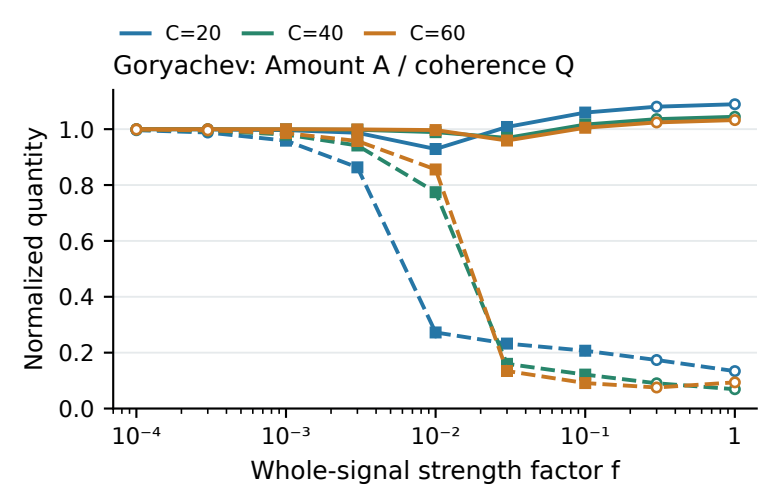
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

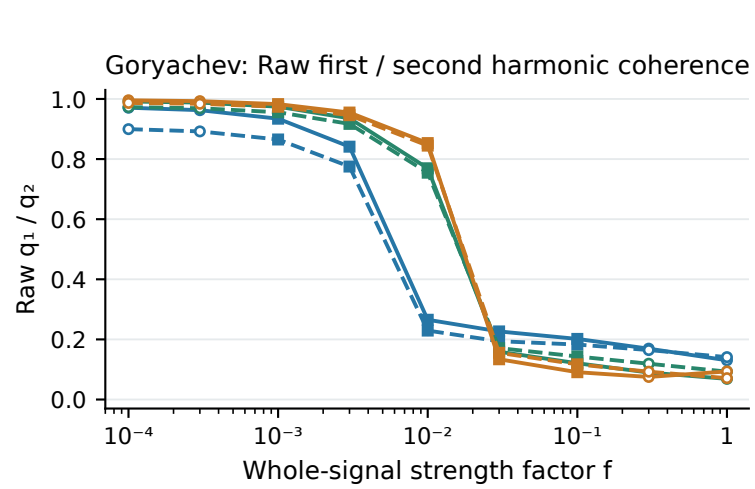
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

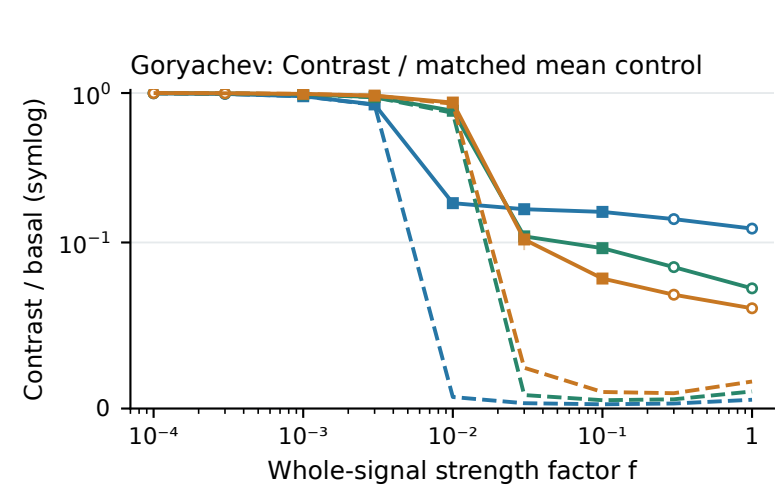


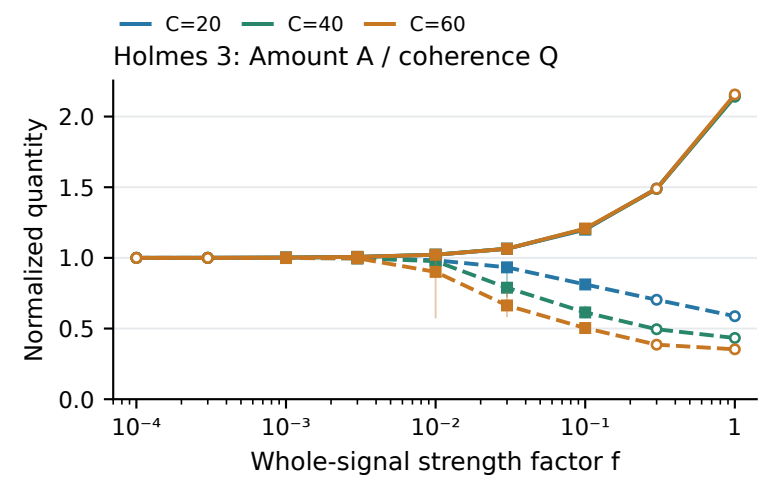


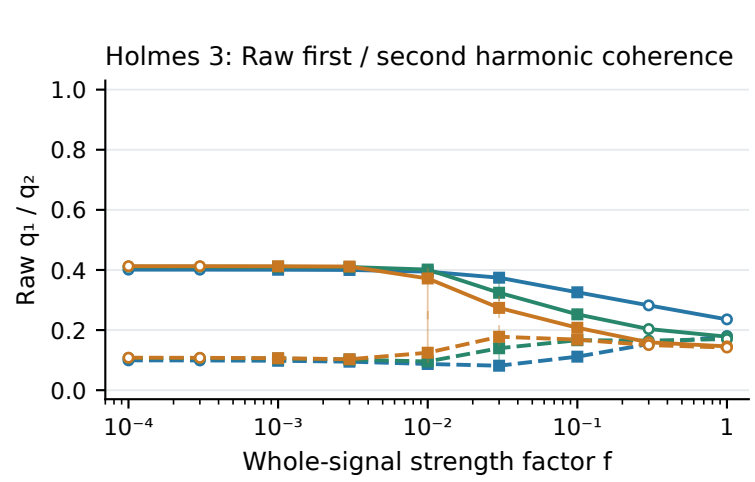


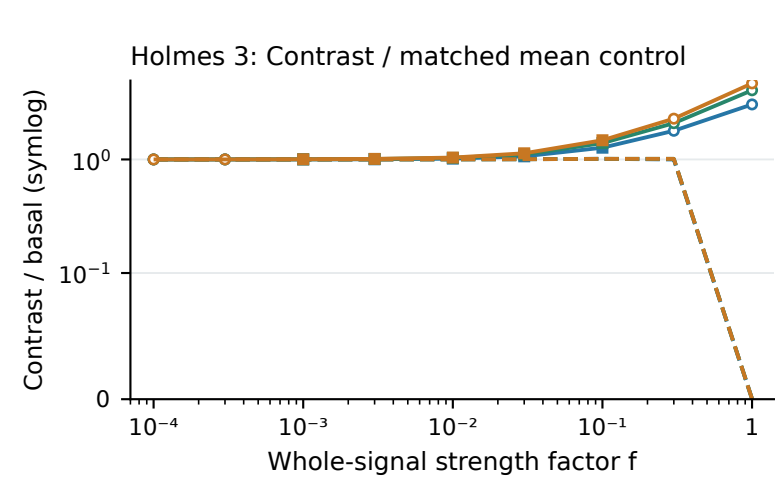


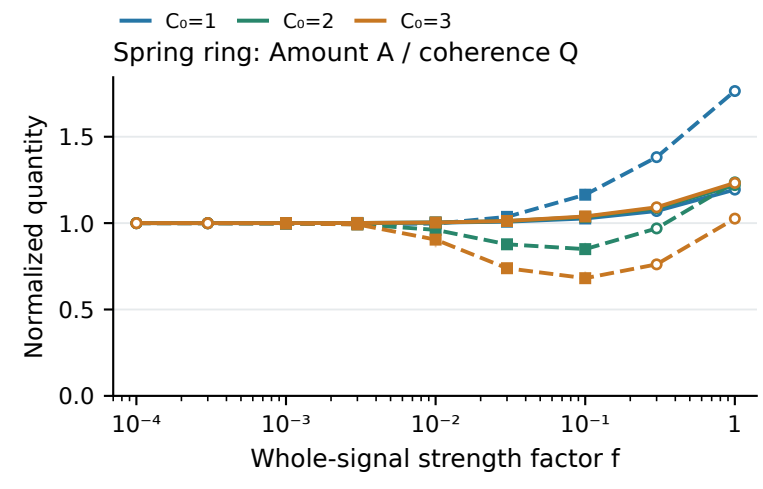


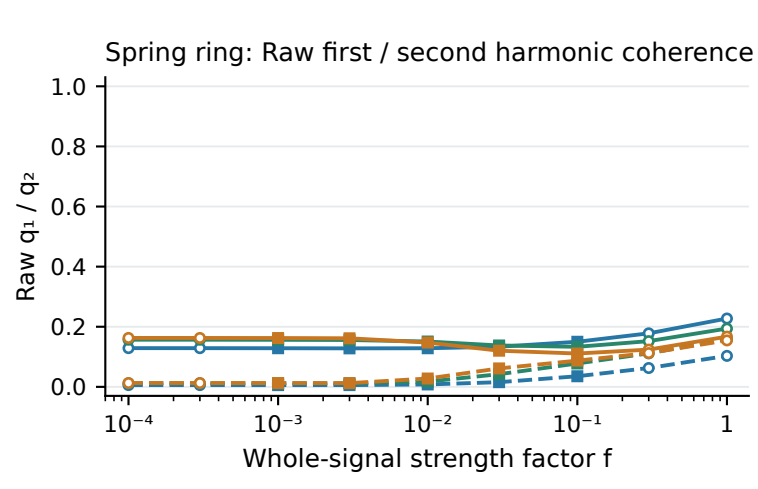


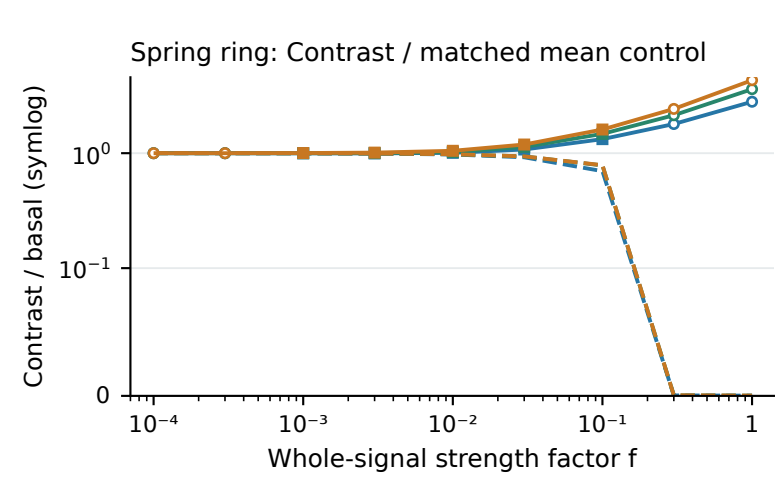












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

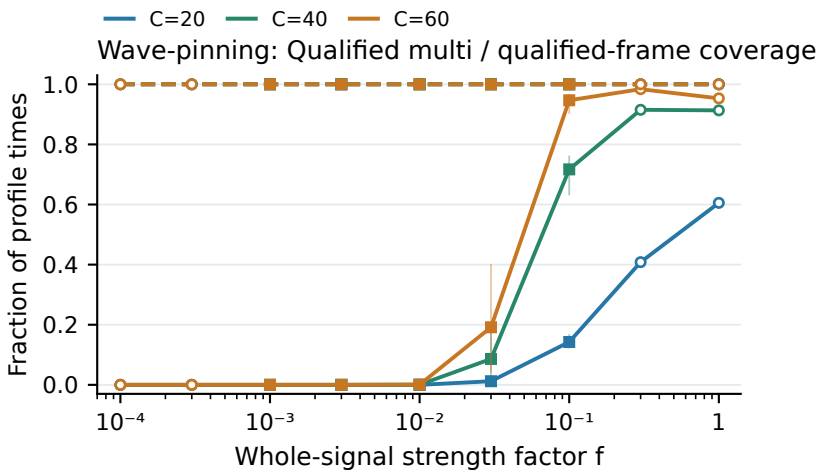
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

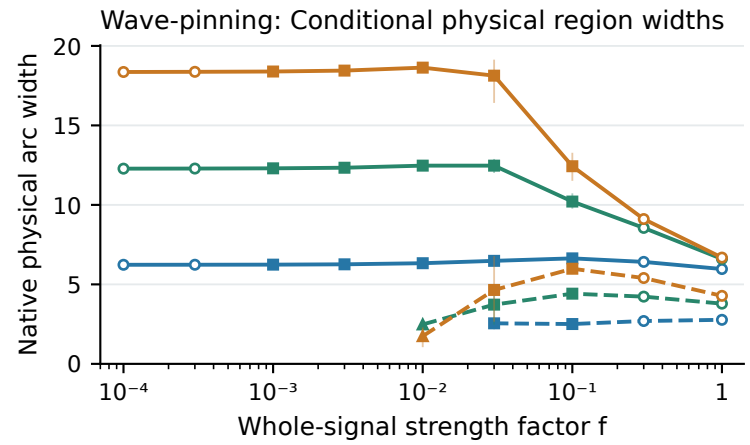
Instantaneous region widths with qualification coverage: all sizes and strengths

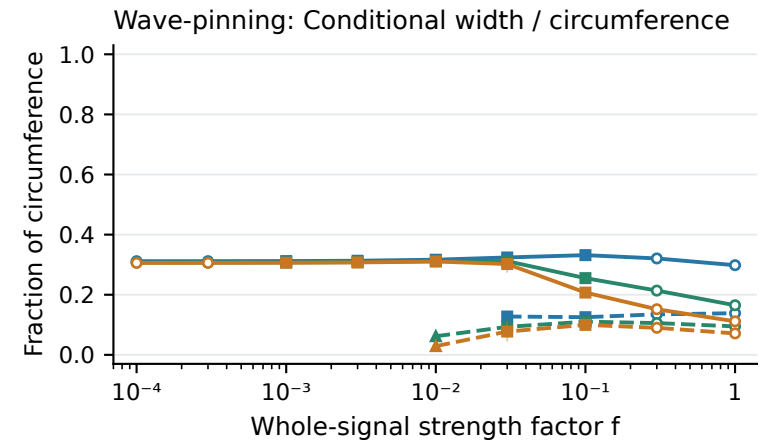
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

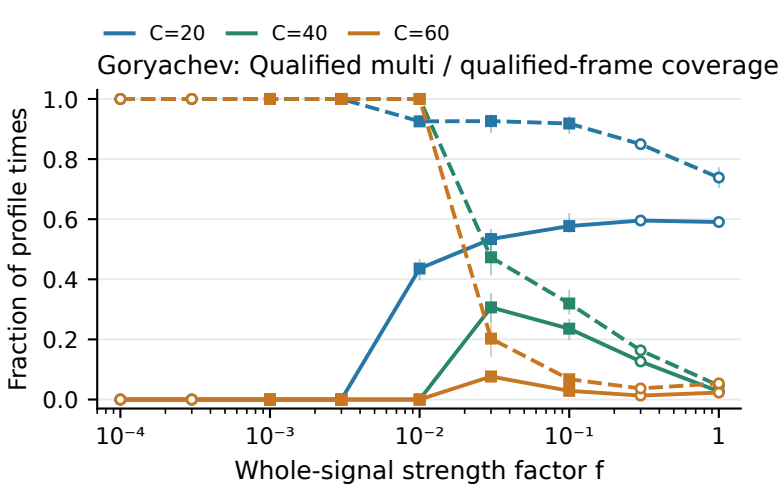
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

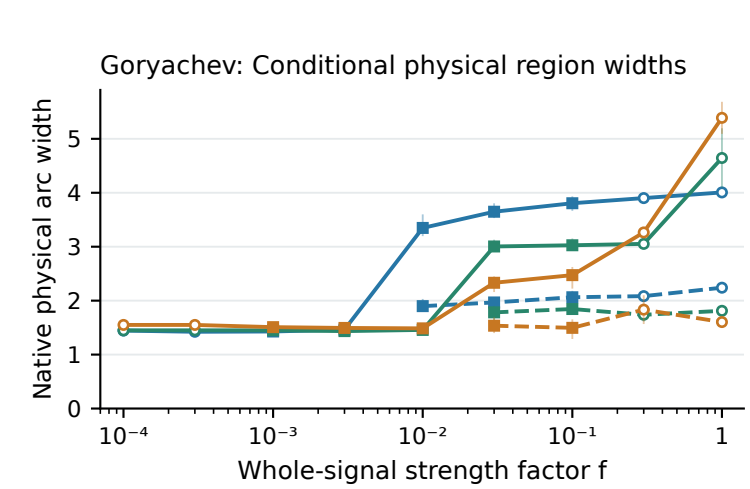
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

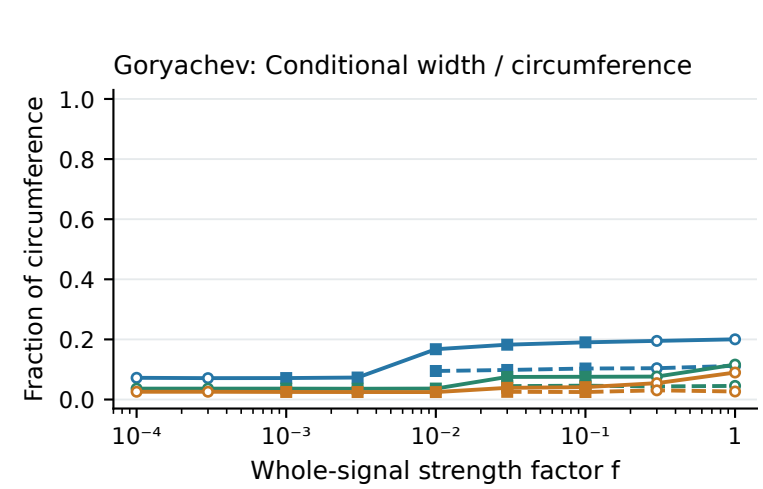


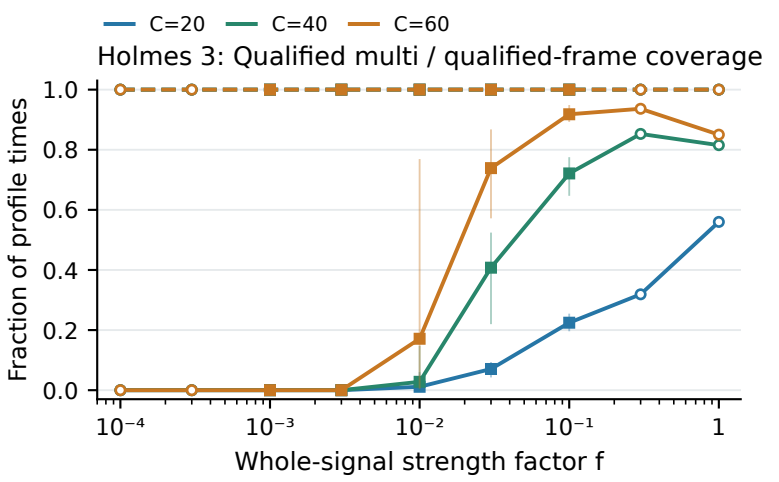


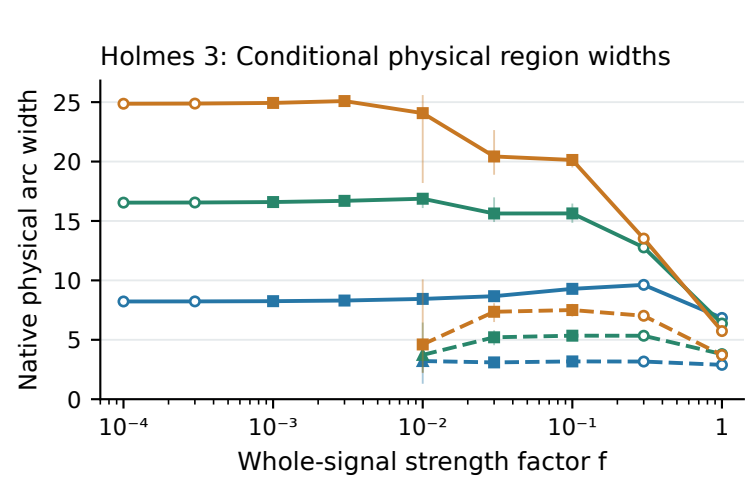


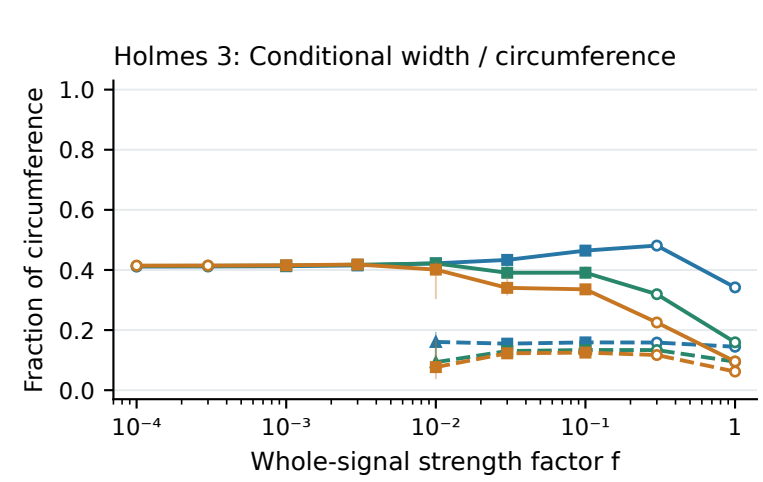


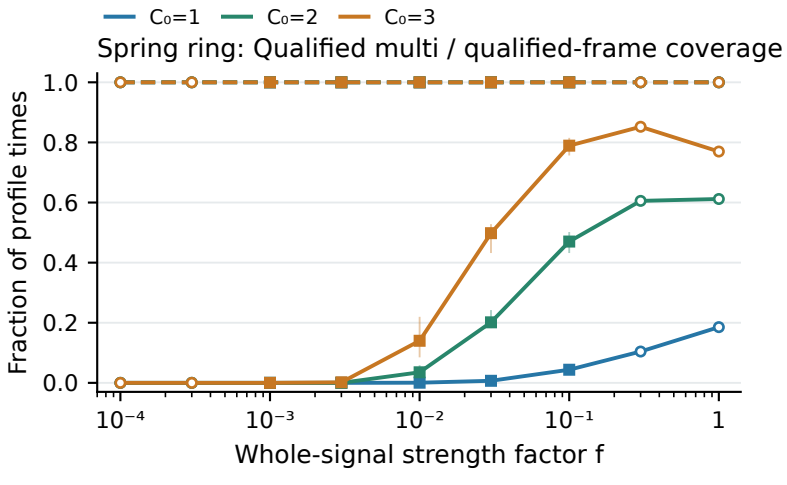


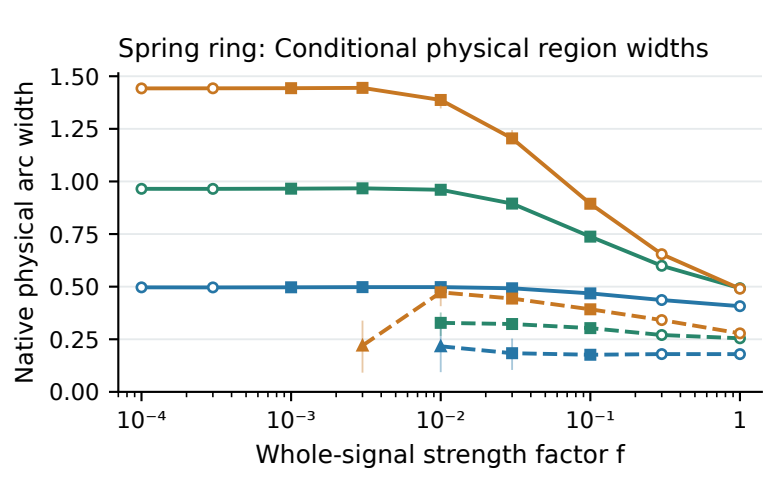


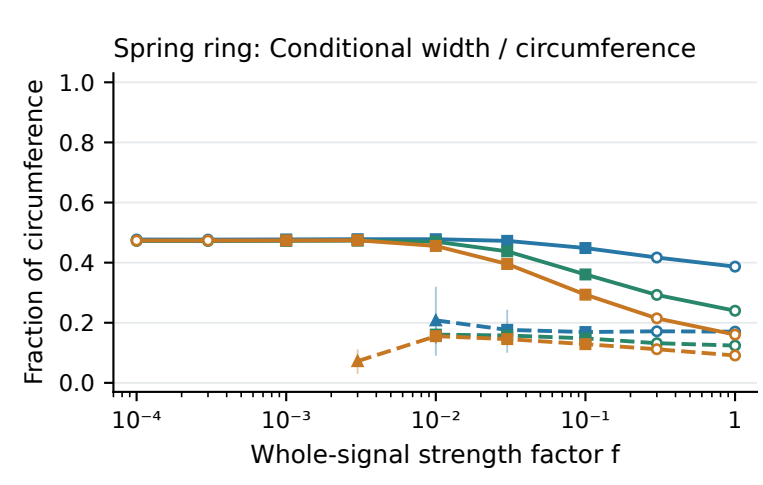












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

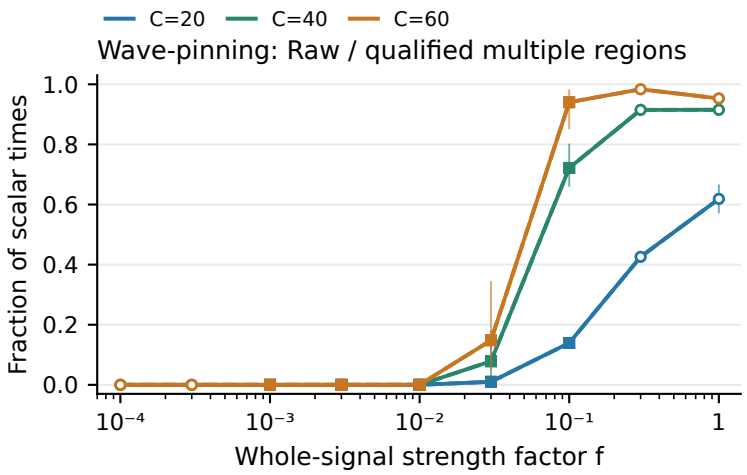
Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

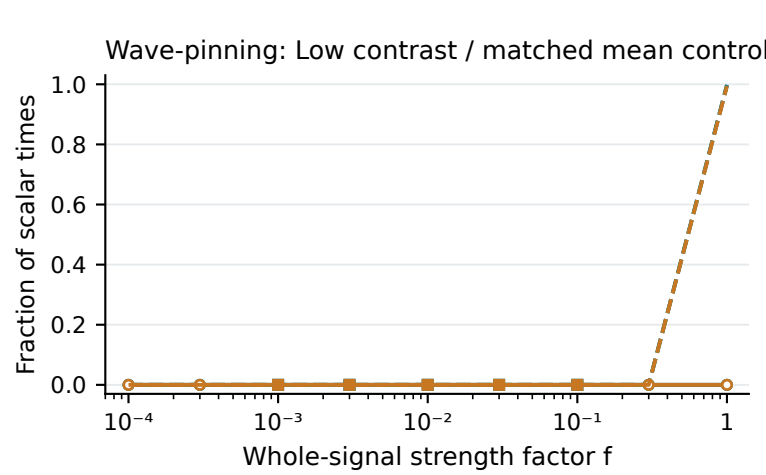
Organization, contrast and global magnitude: all sizes and strengths

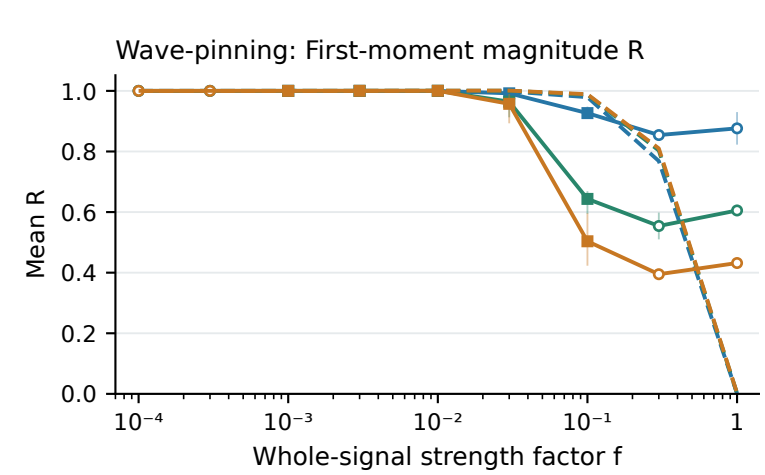
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

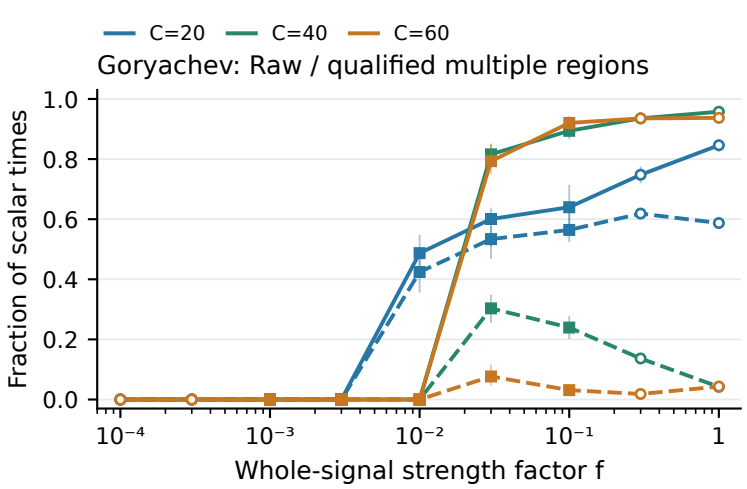
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

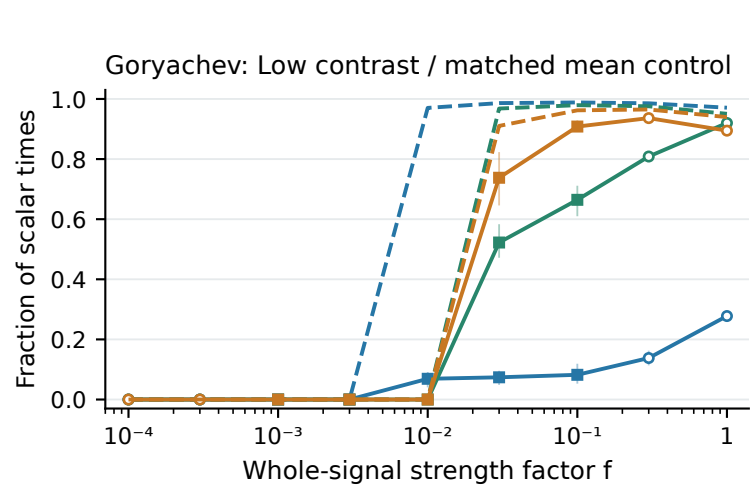
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

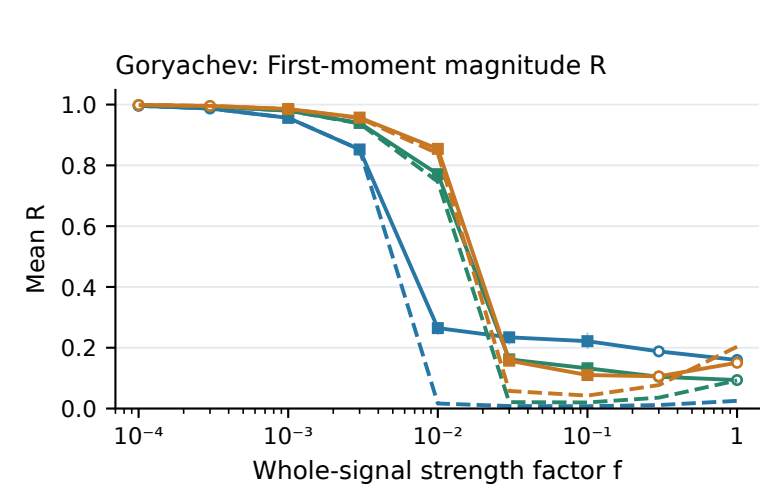


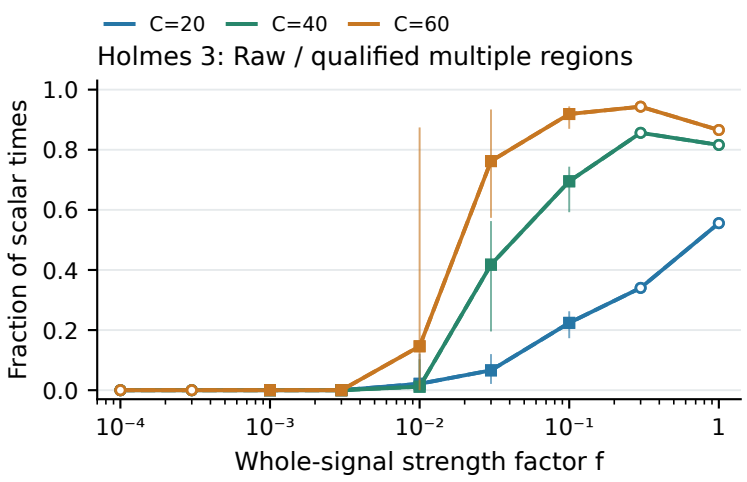


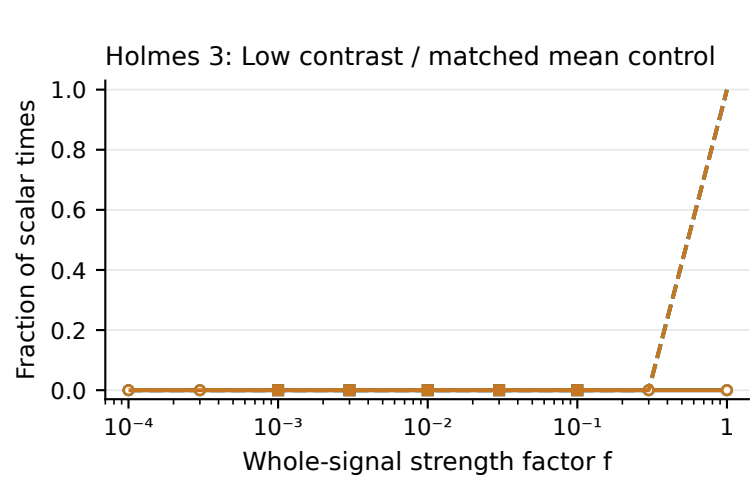


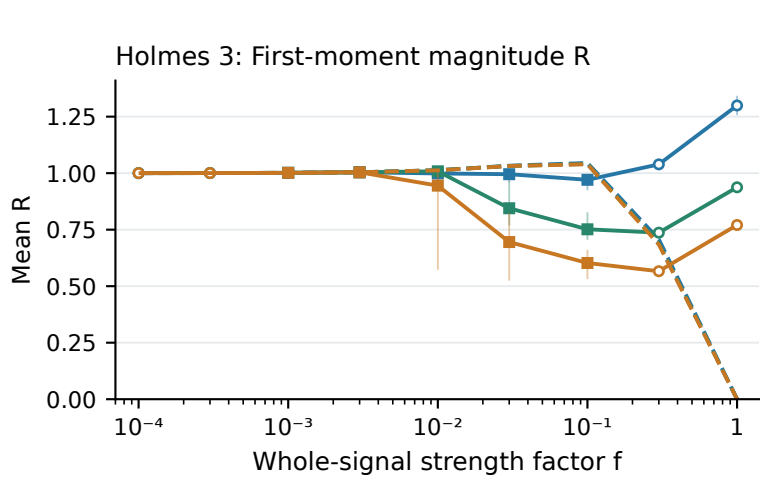


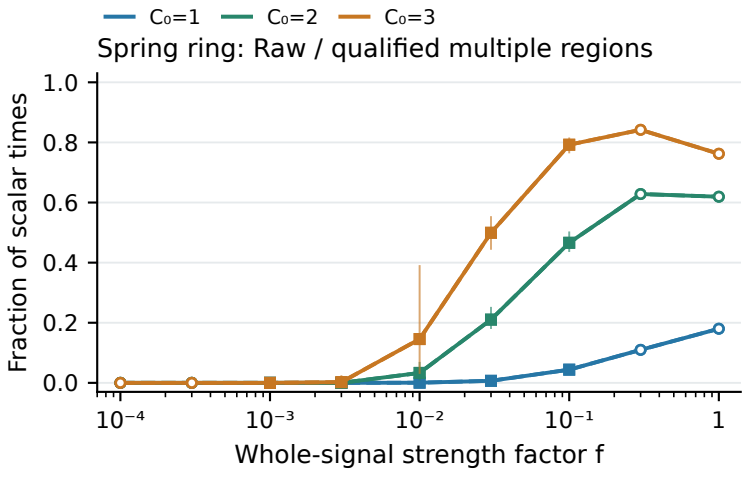


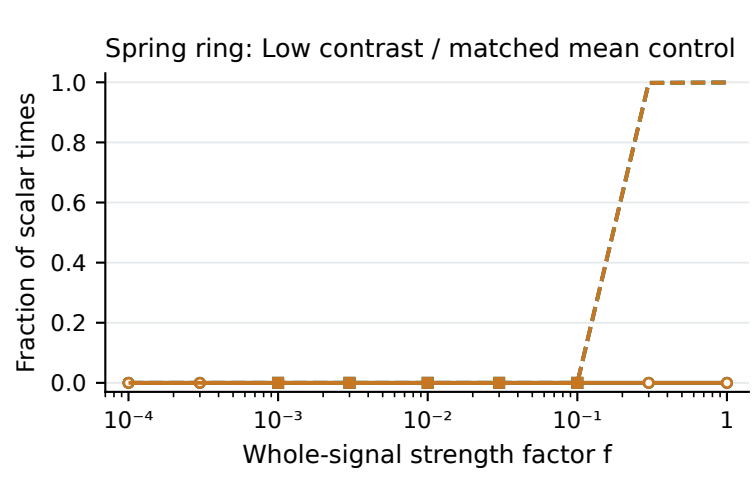


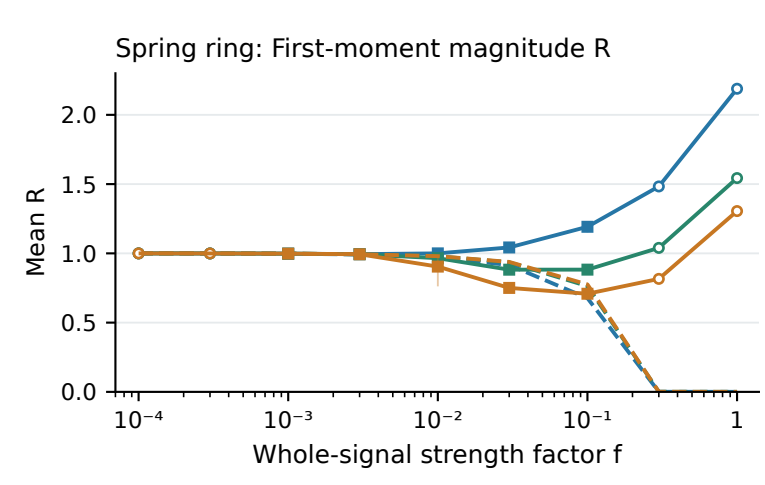












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

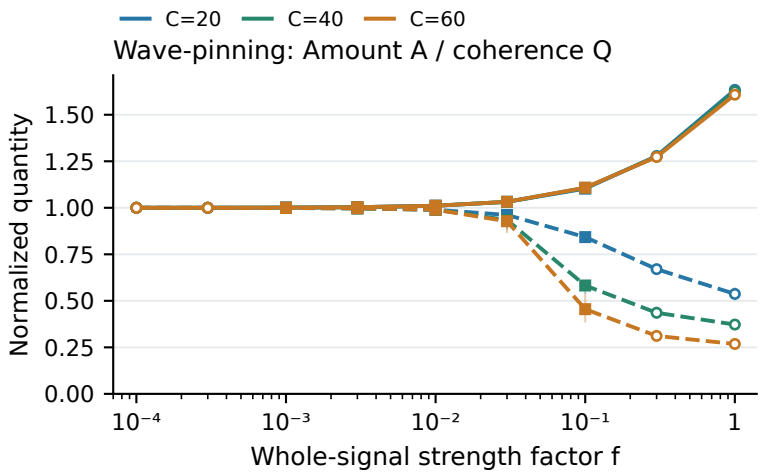
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

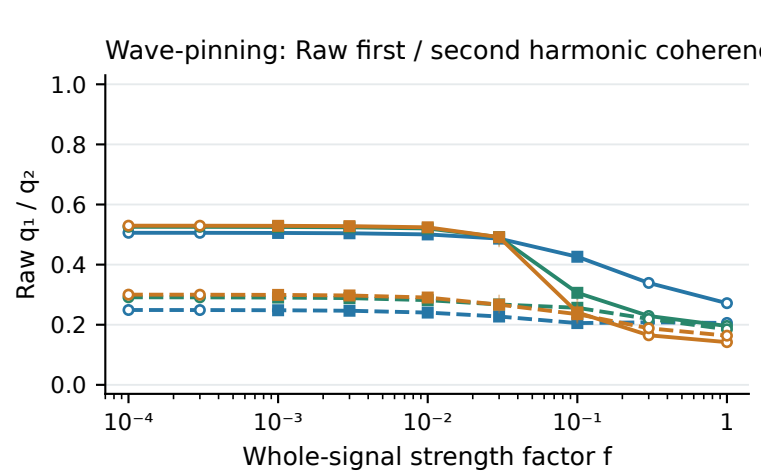
Active amount, coherence and spatial harmonics: all sizes and strengths

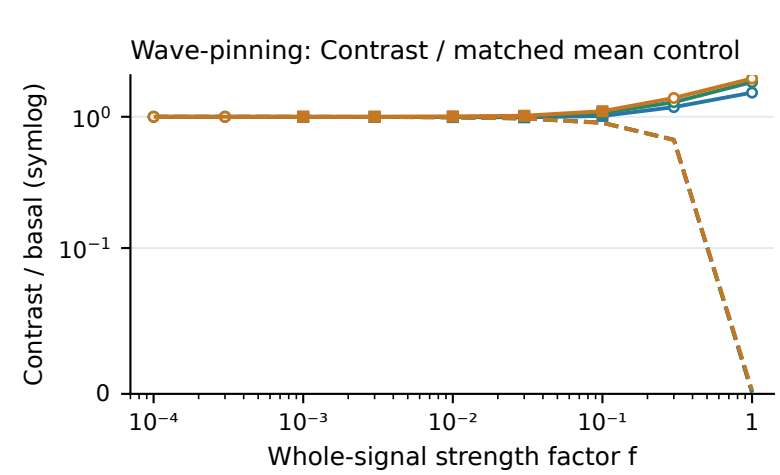
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

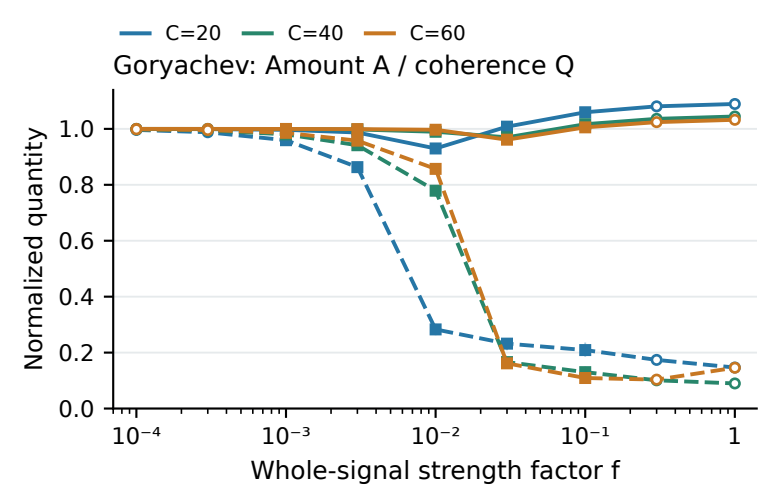
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

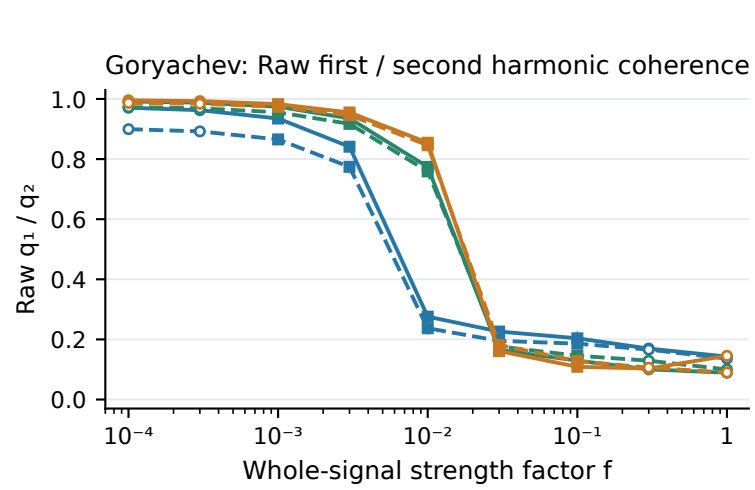
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

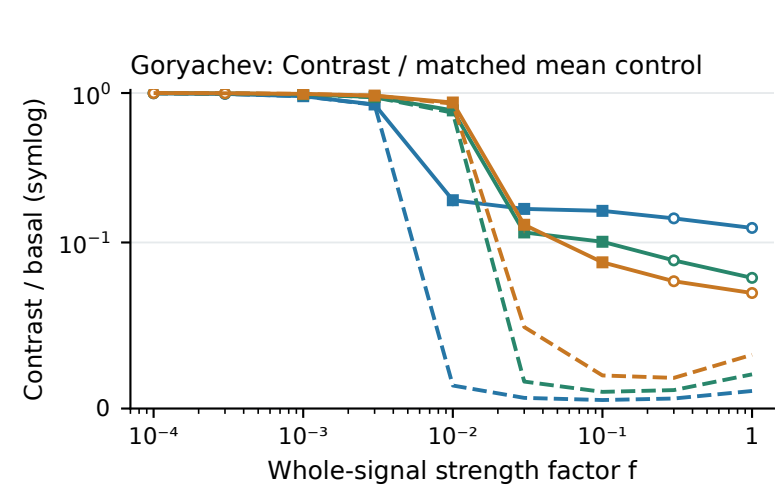


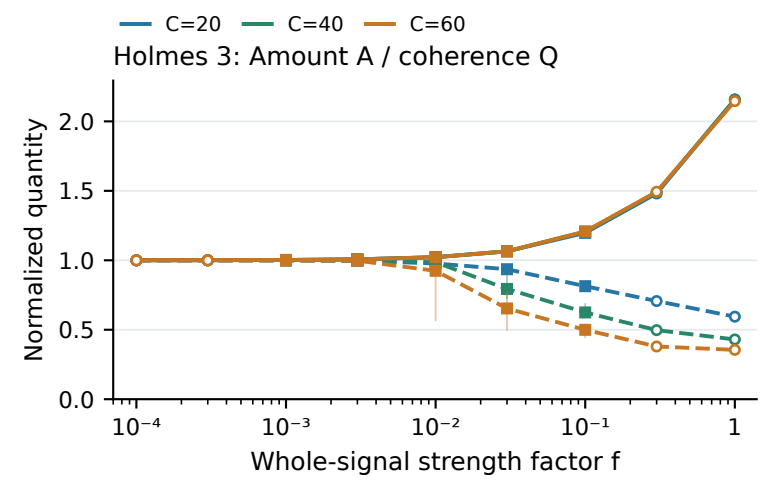


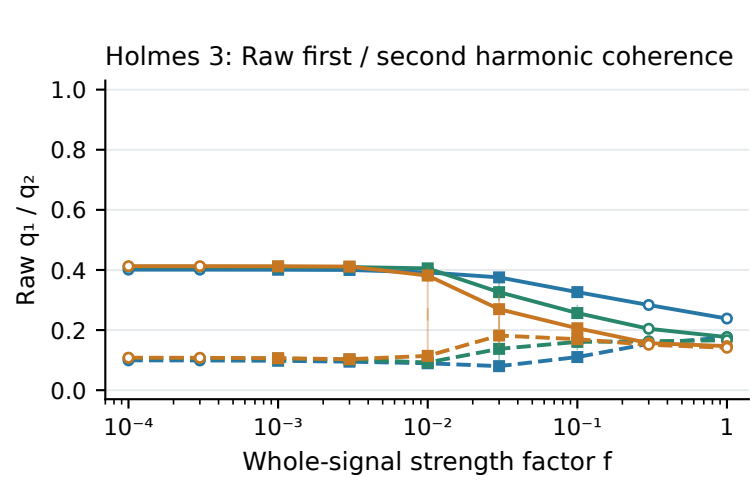


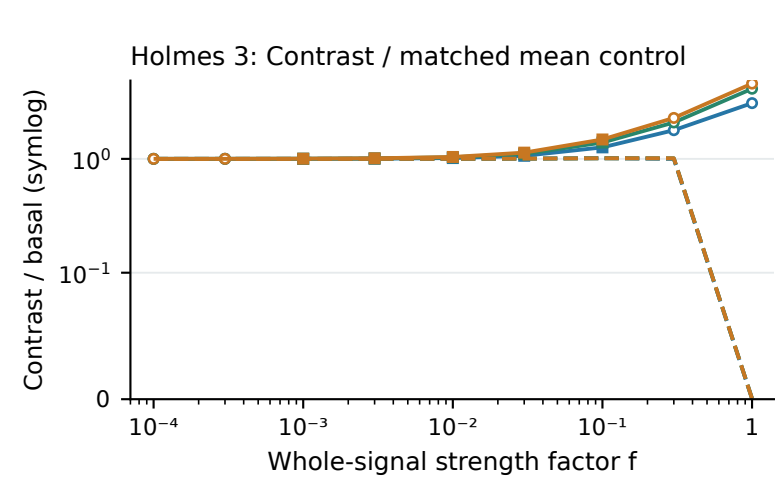


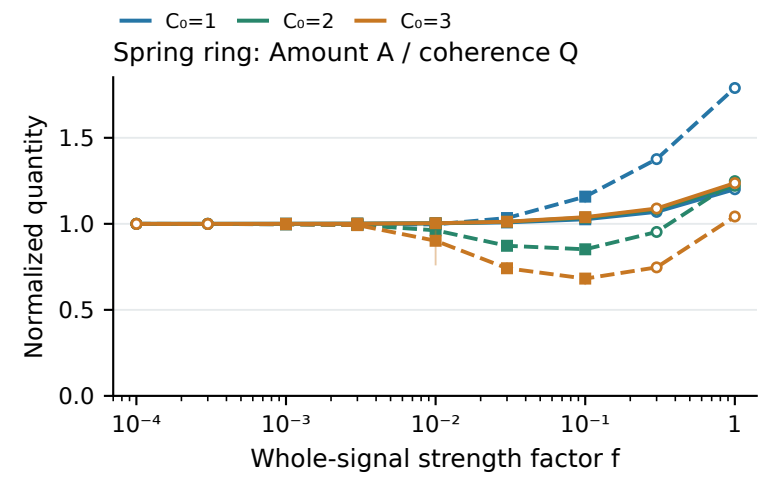


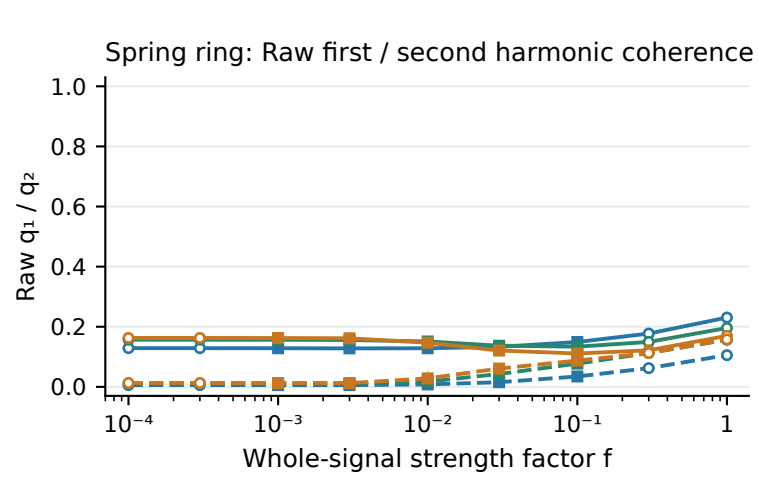


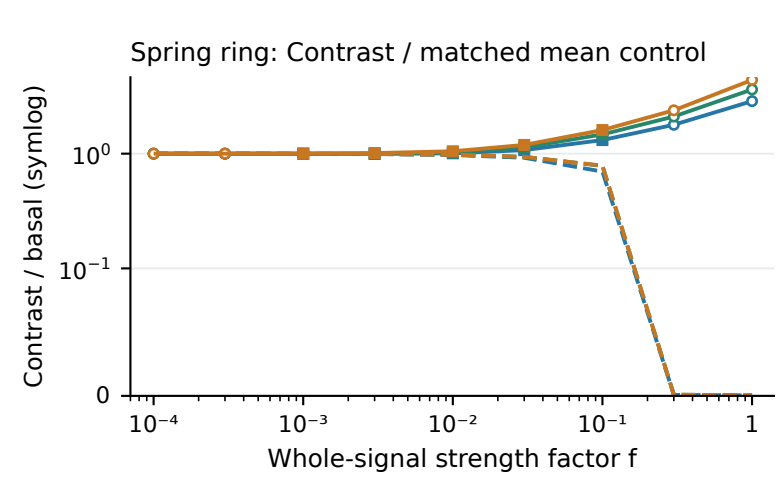












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

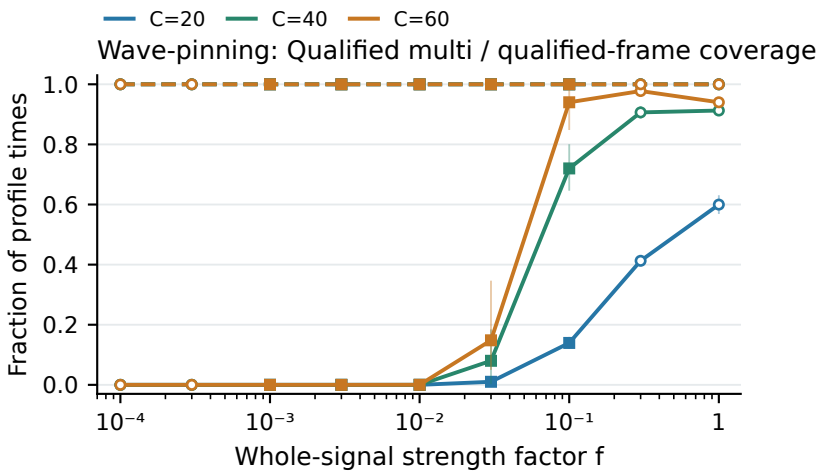
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

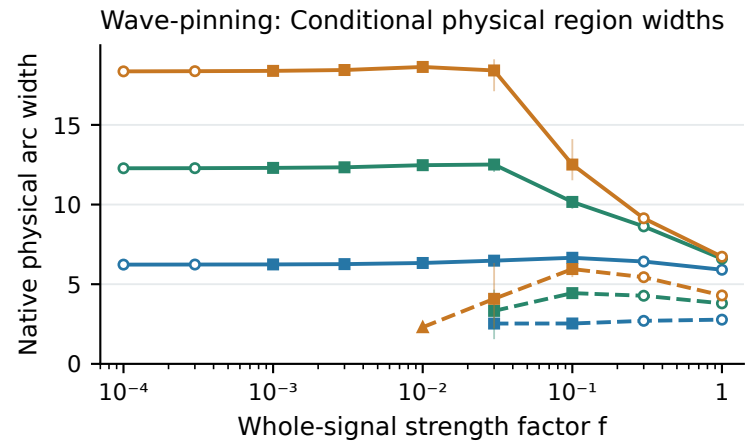
Instantaneous region widths with qualification coverage: all sizes and strengths

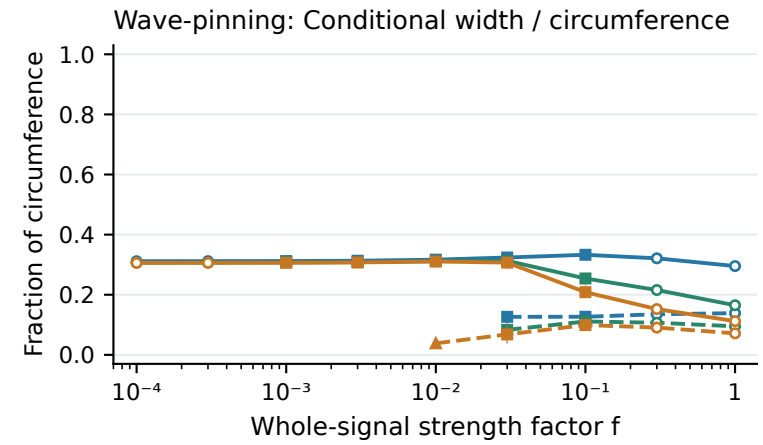
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

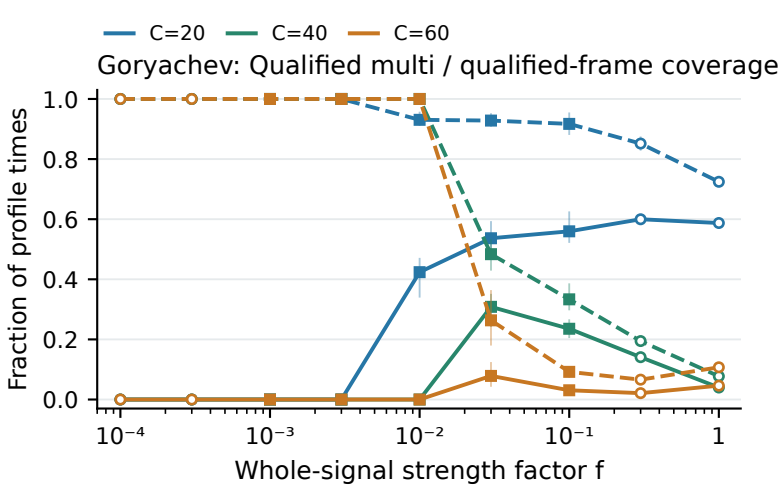
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

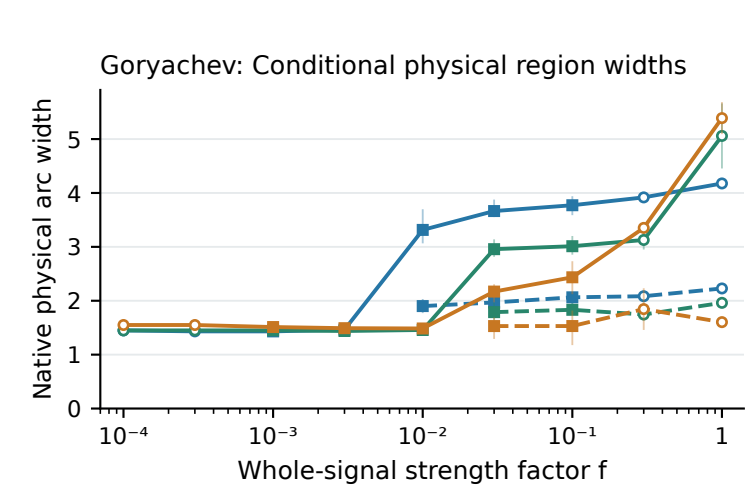
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

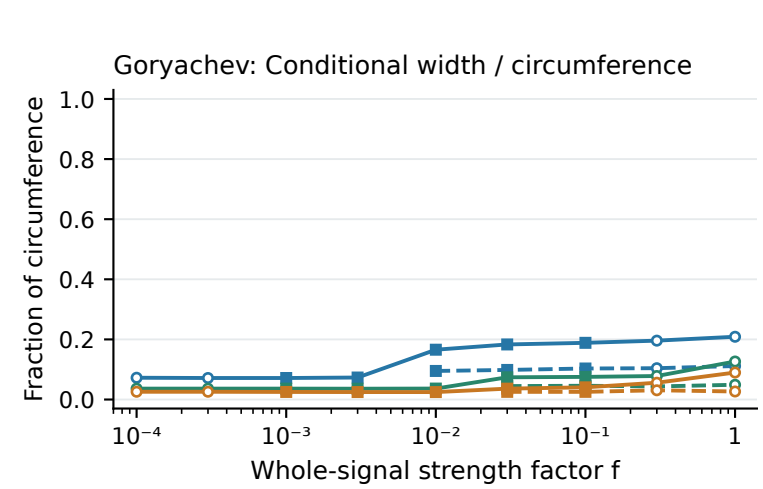


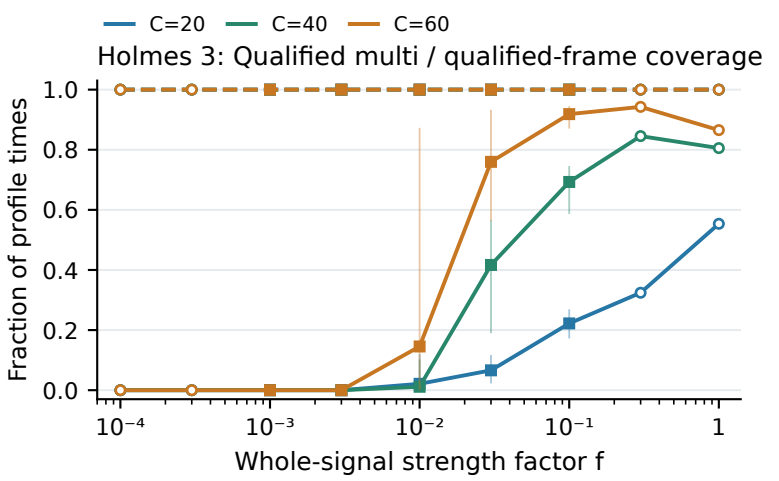


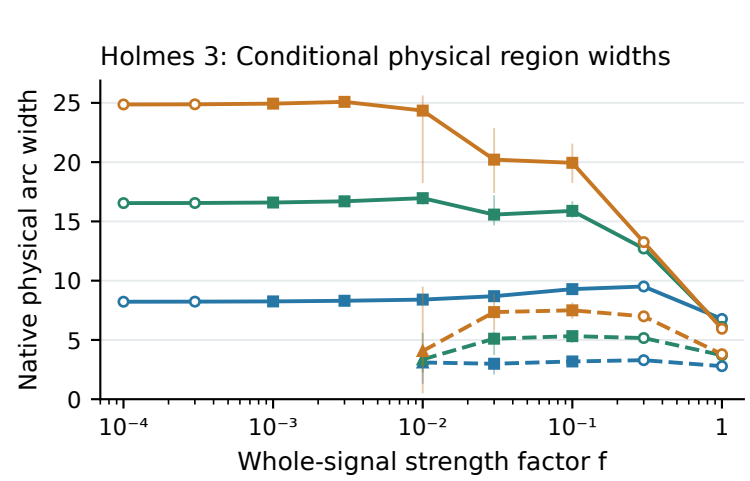


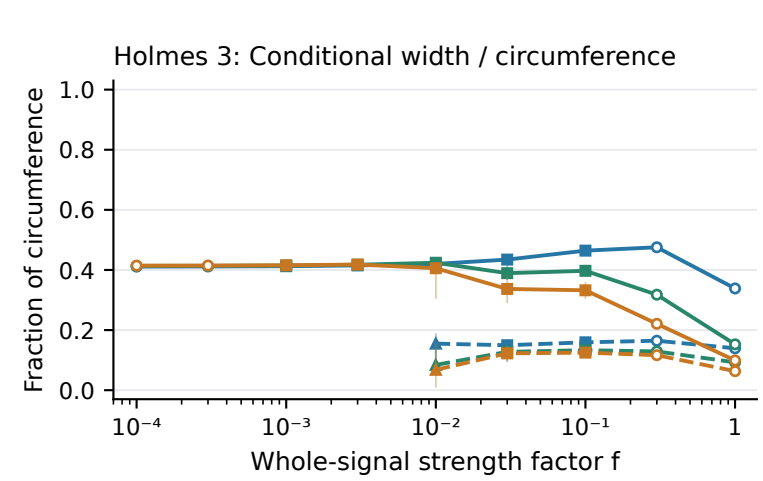


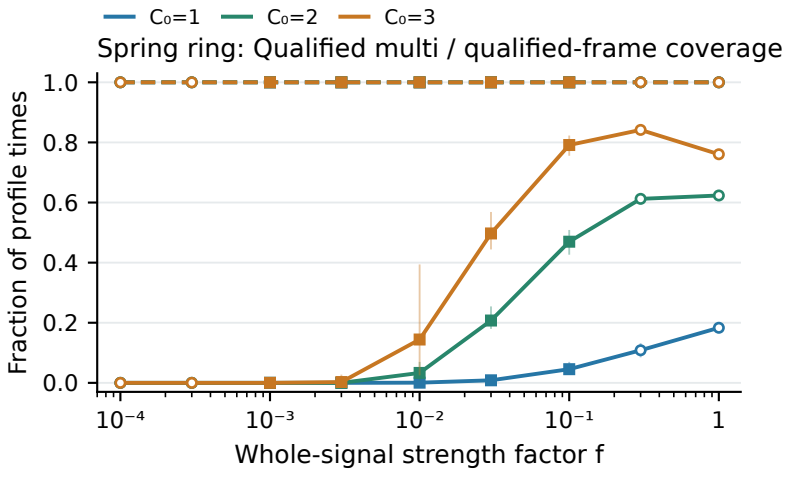


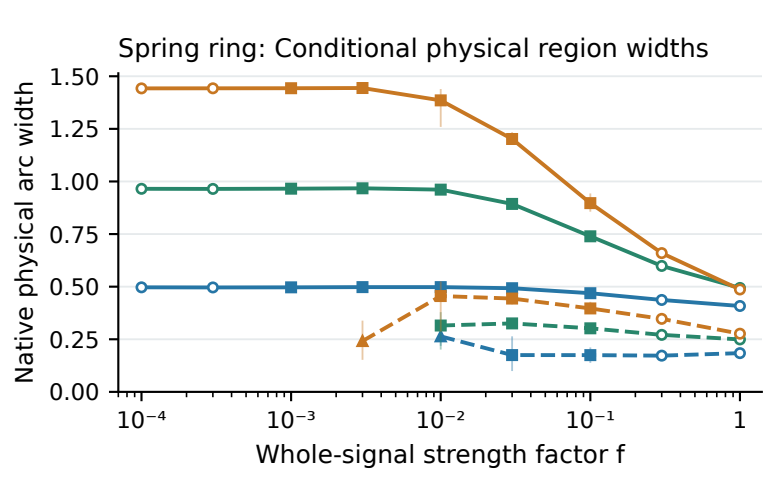


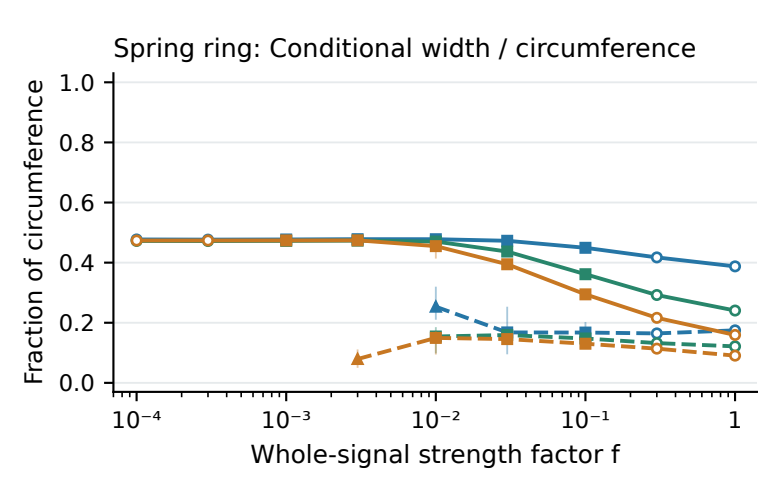












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

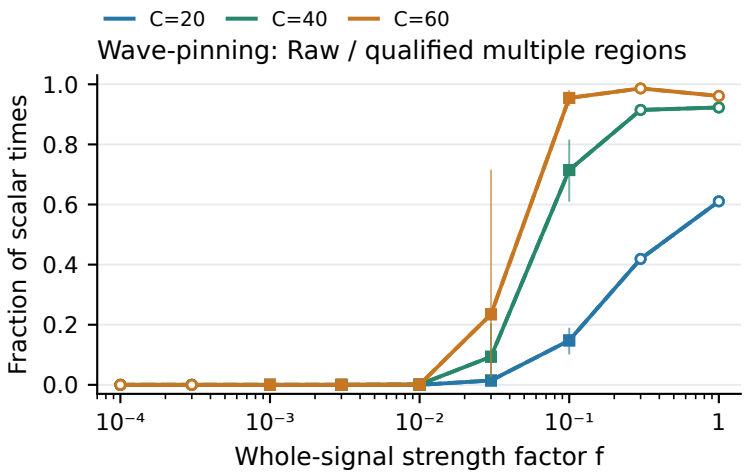
Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

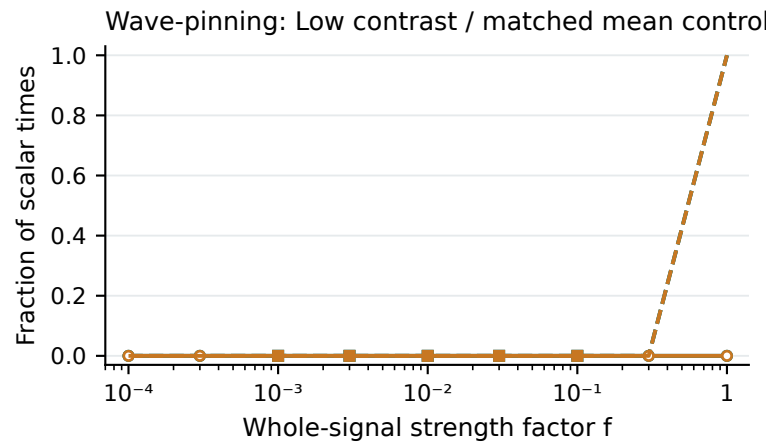
Organization, contrast and global magnitude: all sizes and strengths

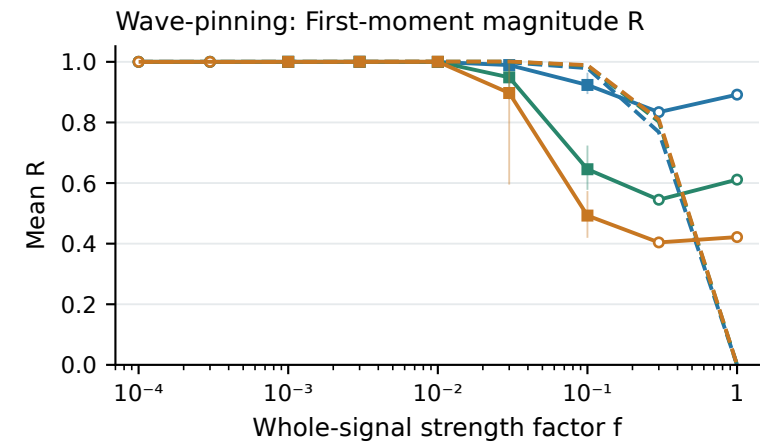
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

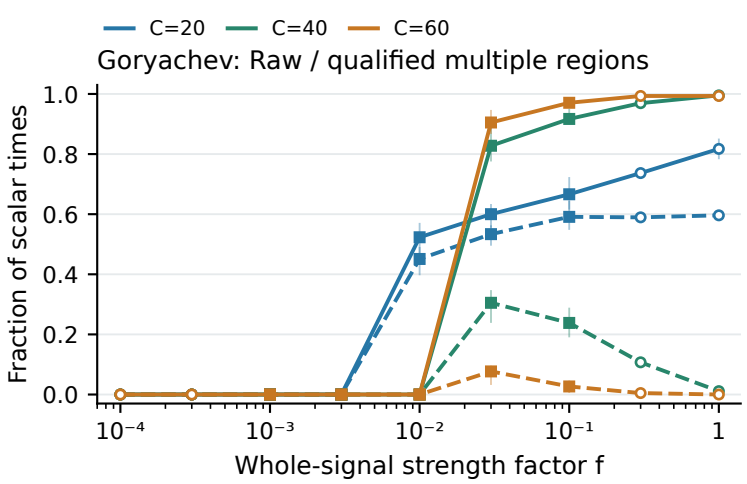
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

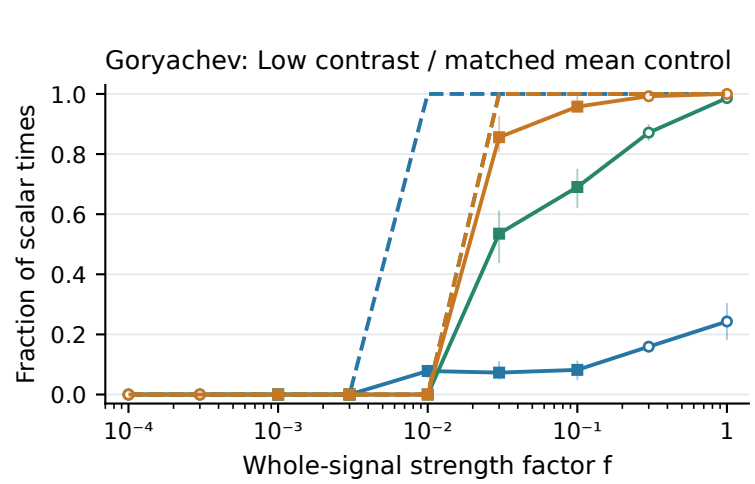
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

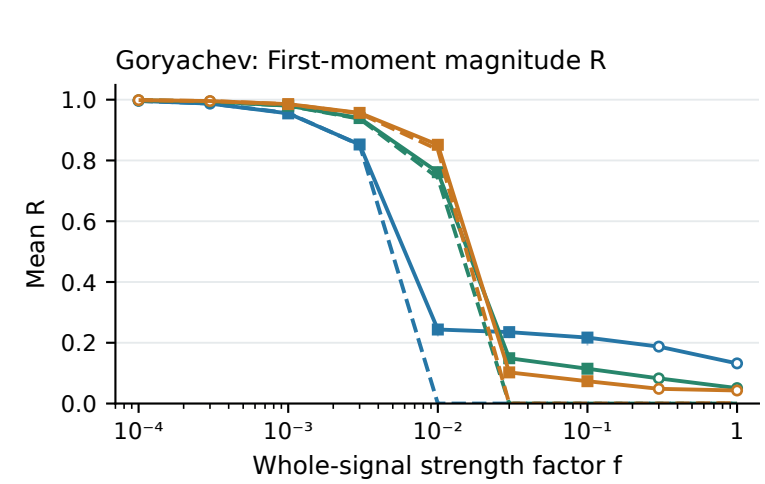


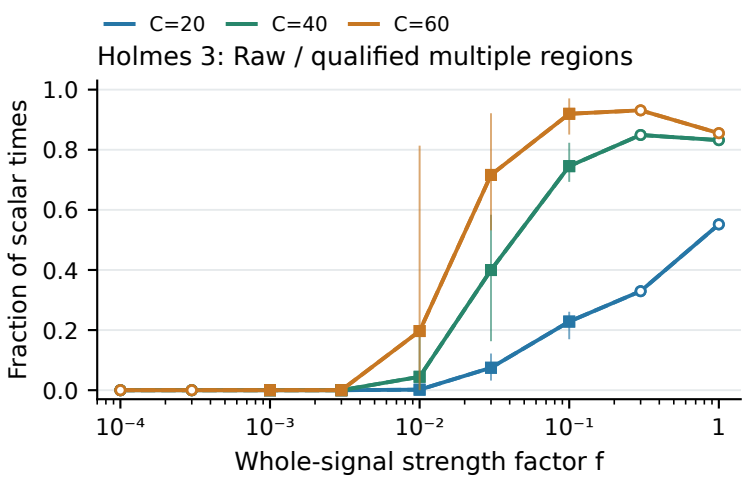


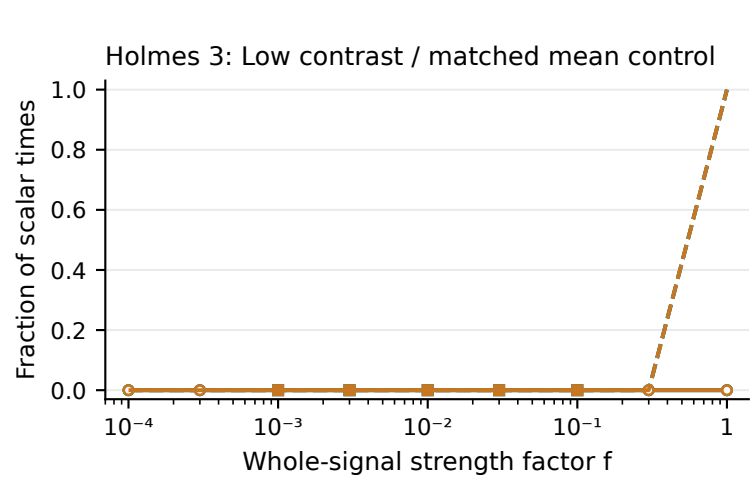


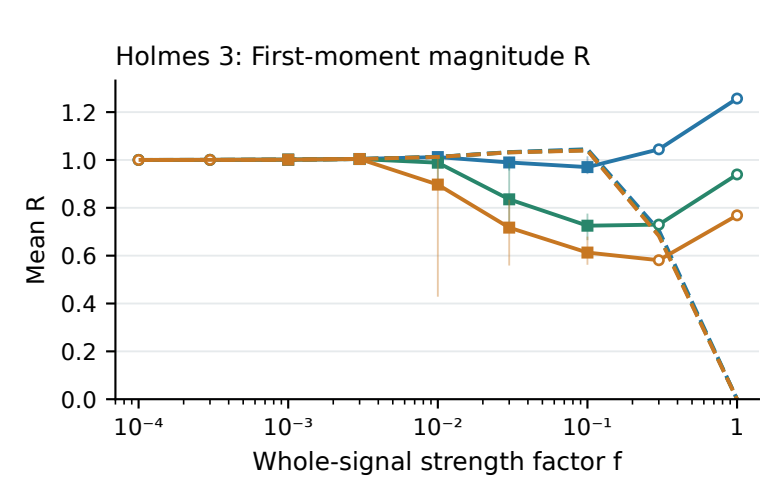


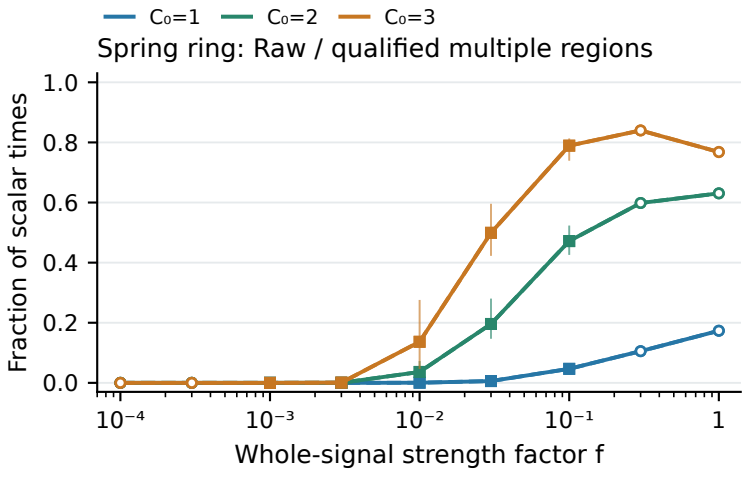


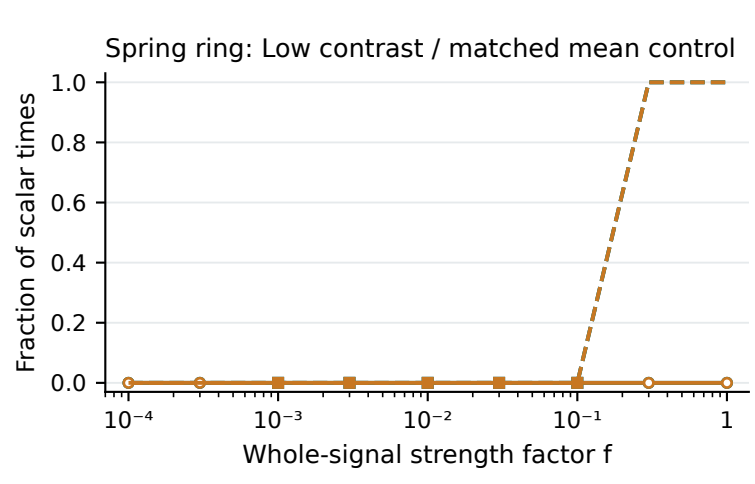


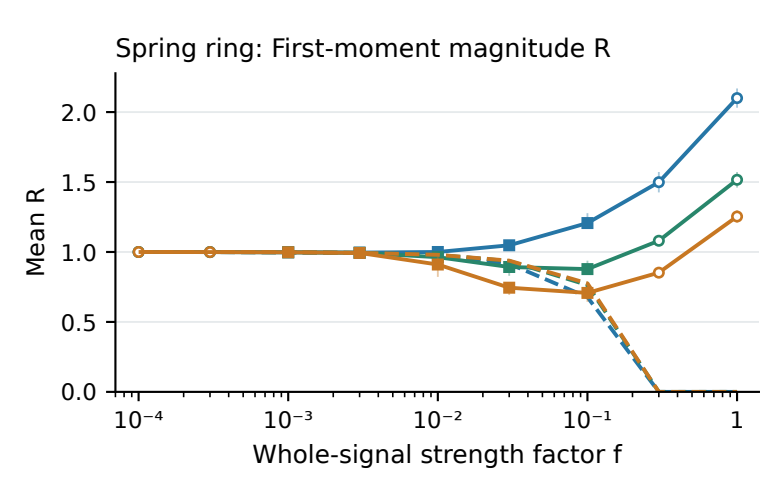












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

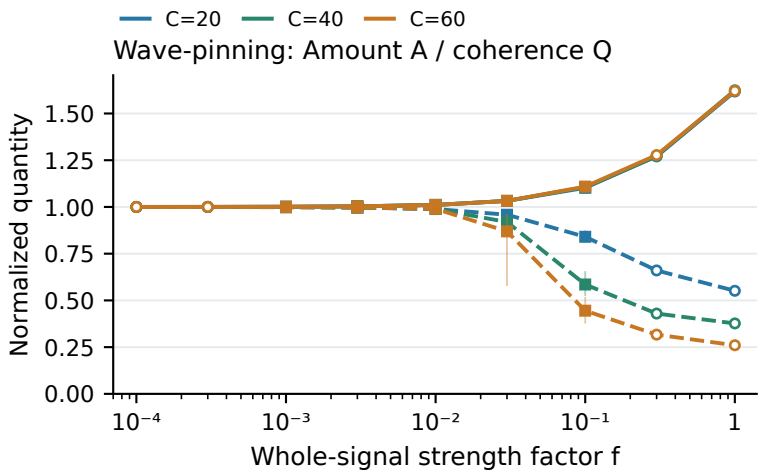
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

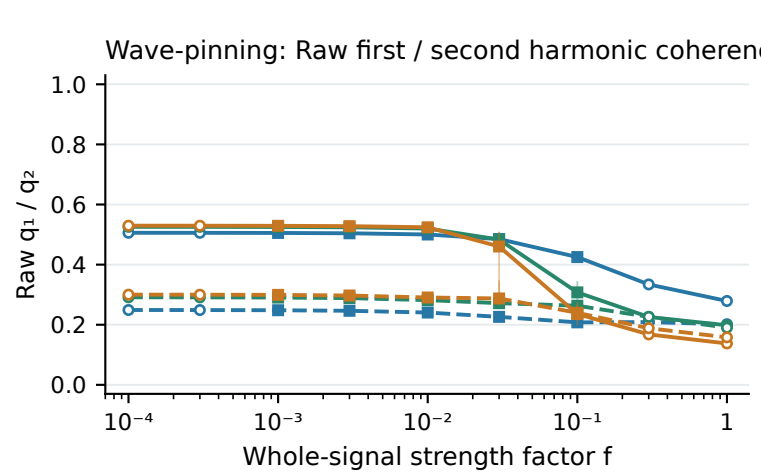
Active amount, coherence and spatial harmonics: all sizes and strengths

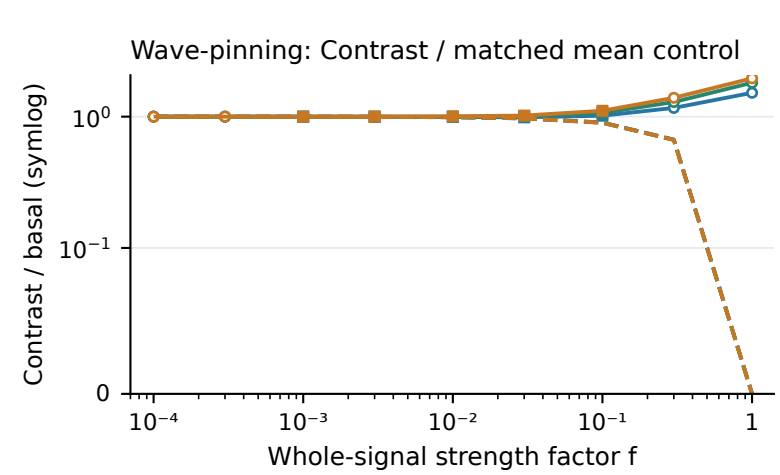
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

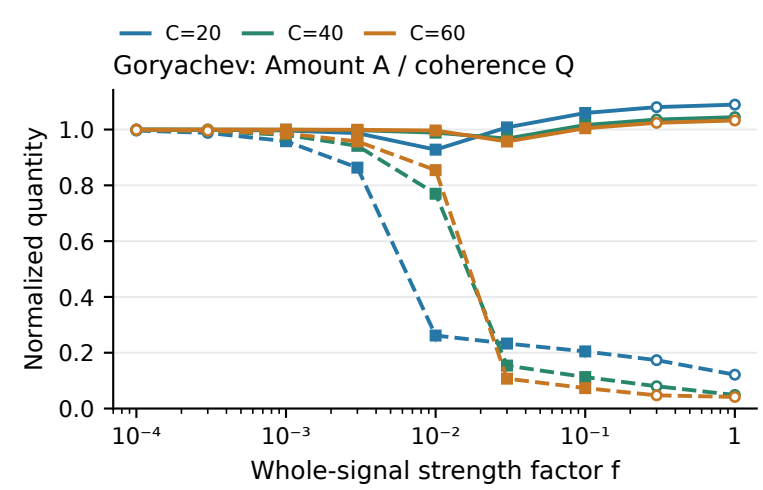
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

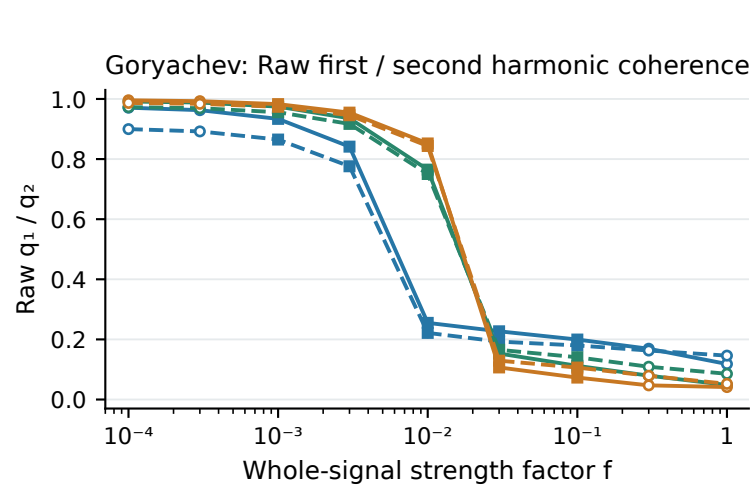
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

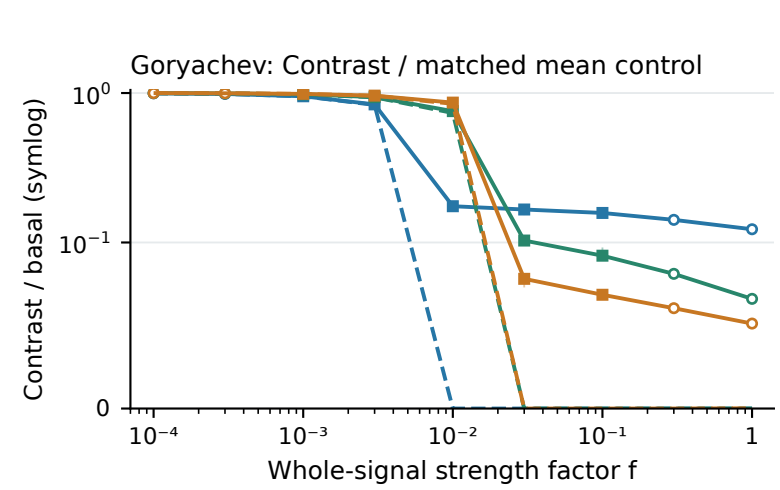


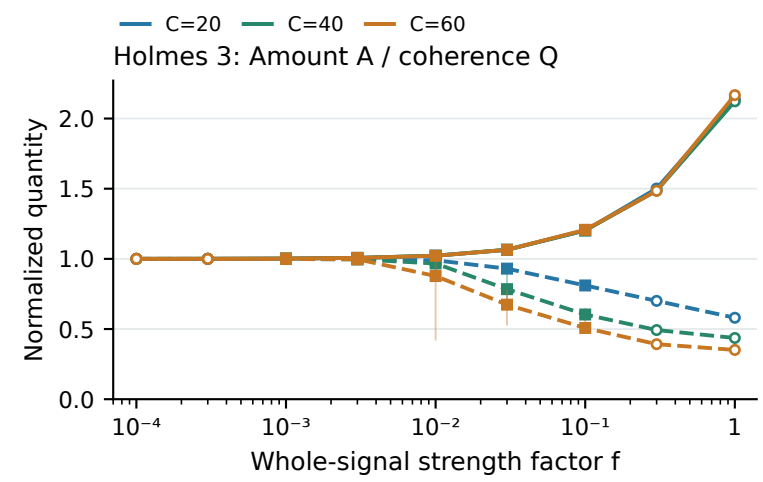


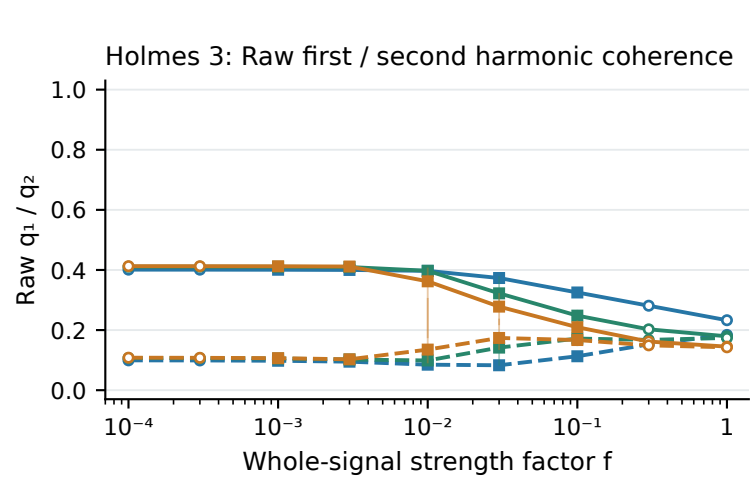


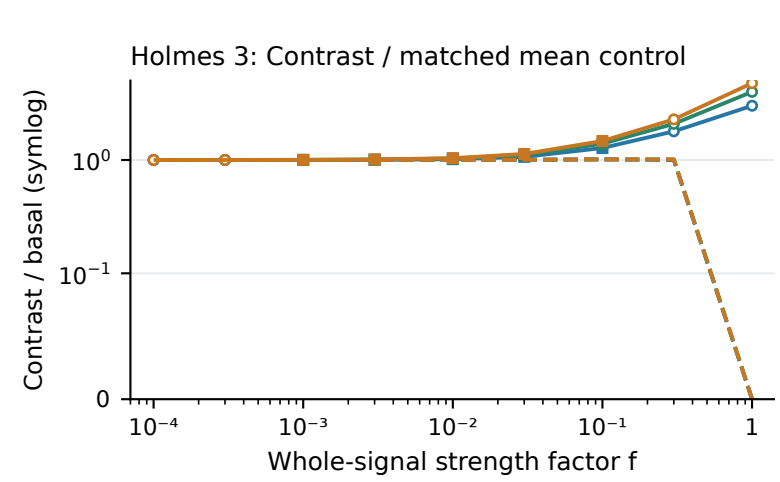


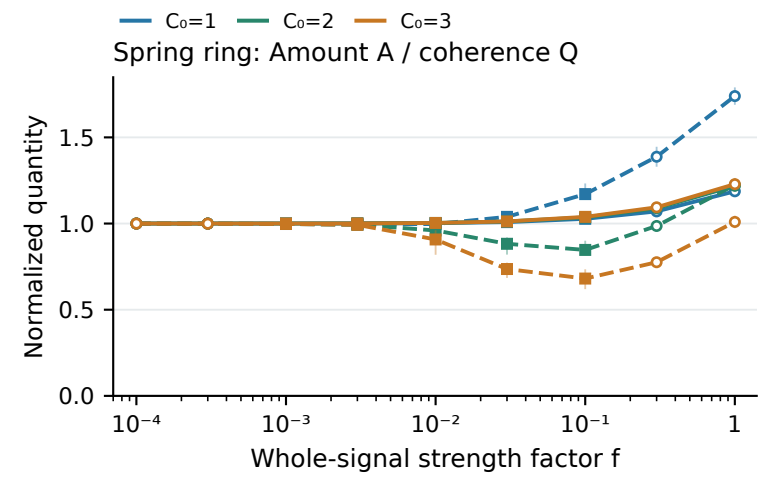


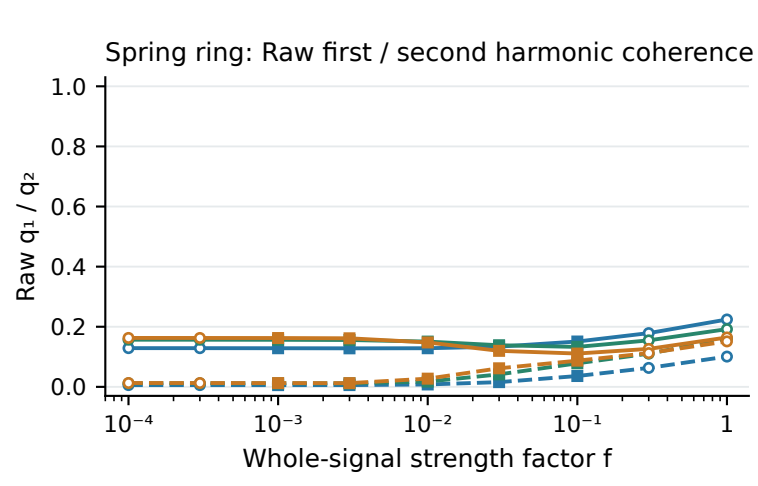


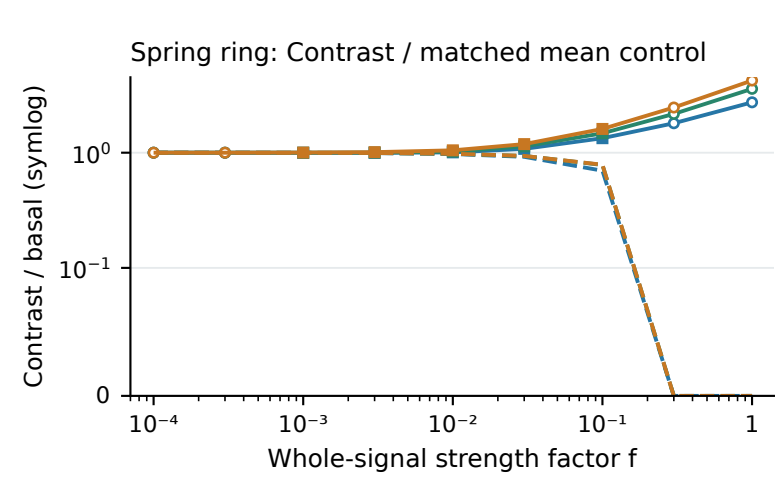












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

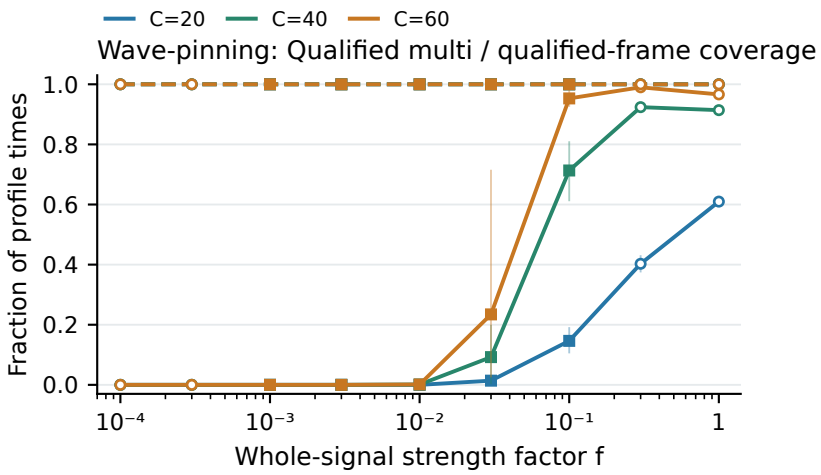
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

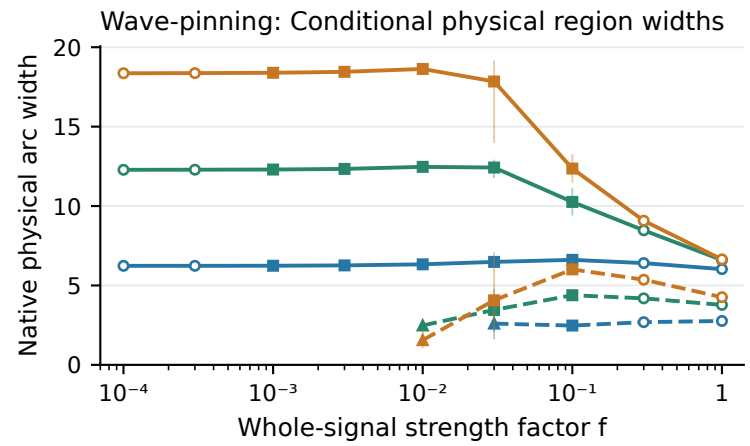
Instantaneous region widths with qualification coverage: all sizes and strengths

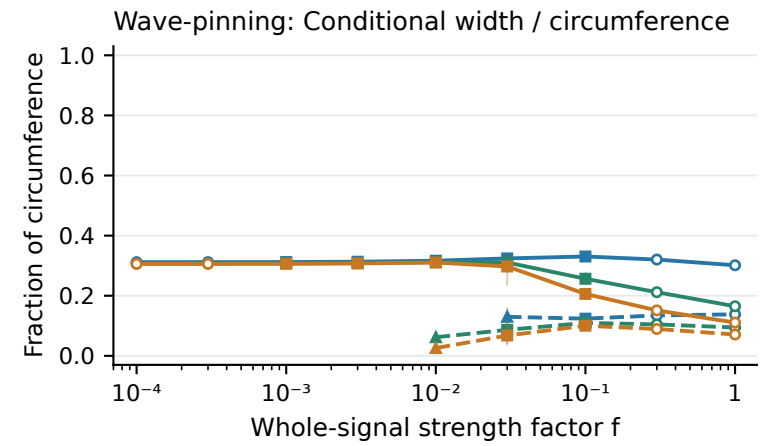
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

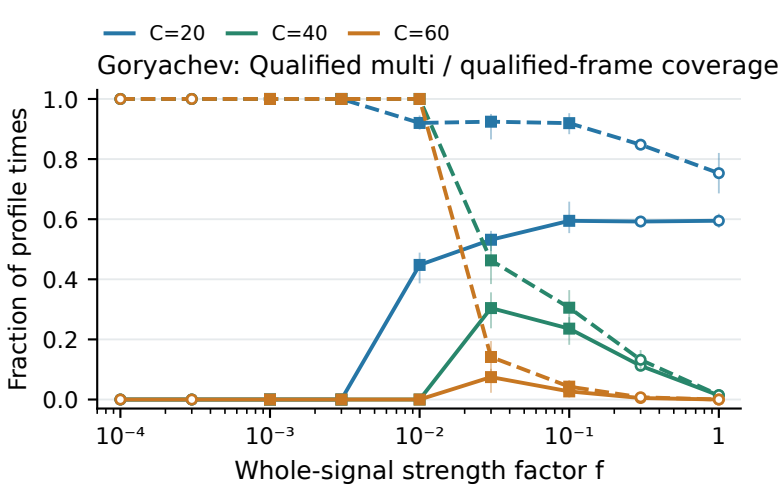
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

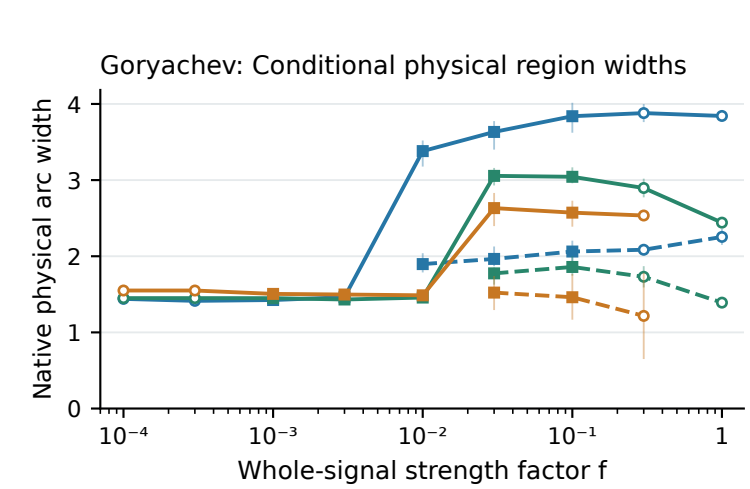
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

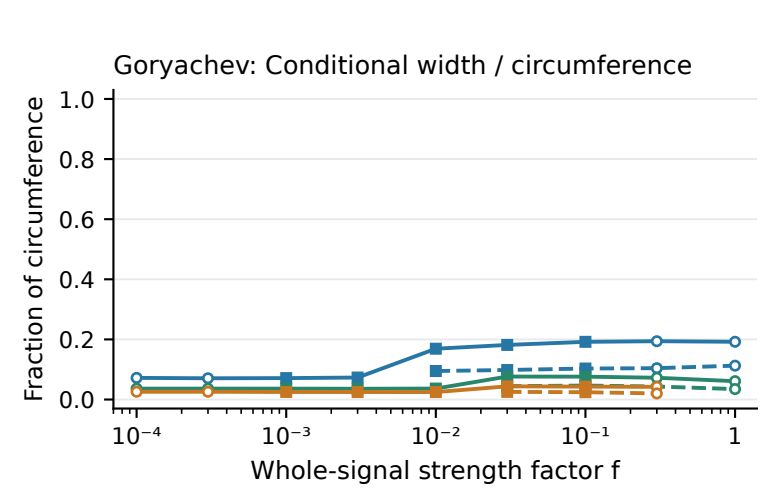


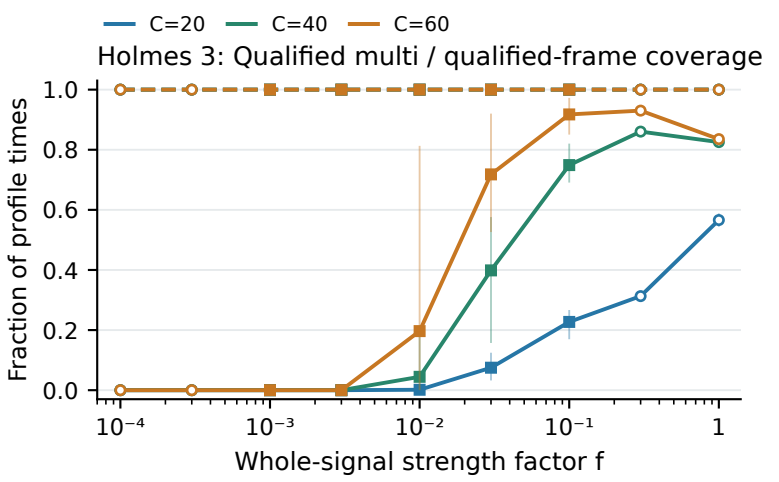


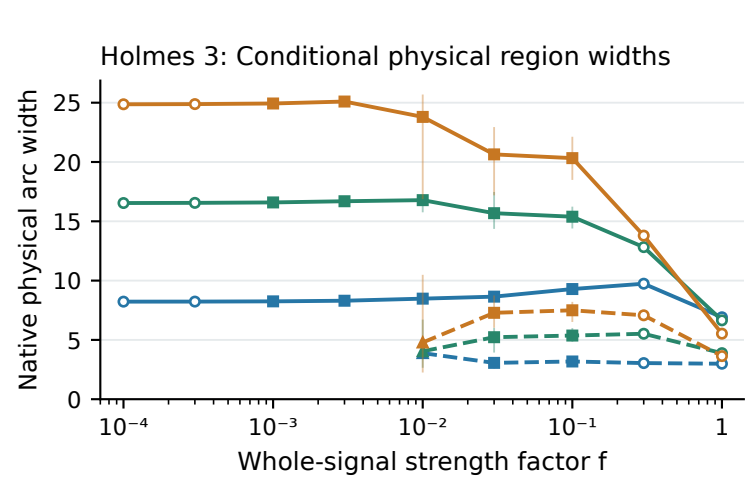


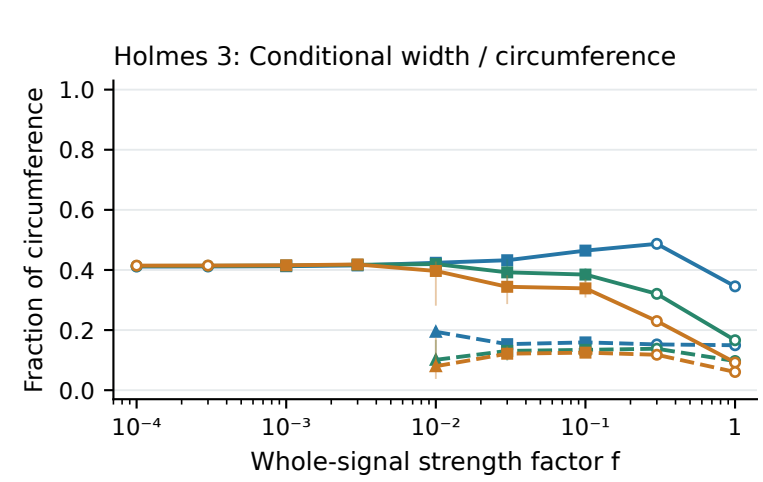


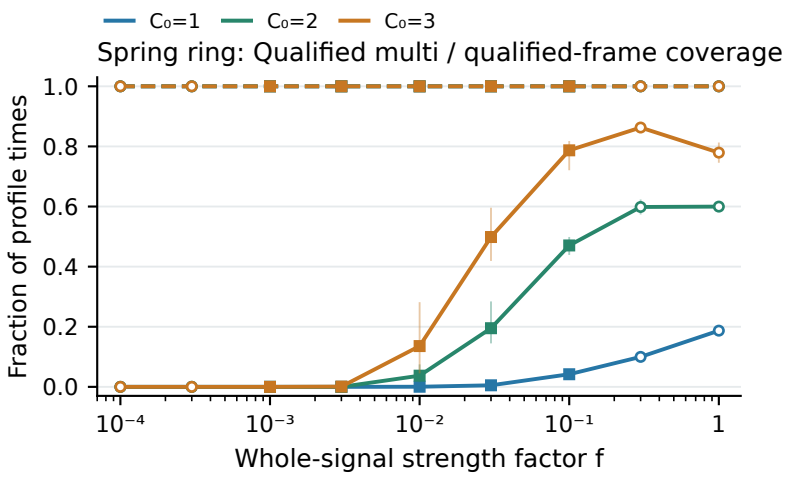


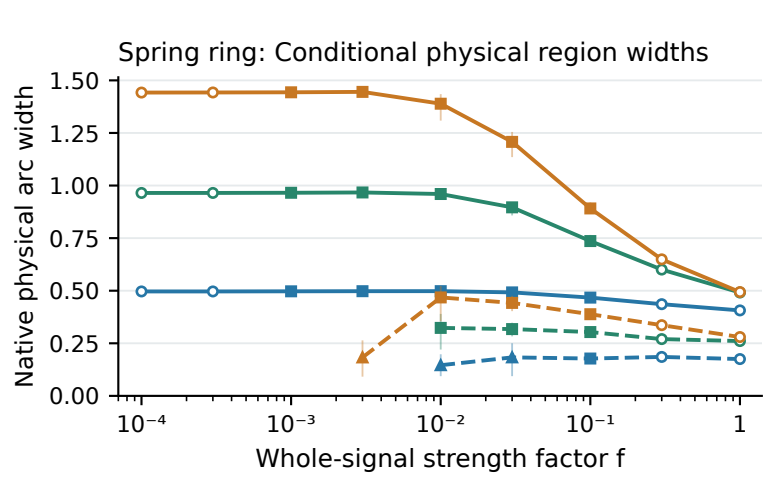


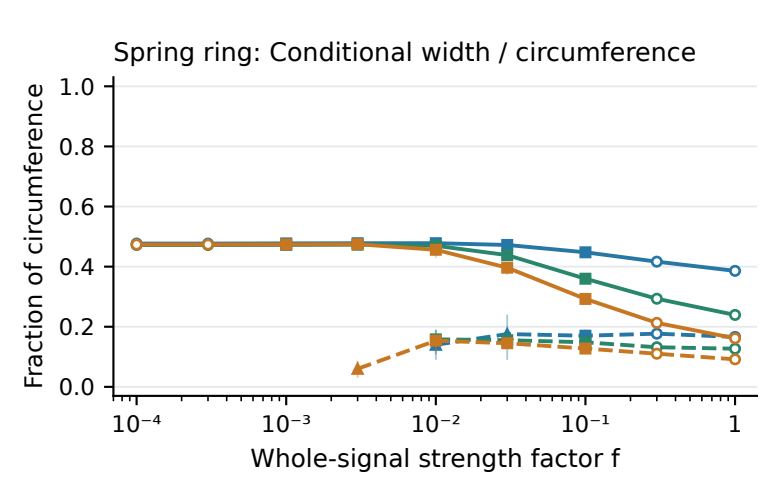












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

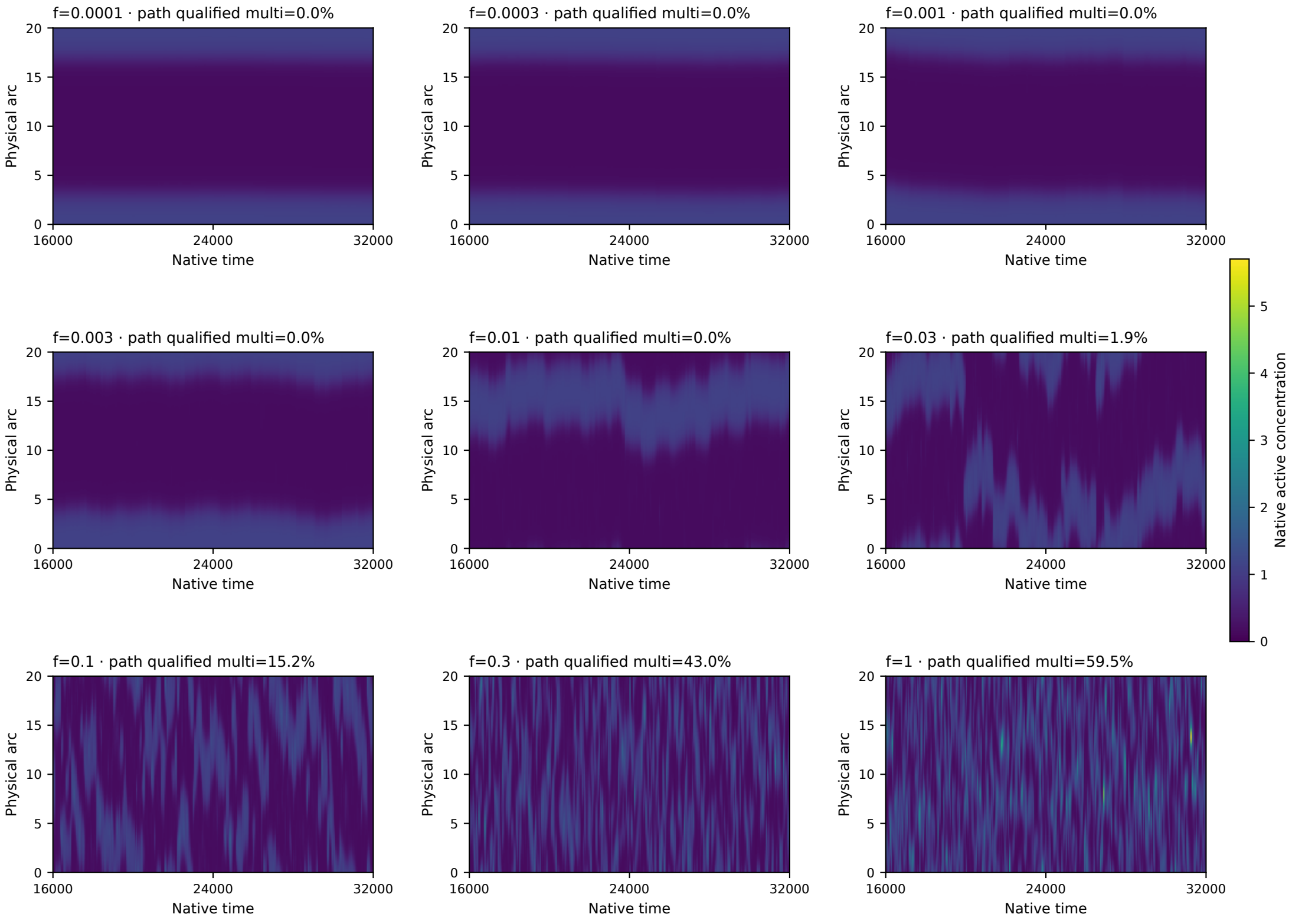
Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

Wave-pinning, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



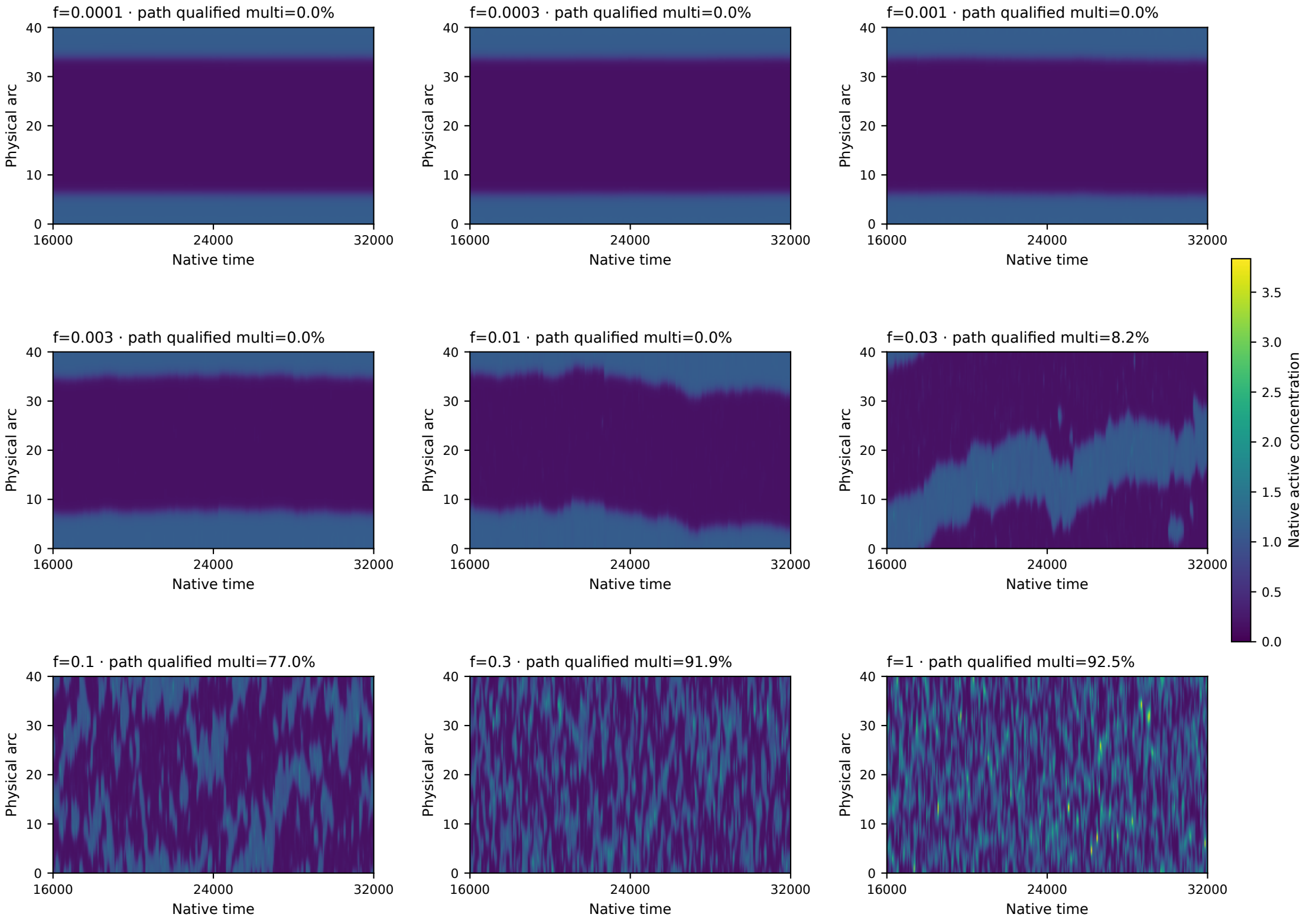
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Wave-pinning, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



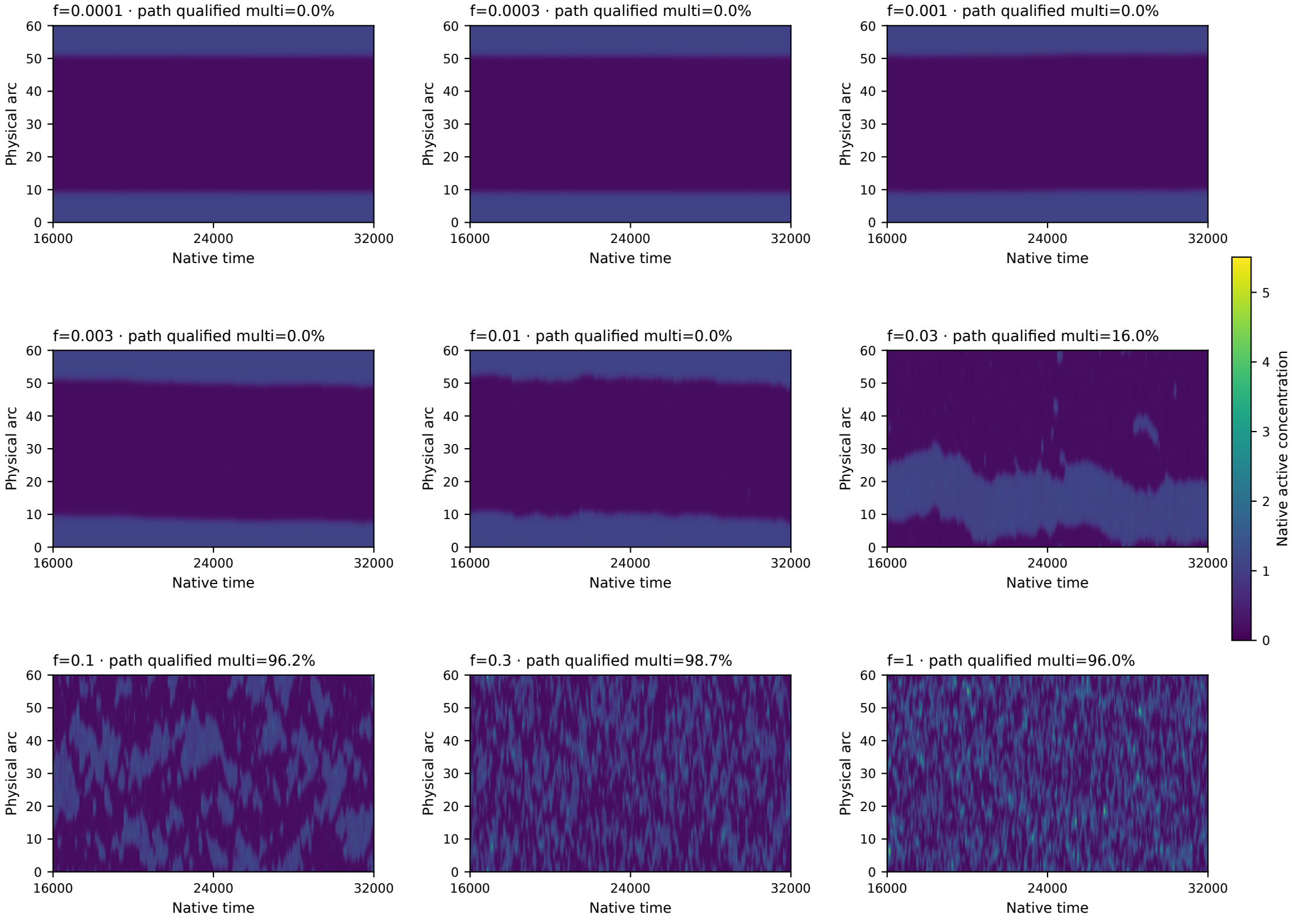
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Wave-pinning, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



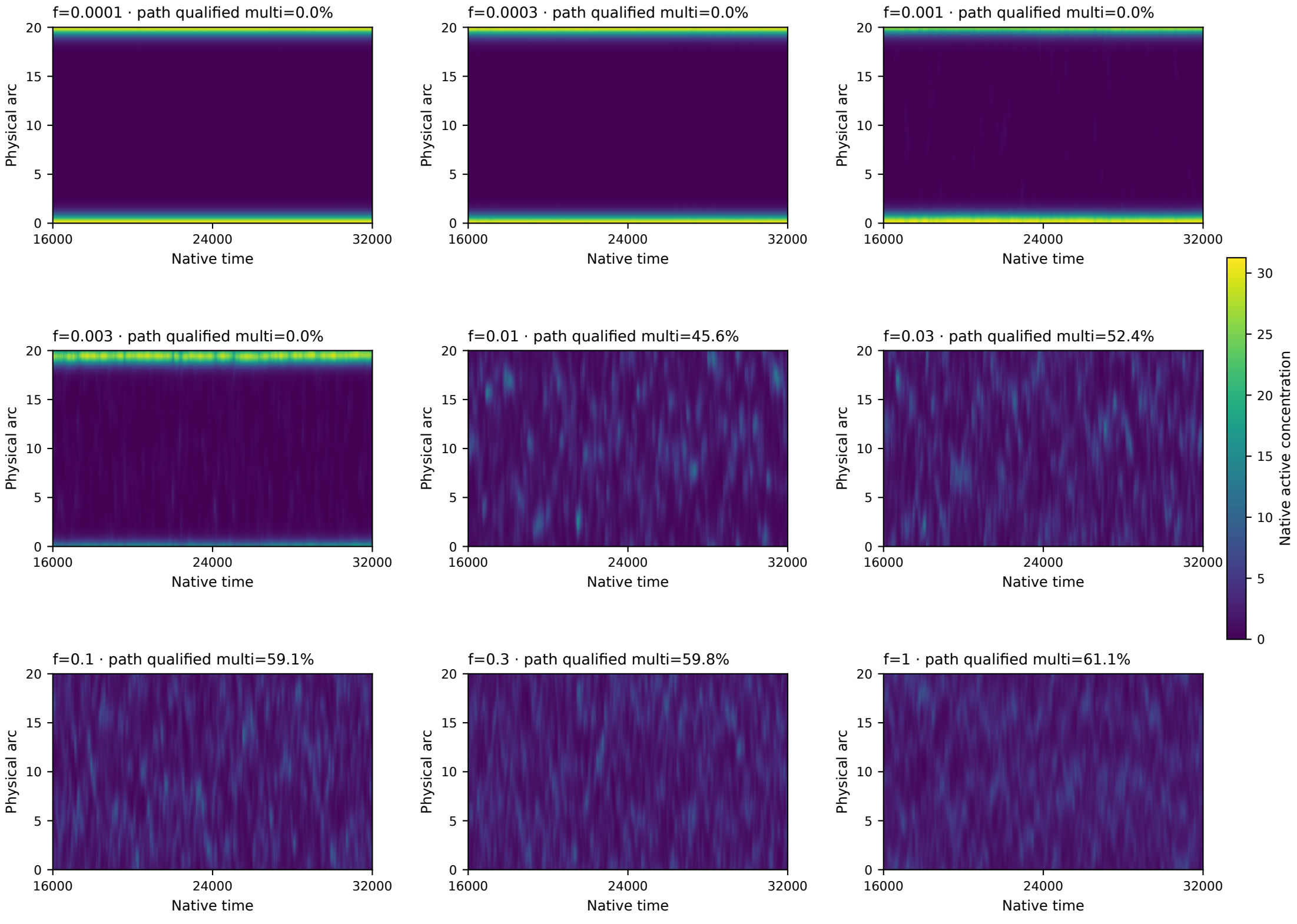
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



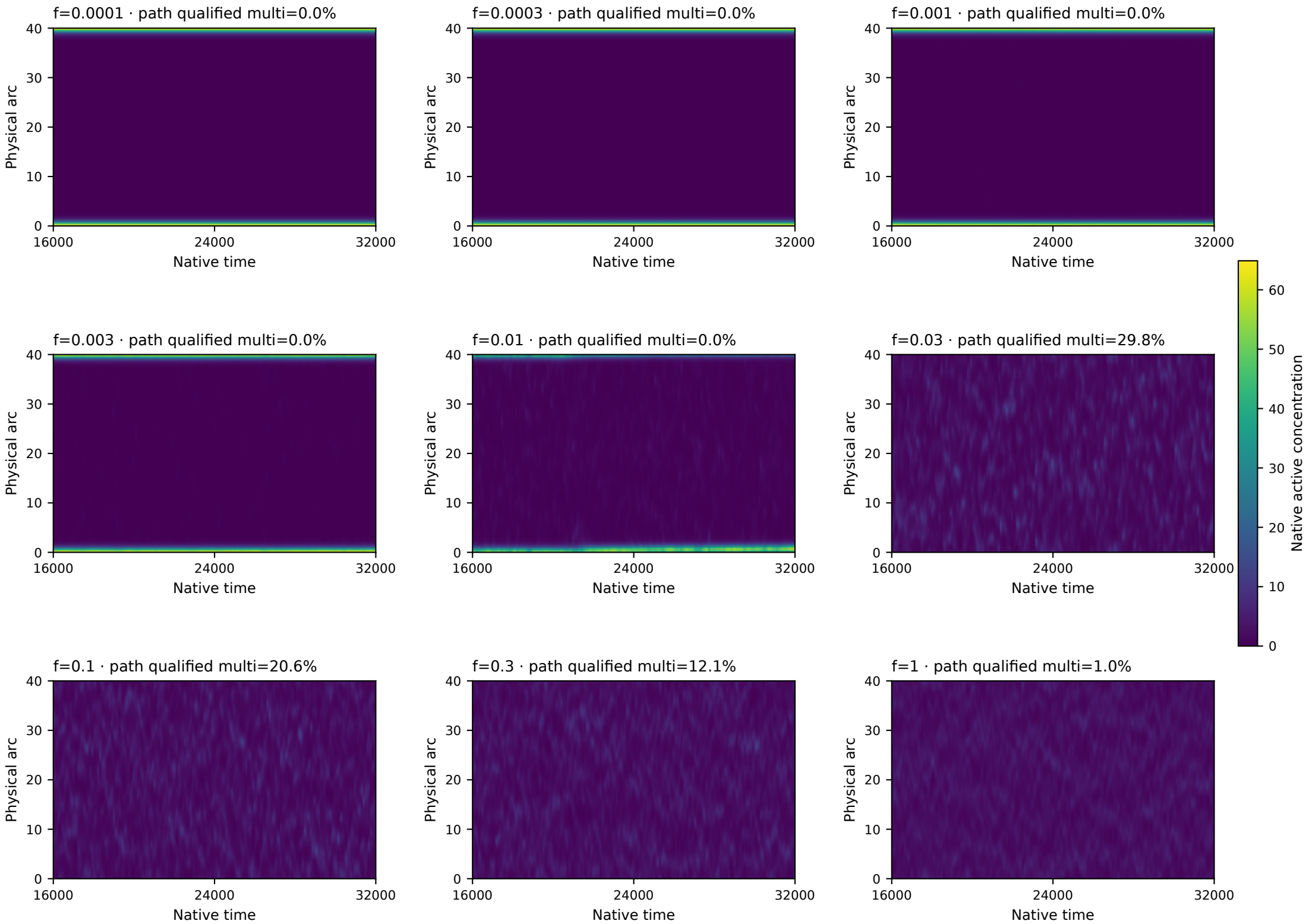
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



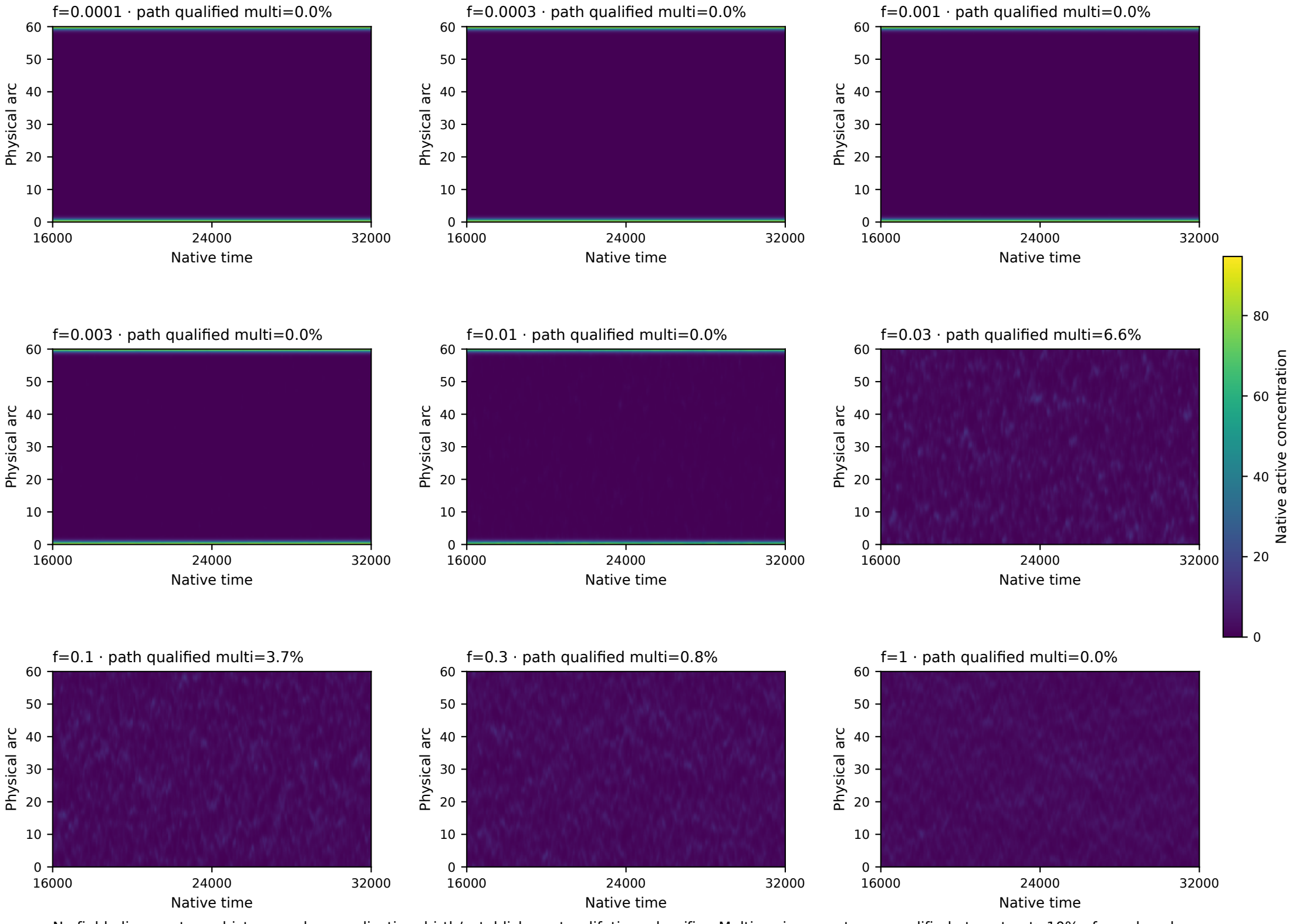
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



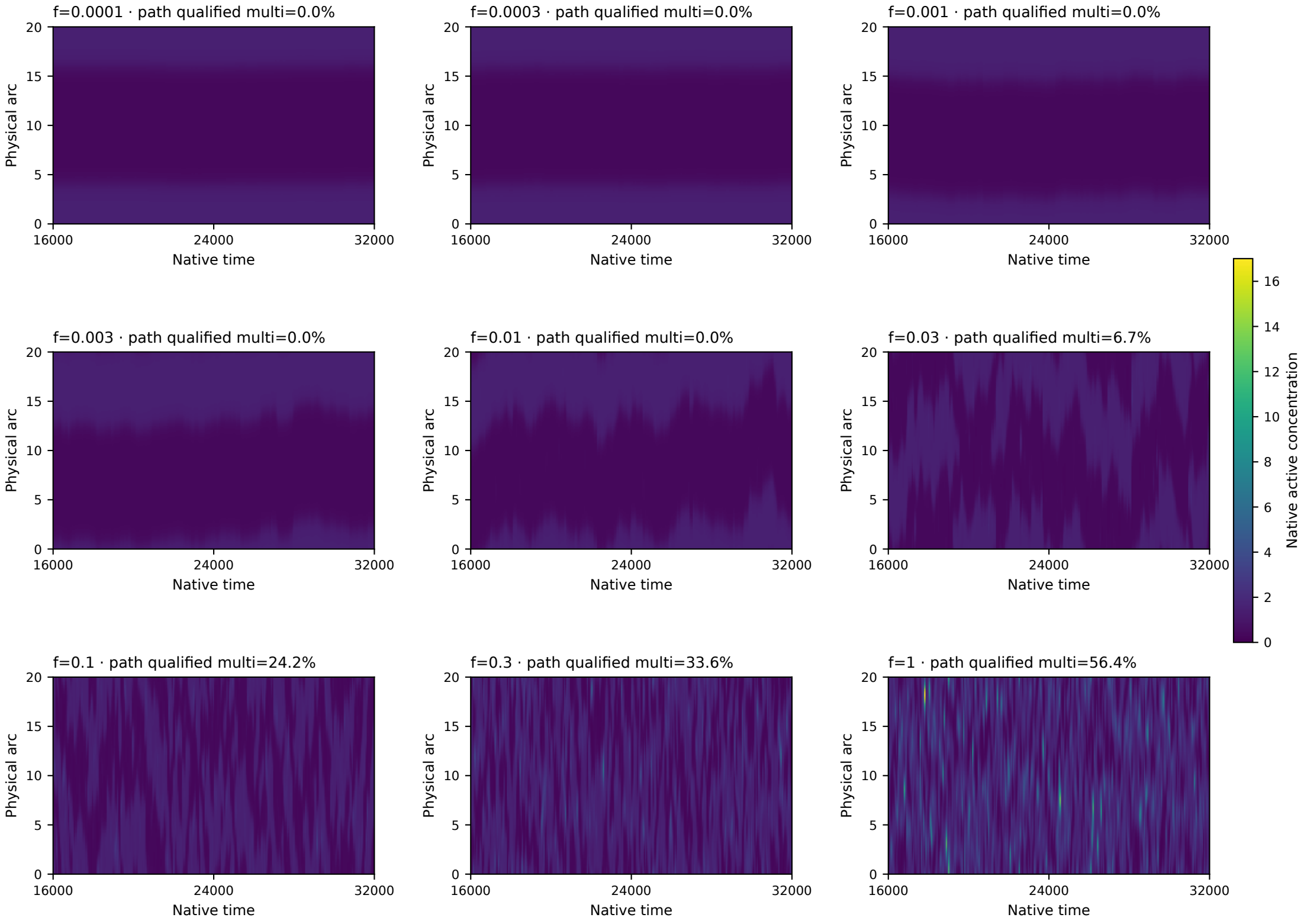
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



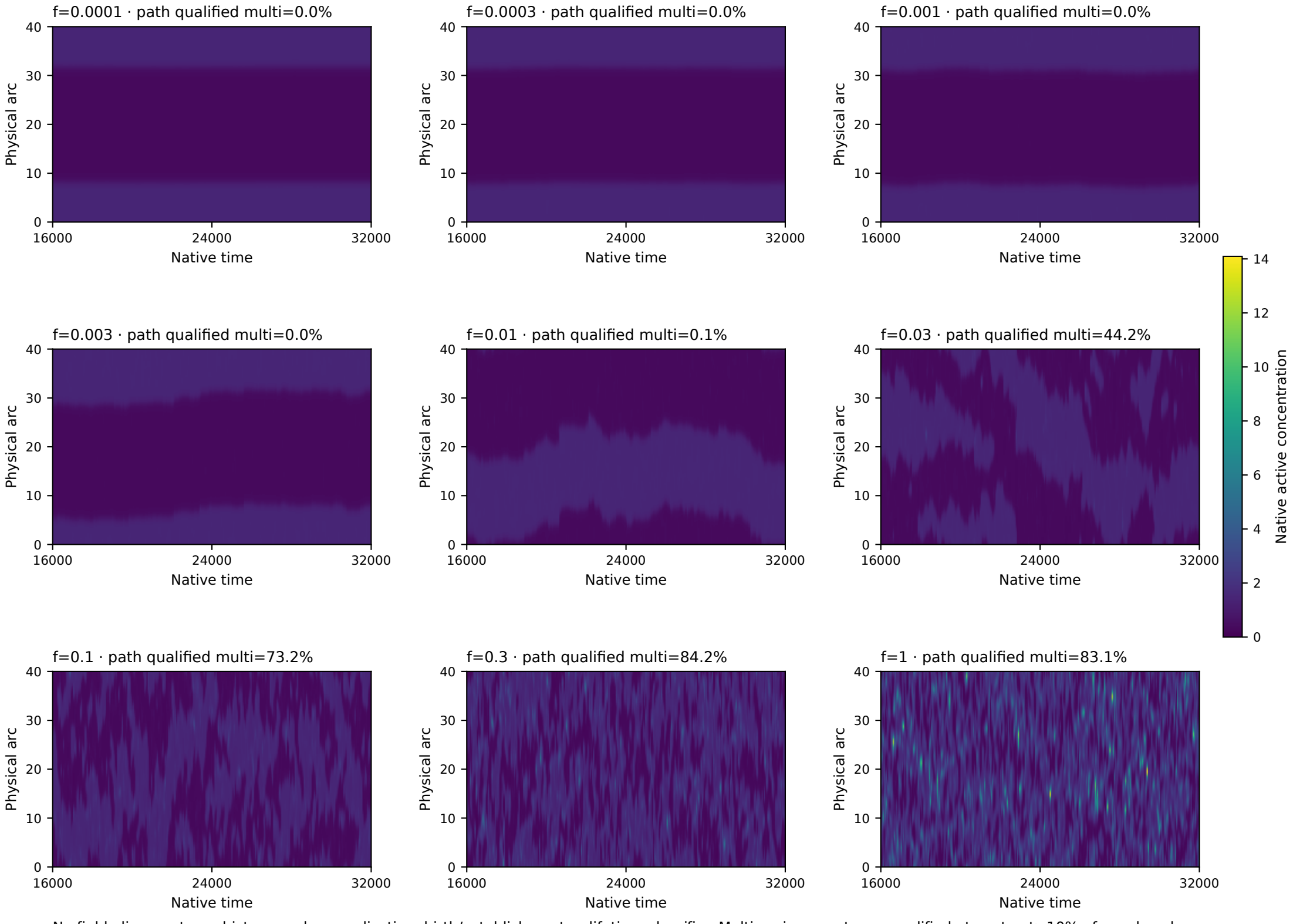
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



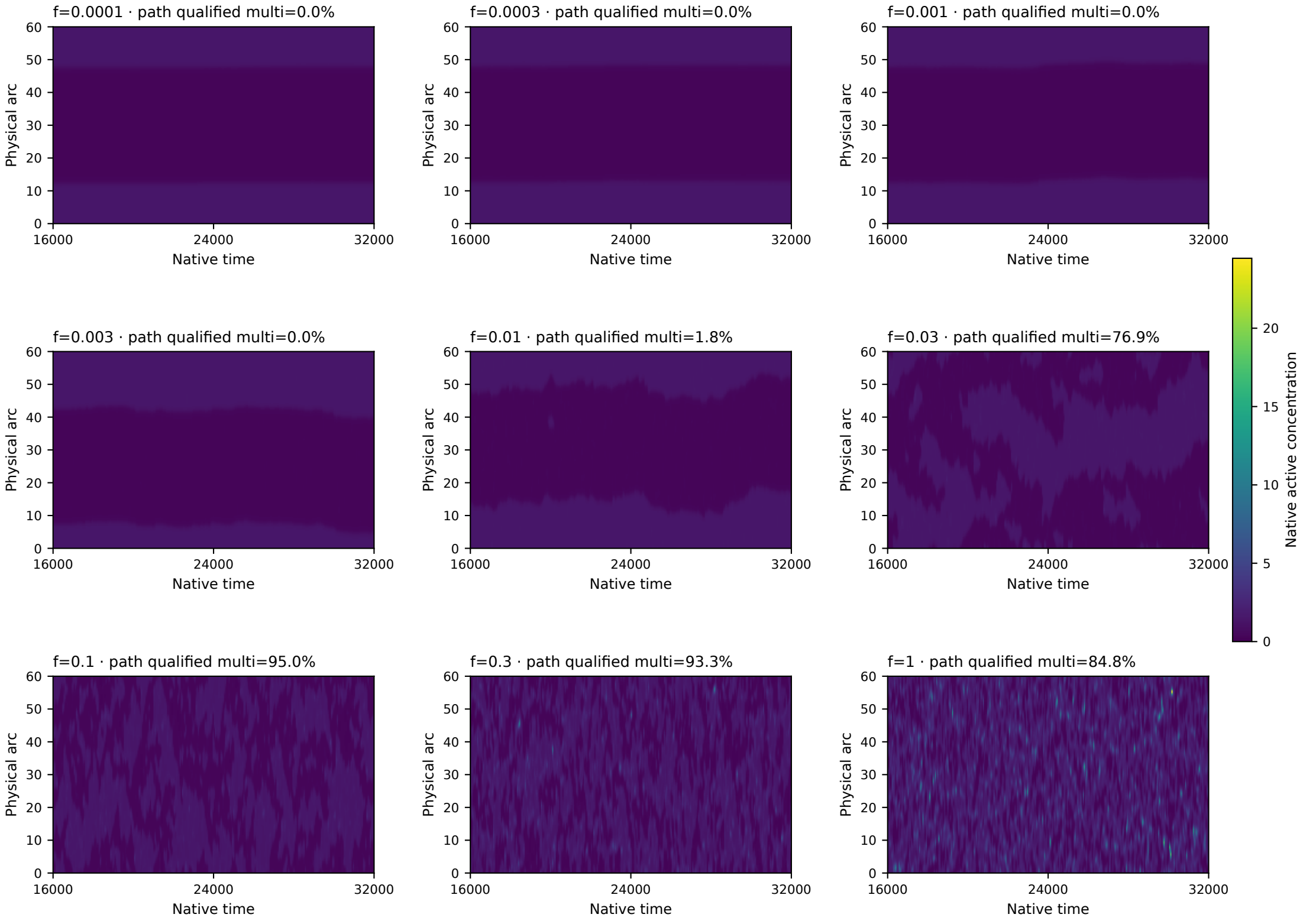
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



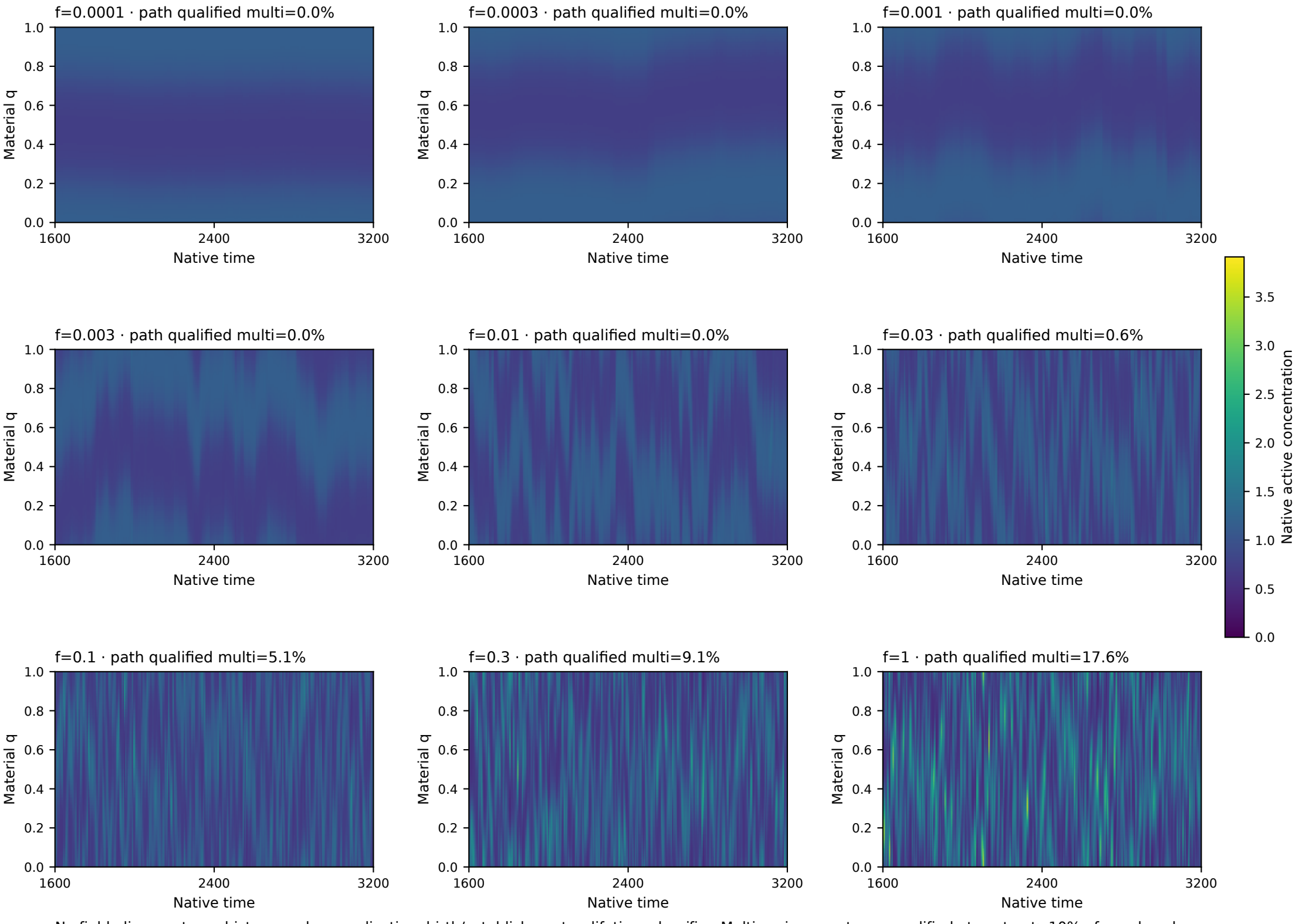
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C_0=1: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



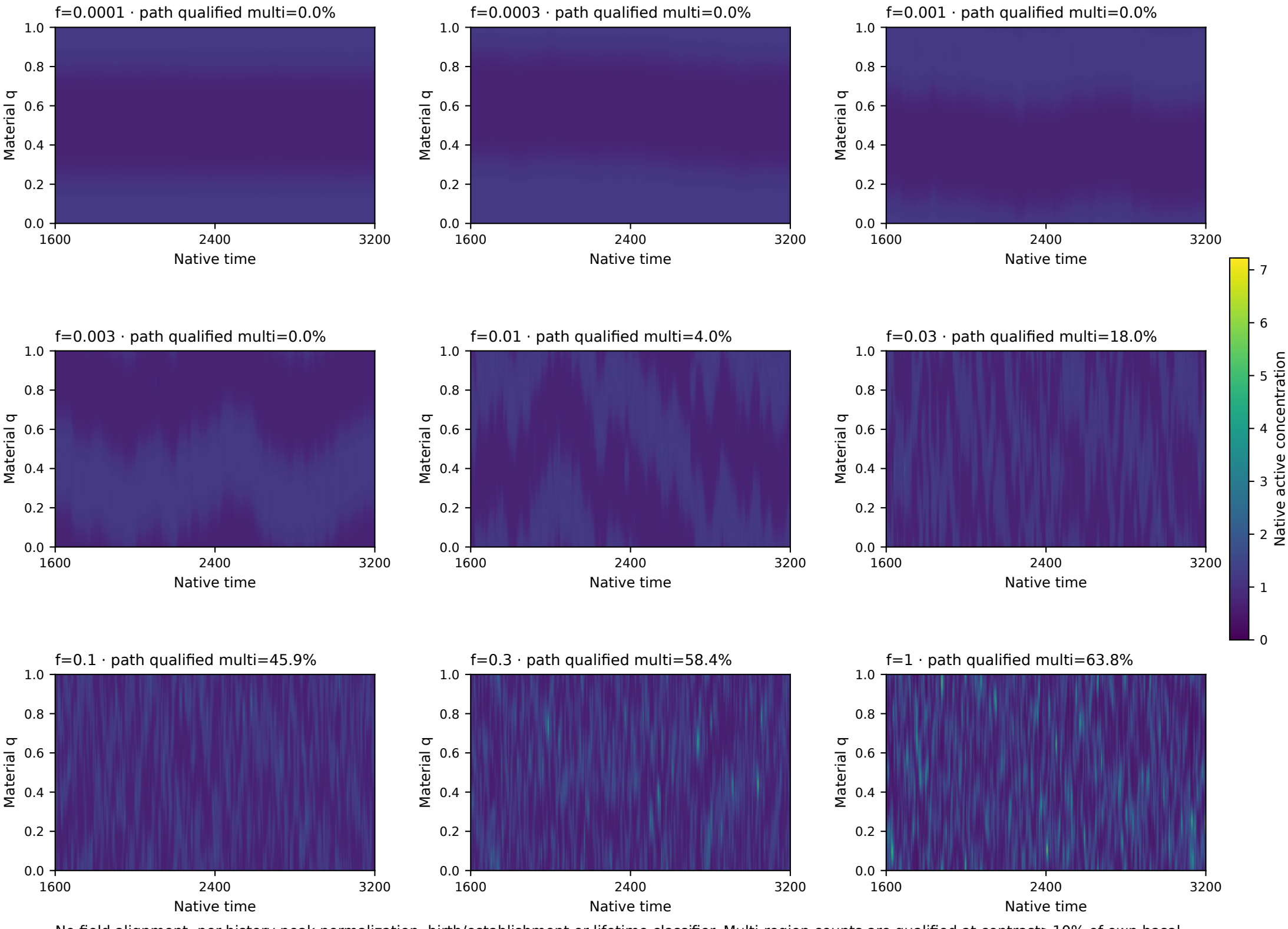
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C_0=2: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



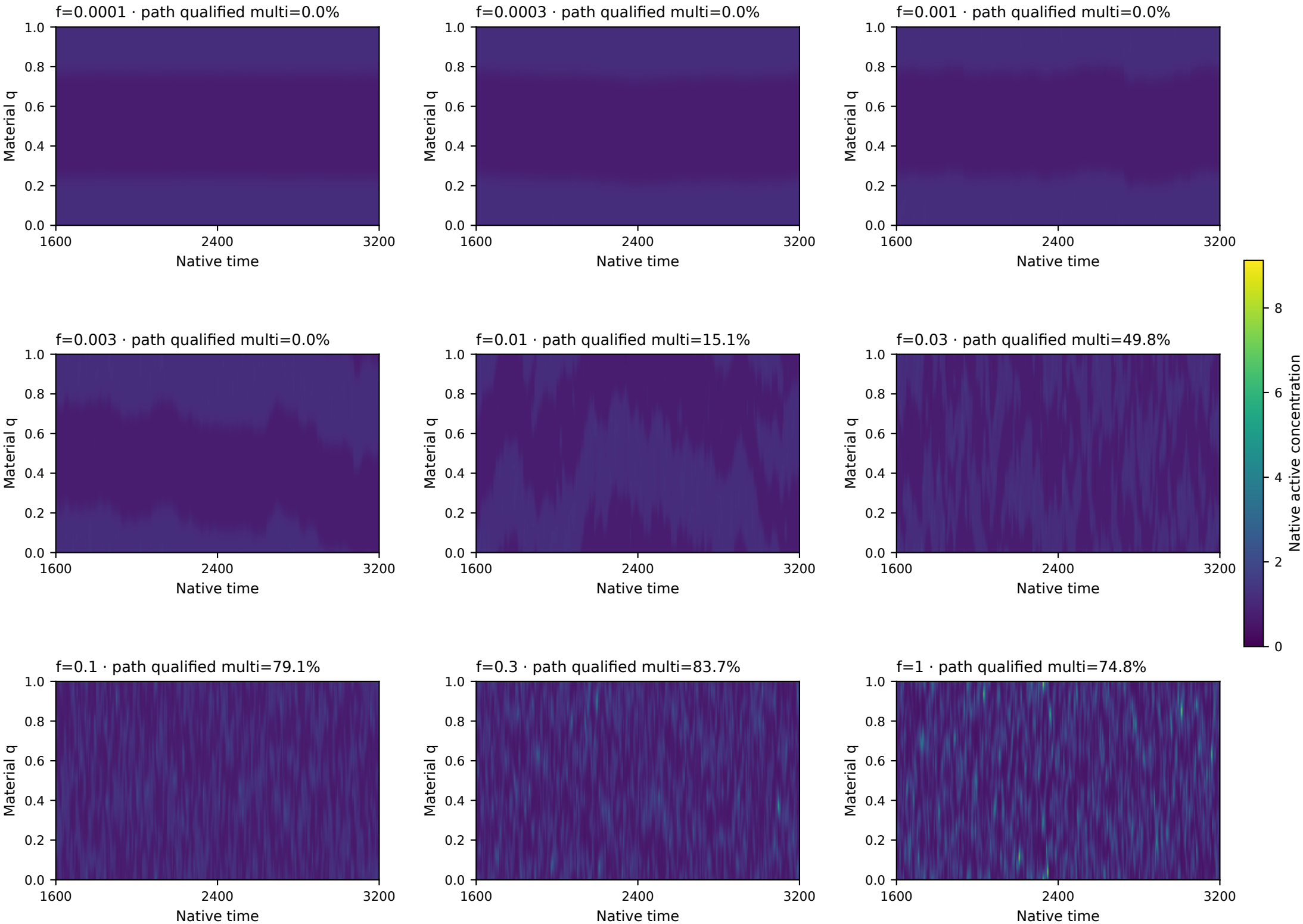
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C₀=3: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

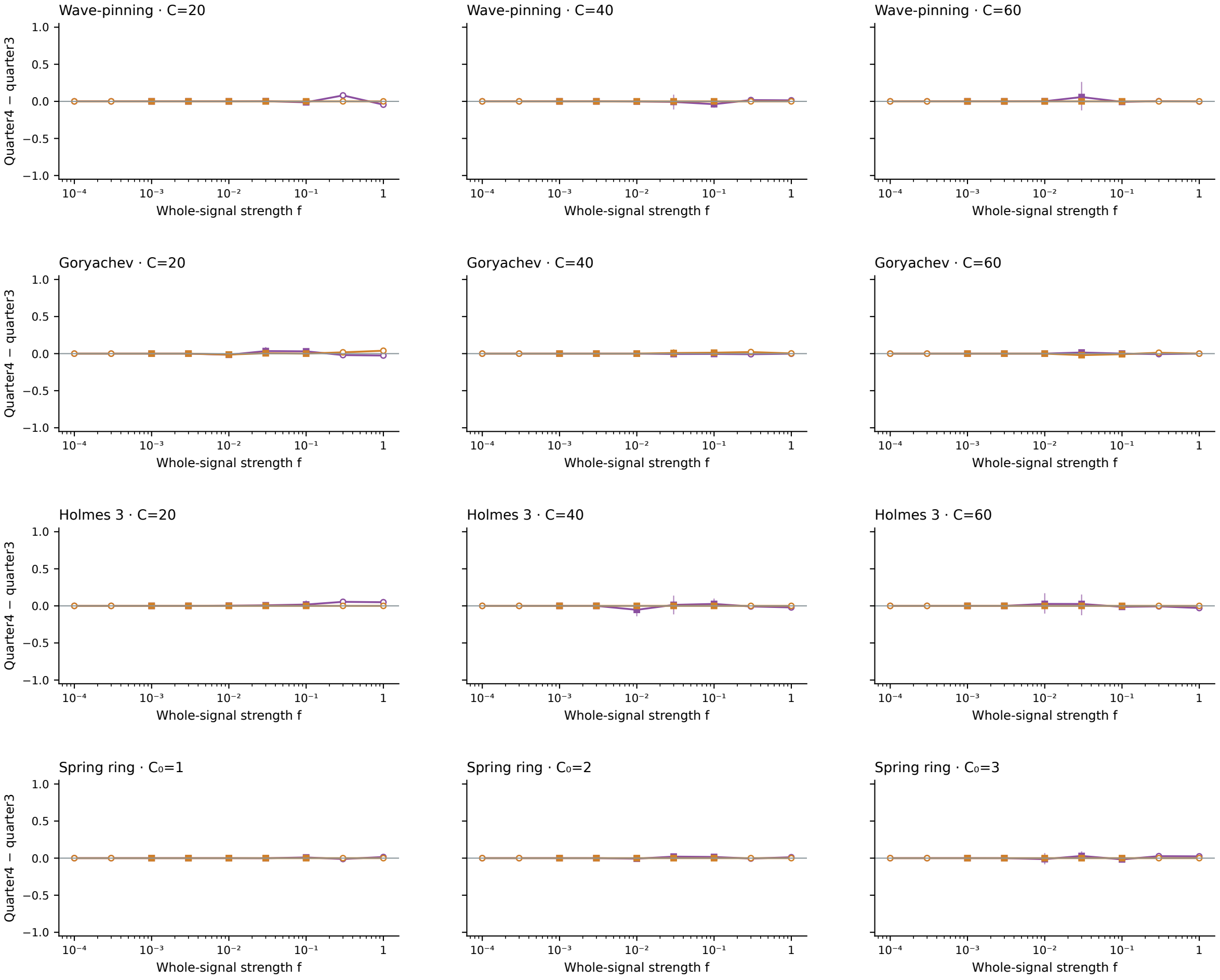
The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Late-quarter organization sensitivity

Quarter4 minus quarter3: chemicals24000–32000 minus16000–24000; Spring2400–3200 minus1600–2400.

Qualified multiple-region occupancy Low-contrast occupancy

Filled squares:n=12; open circles:n=2. Bars:95% whole-seed intervals where n≥3 and direction support≥95%; unavailable directions stay absent.



These are finite-window differences, not a stationarity pass/fail test. Intervals resample independent whole seeds with quarter pairing preserved.

Quarter direction uses shorter native lag2000/200; main lag8000/800 leaves only one pair per seed in a quarter and cannot meet the reporting support gate.

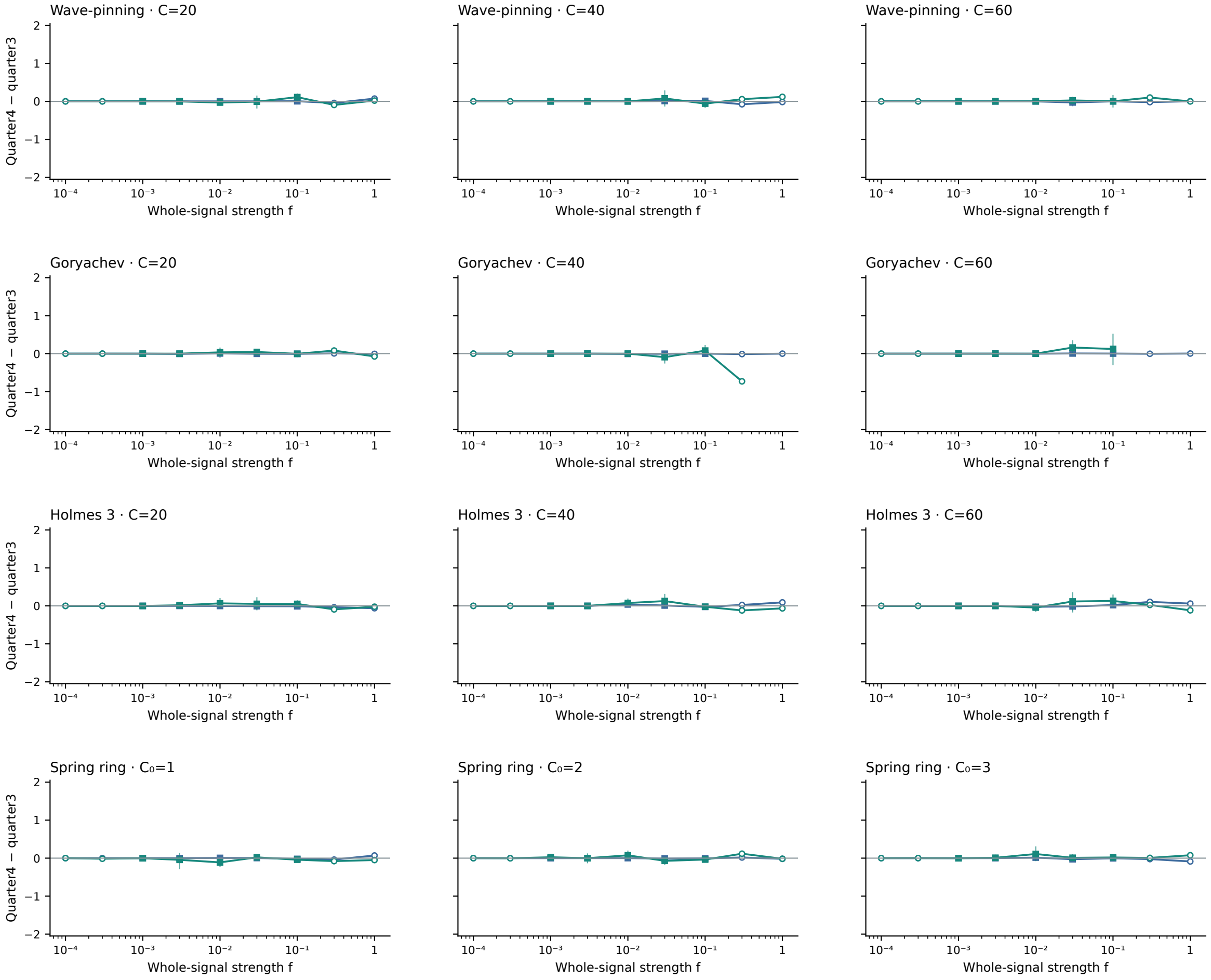
Use the exported quarter-specific pair coverage and seed counts. No continuous region lifetime or migration-velocity inference is made.

Late-quarter magnitude and direction sensitivity

Quarter4 minus quarter3: chemicals24000–32000 minus16000–24000; Spring2400–3200 minus1600–2400.

Mean R Direction cosine at native lag2000/200

Filled squares:n=12; open circles:n=2. Bars:95% whole-seed intervals where n≥3 and direction support≥95%; unavailable directions stay absent.



These are finite-window differences, not a stationarity pass/fail test. Intervals resample independent whole seeds with quarter pairing preserved.

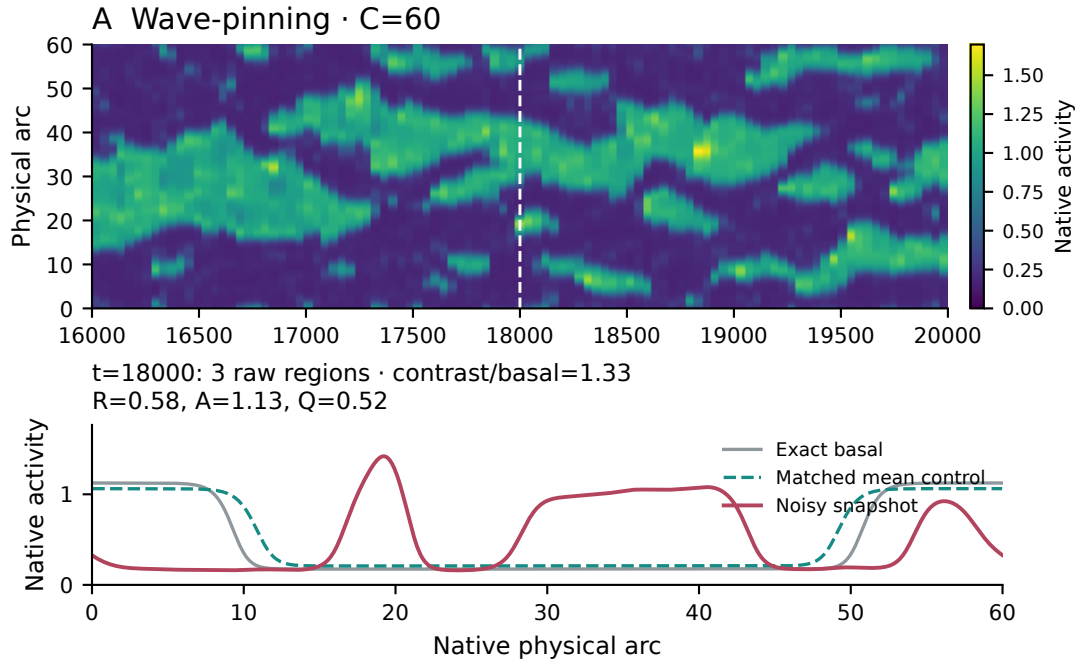
Quarter direction uses shorter native lag2000/200; main lag8000/800 leaves only one pair per seed in a quarter and cannot meet the reporting support gate.

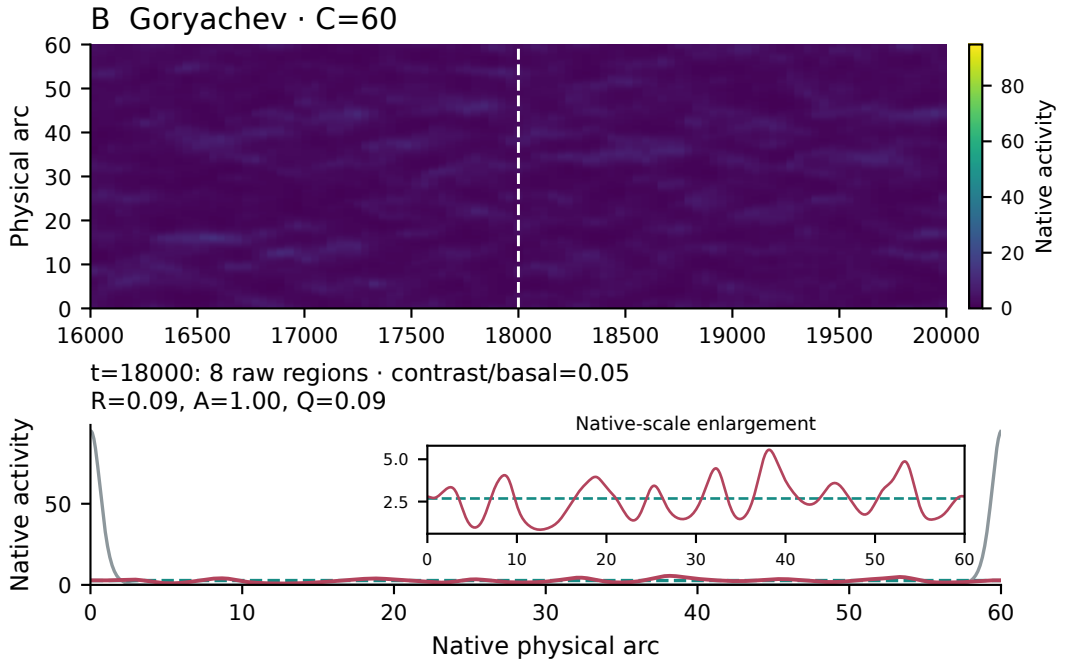
Use the exported quarter-specific pair coverage and seed counts. No continuous region lifetime or migration-velocity inference is made.

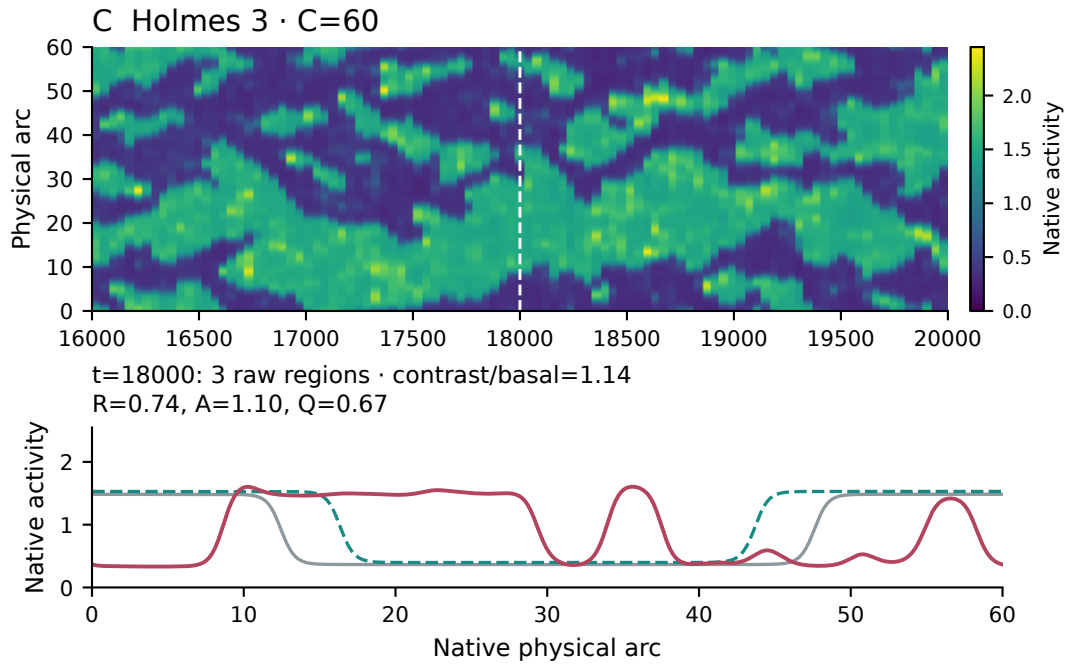
Local organization during a short second-half window

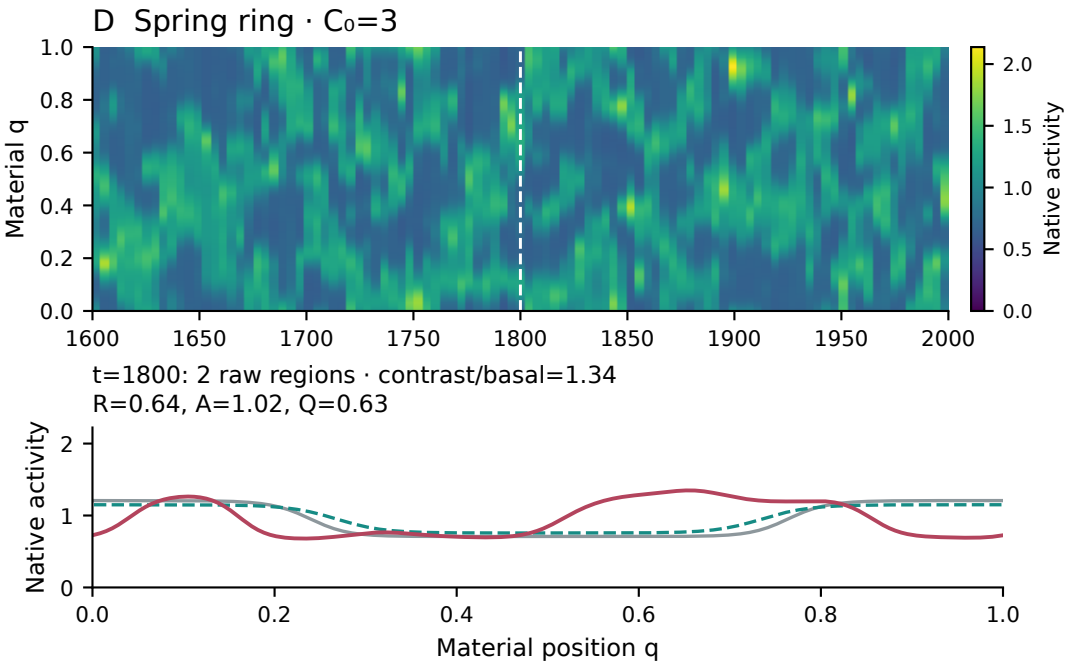
Largest original size; f=0.1. Fixed display window begins at half the run; white line marks the native profile shown below.

Morphology-descriptor medoid from12 independent paths per condition; fixed snapshot time is independent of field values or region counts.









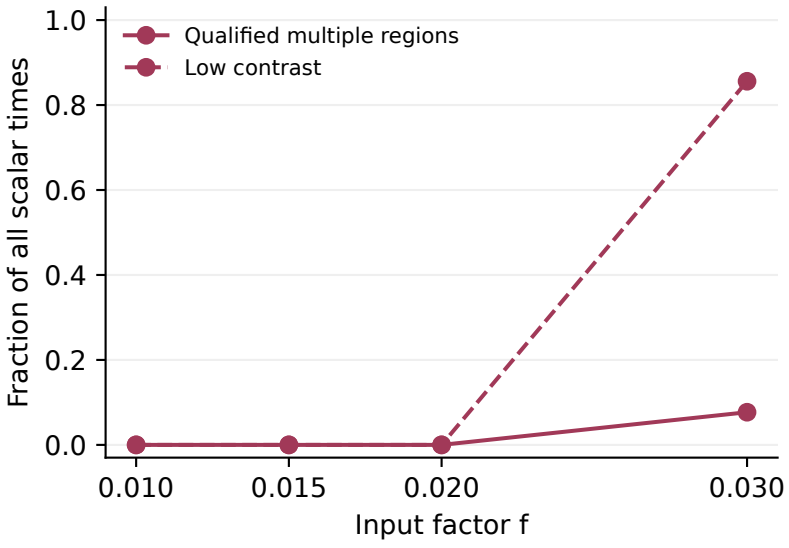
A region is a connected arc above half current contrast; counts need amplitude context. Goryachev's high-input noisy AND mean-control profiles have low contrast.

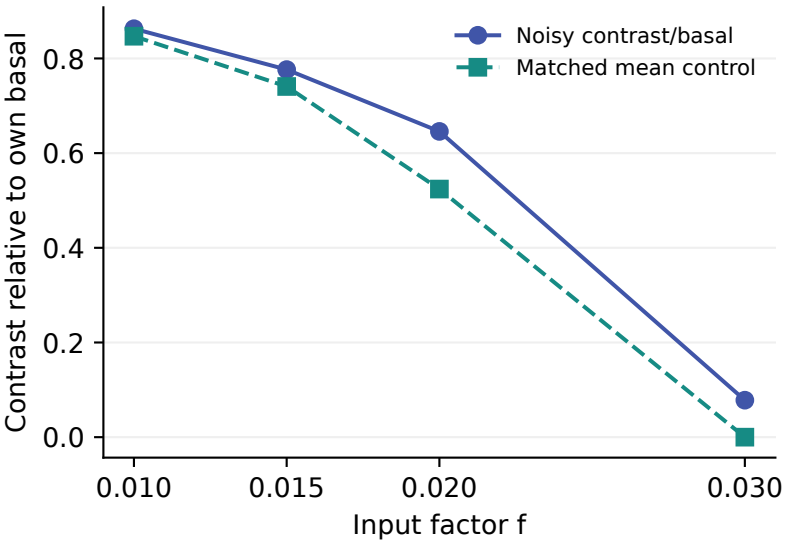
Snapshots are fixed in advance within the displayed windows, not selected for extra peaks. Regions do not establish competition or site lineage; no duration classifier.

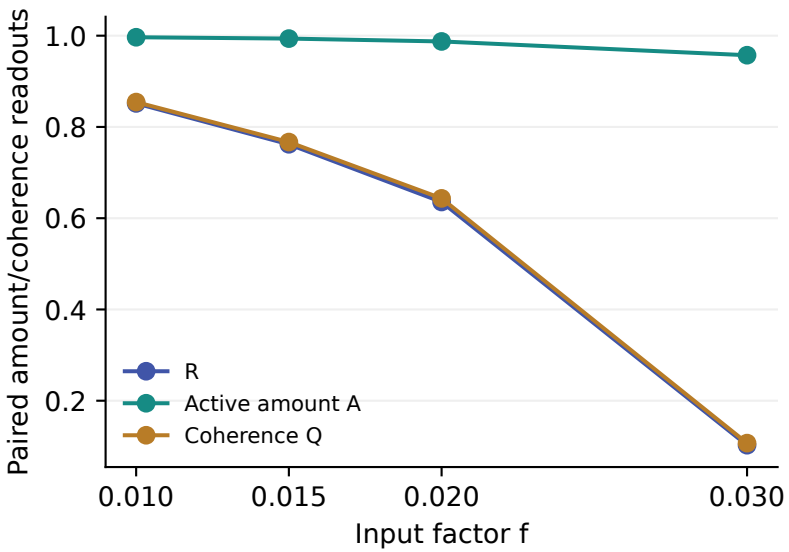
Native activity; no alignment/peak normalization. Saved fields: chemicals40, Spring4 native units; actual C(t) enters Spring M_0/M_1. Laws/clocks differ.

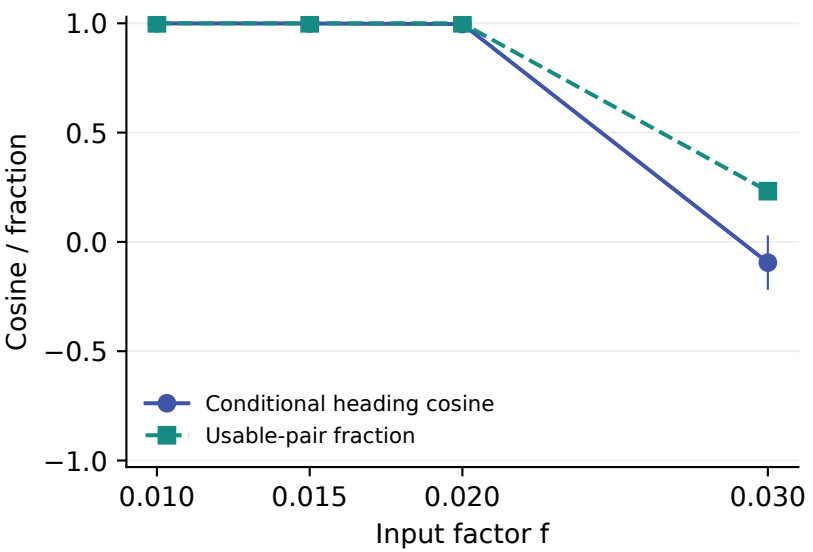
Goryachev C60: targeted refinement of the sampled transition

f=.015/.02 are new; .01/.03 are exact retained anchors. Twelve noise seeds and a matched mean control per strength.







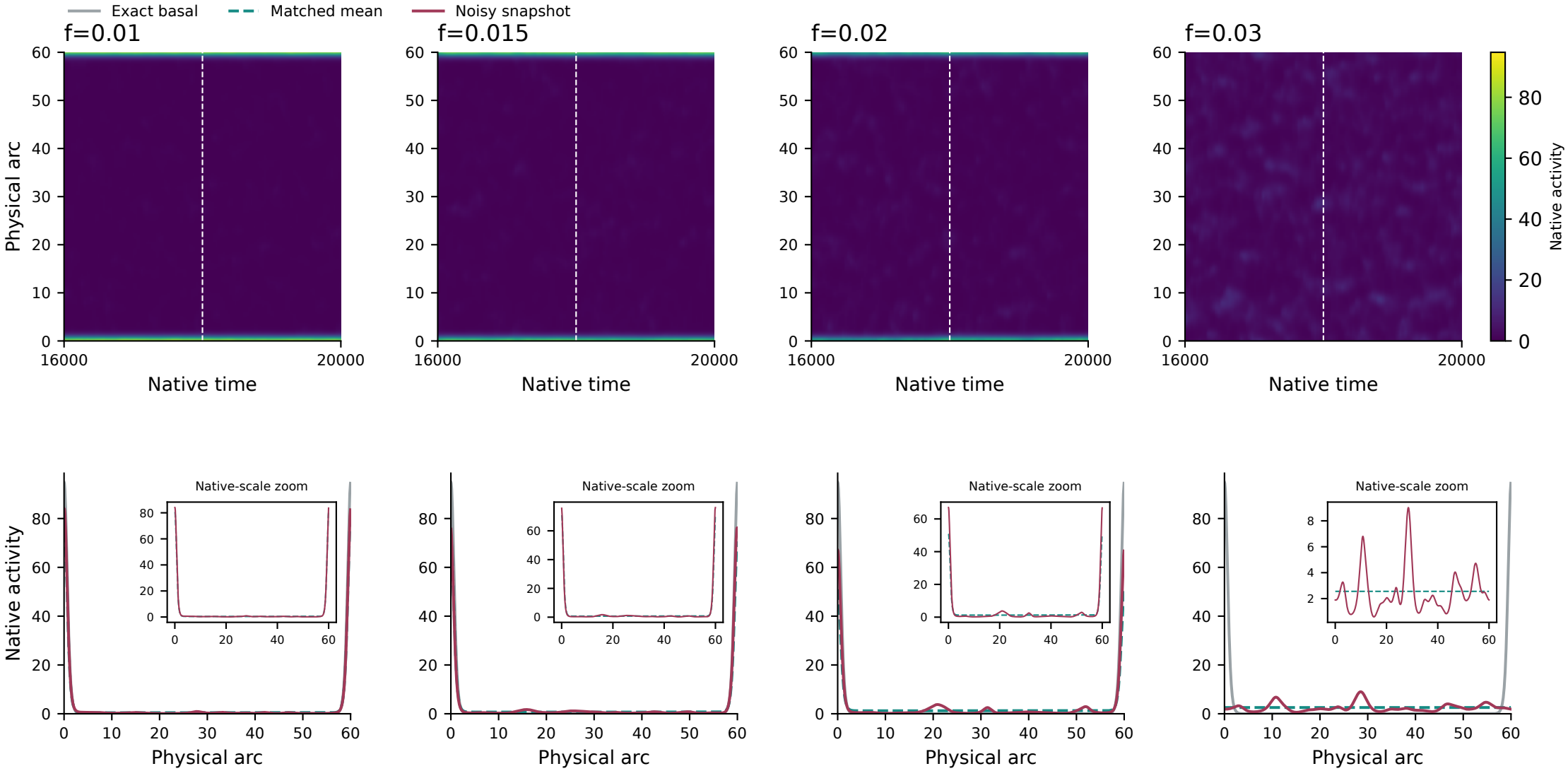


Second-half window16000–32000; 2000 whole-seed pointwise intervals. Direction lag8000 requires endpoint R≥0.1.

No phase boundary, competition, continuous lifetime or migration law inferred. Positive mean and fluctuations scale together.

Goryachev transition: native medoids and fixed-time profiles

Same zero-based native activity scale; fixed window16000–20000 and snapshot18000. Medoids use the unchanged seven-descriptor rule.



White lines mark the exact native profiles below. Insets enlarge native noisy/mean amplitudes, without normalization.
The incumbent crosses the 0/60 periodic seam: its two plotted edge portions are one circular region.